

Dalek faceted-body design and assembly

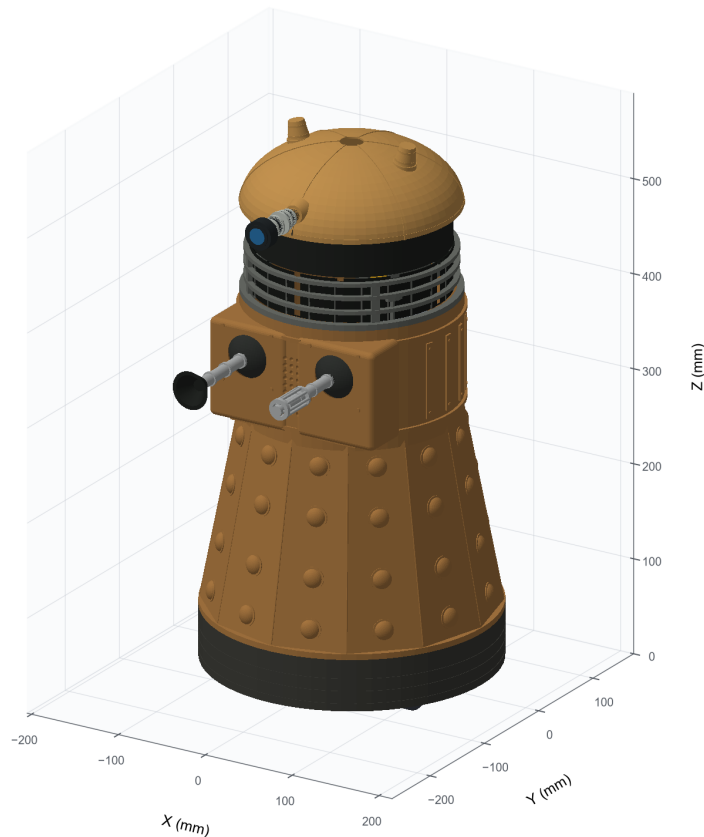
FACET-1 uses ten STL designs for a reinforced circular base, a 213mm skirt with twelve full-height flat panels, a combined 186mm shoulder and neck, smaller concealed arm mechanisms and an adjustable head friction drive. Four printed motor clamps use four screws total; motor ties can replace those clamps. The screw-clamp option needs fourteen prints, and the motor-tie option needs ten.

The robot keeps four TT drive motors, a rear ESP32 display, Wi-Fi access point and network controls, circular arm motion, a fifth TT motor driving the head through a friction wheel, and 24 original MP3 voices. A single printed board platform with legs fits over the original battery or the specified 7Ah/14Ah lead-acid alternatives. The package includes editable CAD, five circuit sheets, both retention options and assembly instructions.

This is a digitally checked design for a first prototype. Physical fit, loaded traction, supply temperature and stopping behavior require the tests in this manual. Do not skip those gates.

Dalek FACET-1 / assembly

Actual STL geometry | Purchased hardware uses nominal envelopes



Actual FACET-1 STL geometry | Dimensions in mm | +Y is rear; -Y is front

CAD assembly view. Purchased parts and internal features may be simplified in the preview.

Using this package

Read the assembly sequence first, then use the exact part-specific mechanical instructions. Read the firmware procedure before wiring or flashing. The electrical research chapter explains the power and pin choices. Full-size circuit sheets and complete bills of materials follow the text.

The source package is in JeremyProffittOrg/robots on main under dalek/. Each drawing and source filename in this PDF is relative to that directory. Original files and firmware:
<https://github.com/JeremyProffittOrg/robots/tree/main/dalek> .

Use the PDF bookmarks to jump between chapters. The final appendix lists parts and wires. The supplied digital verification record reports what was actually checked and what remains unverified.

Assembly and commissioning sequence

Build and test one subsystem at a time. The mechanical chapter gives the exact part filenames, fasteners, bearing fits and assembly order. The electrical chapter and circuit sheets give the exact wire connections. This chapter places those operations in a safe order and defines when each stage is complete. A check is not complete until it has been performed on the physical robot.

1. Inventory before printing

Read the whole manual and open the CAD preview. Inventory `bom/electronics.csv`, `bom/hardware.csv`, and `bom/printed-parts.csv`. FACET-1 has ten STL designs. Print the shared pitch carrier twice and the optional motor clamp four times, giving fourteen printed pieces. The motor-tie option omits all four clamps and needs ten prints. The skirt is one 213 mm print with twelve full-height flat panels and a single straight taper; the shoulder and neck are one 186 mm print. Select one battery and its matching charger. Keep purchased screws, bearings, retainers and spacers in labelled bags.

Check the actual board: it must be the original TTGO T-Display ESP32 with the 1.14-inch screen. Inventory four MG92B servos, five TT3777 motors, five 3766 wheels, three DRV8833 boards and the specified battery. Four motor/wheel pairs propel the base; the fifth turns the head by friction. Obtain the exact wheels before accepting either fit. Check board and servo part numbers against the drawings.

Tools: a digital calliper, metric hex keys, small Phillips drivers, flush cutters, a deburring tool, soldering iron, solder, wire stripper, crimp tool for the selected connectors, heat-shrink tube, digital multimeter, a current-limited bench supply, and a scale. A contact temperature probe and an oscilloscope improve the power and noise tests. Use eye protection when cutting wire or drilling. A threaded fastener, bearing, motor, battery or connector is purchased hardware, not a printed substitute.

Done when every required purchased item is present or its absence is recorded, the board variant is confirmed, and the tools can measure voltage and current. This is an inventory gate. It does not require assembling or energising anything.

2. Prove fit with small prints

Print `07_pitch_carrier.stl`, `10_head_motor_carriage.stl` and one `11_motor_clamp.stl` if using clamps first. Use them to check the real servo case, ears, supplied horn, TT case, sliding slots and screw fits. The servo shaft offset and ear thickness are measured fit gates. Then print `01_base.stl` and check the motors, fully seated wheels and selected retention method. Check each subsequent part's screw holes, bearing seats and board slots. Clean the first-layer edge before concluding that a dimension is wrong. Do not force a bearing or crush a board into a tight print.

If a fit needs adjustment, change the relevant clearance parameter in the OpenSCAD source, export that part again, and test it again. Do not scale the whole model in the slicer. Record the material, nozzle, layer height and clearance that passed. A printer-specific compensation may be needed, but it must preserve the joint centres and part lengths.

Done when fasteners pass without splitting the part, nuts retain their flats, the motor cannot shift in its holder, servo bodies seat without force, and the screen is visible with the USB connector accessible. The rear holder must retain the PCB by its edges without pressing on the glass.

3. Print and label the remaining parts

Follow the orientation and quantity fields in `printed-parts.csv`. The body order is 01_base, 02_skirt, 04_shoulder, and 06_head; identifiers 03 and 05 are retired. Print each complete body section upright at 100% scale. Print 12_electronics_platform.stl deck down with its four legs up, as exported. Its supports must be removed from every foot, key and fastening groove. Print four identical hook clamps only for the motor-screw option. The taller skirt and combined upper shell keep overall height at 563.8 mm.

Use the H2D single-nozzle 0.4 mm profile for the circular base. Keep its floor flat on the bed. The 300 mm diameter plus an 8 mm brim allowance needs 316 x 316 mm. Use the verified left-nozzle placement at X162.5,Y160, within X0..325 and Y0..320. The package's `scripts/slice_h2d.py` uses an isolated copy of the installed settings and records the actual model, support and generated brim bounds. Do not use the combined 350 mm bed width as a single-nozzle limit. Use the specified PETG profile for the base and structural parts. Enable the recorded automatic supports and inspect the sliced adhesion paths before printing; a requested brim width does not prove that the slicer generated a brim around a supported feature. Remove supports fully from nut traps, screw holes, grooves and bearing seats. Keep paint out of the stacking registers and moving fits.

The FACET-1 skirt slice uses 1009.47 g of PLA including supports and brim and predicts 23 hours 55 minutes 37 seconds; allow at least 1.12 kg with a 10 percent reserve. The unchanged combined upper shell uses 1235.49 g of PETG; allow at least 1.36 kg. Each job exceeds one 1 kg spool. Arrange compatible filament changes or sufficient spool capacity. The complete clamp-option package predicts 4180.82 g of filament, 2473.98 g of installed printed plastic and 124.28 hours of sequential printing. Filament purchase allowances are reserves, not predicted consumption. Match each printer job to its released mesh hash.

Weigh each group after support removal. Record total printed mass and the eventual fully assembled mass. Do not use slicer infill percentage as a direct mass estimate for thin shells: most of a thin shell may already be walls. The validation report's solid mesh volume is a useful conservative bound for fully filled material, not a slicer timing estimate.

Done when every manifest quantity is made, identified, free from cracks and checked against its mating piece. Do not close the cosmetic shell yet.

4. Assemble the chassis without power

Follow the base and motor-retention steps in `mechanical.md`. Perform the physical proof-load test below before adding motors or electronics. Inspect the perimeter, floor, ribs and pocket walls. Retain the four standard M4 base-joint nuts in their underside pockets first. Each motor then uses one printed hook clamp with one M3x16 screw, washer and nut, or two longitudinal 4.8 mm ties. This is four motor-clamp screws total. Tighten only until the motor is secure. Do not put liquid thread-locker on printed plastic.

Seat all four motors in the specified direction. Give their thin motor leads strain relief. Press each wheel onto its TT output shaft while supporting the gearbox body. The wheel's hub fit provides retention; do not drill the motor shaft. Check that tires cannot rub the bumper, a bolt, or a motor cable at any rotation. Check that the assembled chassis does not rock on a flat surface.

With the skirt off, place the disconnected battery on a 2 mm pad in the base cradle. Fill the spare space around smaller packs with firm nonconductive foam. Retain the pack with two separate padded ties, at least 500 mm long, 4.8 mm wide and rated at least 222 N. Keep terminals and wires out of the tie paths. Build the single board platform on the bench using the real PCB templates, its deck slots and 6 mm nonconductive spacers. Start with M2.5x20 PCB screws, or six M2x20 screws for the Pololu boards; trim and deburr to leave at least two full exposed threads and at most 3 mm projection below the deck. Board ties over clear insulated edges are an alternative.

Lower the populated platform over the battery onto its four rail seats. Use eight M3x12 screws with washers and nuts, or eight independent foot ties. The screw option uses the specified 2.5 mm L-key along Y from the side; the wheels can remain fitted. Keep terminal hardware below base-local Z130 and wire bundles inside the deck outline. The deck underside is Z140, its top is Z146, and PCB undersides are Z152. Before lowering the skirt, seat its four upper-joint standard M4 nuts from below and test each thread. Keep washers out of those pockets. After electrical bench tests, lower the whole skirt around the chassis and retain it with four M4x20 screws and screwhead washers into the captive base nuts. The platform does not pass through the skirt's smaller top opening.

Done when all wheels rotate by hand without body contact, motors cannot slide, the chassis rests evenly, and the battery can be removed without cutting wires. The mechanical chapter defines the exact floor and wheel clearances.

Before installing electronics and motors, perform the base proof-load check. Support the four motor-pocket floors on equal-height rigid blocks. Apply a distributed load equal to twice the intended complete robot mass to the actual battery and body-stack load areas for ten minutes. For a 4 kg finished robot, the proof load is 8 kg. Use the actual planned mass including hardware; the earlier 3 kg target is not a tested load rating. Keep all motors, wheels and the battery out of this test. Inspect before loading and after unloading. Fail the base if it cracks, a layer separates, a fastener boss loosens, or measurable permanent distortion prevents a flat fit or free motor insertion. This is a required physical builder test, not a strength result measured during design. A static proof test does not establish impact or fatigue life.

5. Build the power harness on the bench

Make the harness from the five circuit sheets and `electronics/wiring.csv`, using the listed wire gauges and connector ratings. Keep the single 7.5 A main fuse F1 within 100 mm of the intact Bioenno PP30 or the SLA positive terminal, including any adapter lead. Use matching insulated F2 contacts and a correctly polarized PP30 adapter for the specified SLA packs; an existing F1 pack needs F1 contacts. Keep the existing branch fuses and motor limits. Put the physical actuator switch in the feed to both movement converters. Keep the logic branch separately powered from MAIN. Terminate returns at the common ground point. Do not use a solderless breadboard or loose Dupont contacts for motor, servo or battery current.

Label every connector at both ends. Use a different keyed connector, or a clearly different connector size, for battery voltage and 5 V. Insulate exposed terminals and add strain relief. Confirm electrolytic capacitor polarity. Fit the motor noise capacitors close to the motor terminals. Never place a diode directly across a bidirectional motor as though it were a relay coil.

With the battery disconnected, check continuity of every wire against the net list. Check for shorts between each supply and ground. Inspect the physical switch and fuse ratings; an AC-only current rating is not a sufficient DC switch rating. Check the ADC resistor values with the meter before connecting an ESP32 input.

Done when every net has been checked, no supply is shorted, the battery connector polarity is correct, and the actuator switch interrupts both movement branches. Keep all downstream boards disconnected for the next stage.

6. Check the three supplies before connecting boards

Energise the harness from a current-limited bench supply at a voltage in the battery's operating range. Check each converter's unloaded output. Each 5 V branch must be within its specified tolerance and suitable for the attached devices. Then test each branch with an appropriate known load. Do not connect the battery voltage to a 5 V rail to make a failed converter test pass.

Confirm the actuator switch makes the motor and servo outputs fall to zero after stored capacitor charge has decayed. Confirm the logic output remains available. Check the external pack divider voltage against the calculation in the electrical chapter. At the maximum charge voltage it must remain below the ESP32 input limit. A

reversed divider is a board-damaging error.

Done when the motor, servo and logic rails pass unloaded and loaded checks, the pack divider passes, and the physical actuator switch has the intended effect. A warm regulator alone does not prove a safe load; measure its temperature and voltage under load.

7. Flash and start the controller on USB only

Disconnect the battery and external controller 5 V supply. Leave all actuator supplies off. Connect the T-Display to the computer using USB. Do not back-feed the USB supply from an external 5 V source. Use the complete flashing procedure in `firmware.md`, including both the program and the LittleFS image. A program upload alone does not install the audio files.

After boot, check that the rear display is readable and shows network and safety status. Join the robot's access point using the credentials shown by the controller. Open the displayed local address. Check that the interface loads with the phone's Internet access disabled. It must display the sound list and controls without fetching a web font or script from a public website.

Store home Wi-Fi settings only through the local setup form. Use a 2.4 GHz network supported by the ESP32. Check the displayed station address after joining. If the home network is unavailable, follow the fallback procedure. Do not put Wi-Fi passwords in source files, a screenshot, this manual, or Git.

Done when the firmware and filesystem load, the TFT is readable, direct AP control works, station setup works on the intended network, and startup does not arm movement. Hardware-dependent status may correctly block arming until the next stages are complete.

8. Verify logic and hardware disable states

With USB removed, reconnect the external logic supply and the low-voltage control harness. Keep movement power off. Confirm the ground is common before connecting signal wires. Check the PCA9685 address and the buffer connections. Verify the motor sleep line is low and the servo output-enable line is in its disabled state at reset.

U8, the four arm-signal buffers, operates from 5V_SERVO. U9 routes head PWM and operates from 5V_MOTOR. Each signal and supply must match circuit sheet03. GPIO2 has a 10k pull-down and supplies 20kHz head PWM. PCA channels 4/5 are static direction enables; never attach servos to these channels. Check that all three driver SLP pins and GPIO2 stay low during reset. No 5V power cable may connect to the ESP32's 3.3V pin. Compare the ADC display with a meter and apply its calibration factor if needed.

Done when reset leaves actuators disabled, the controller detects the actuator switch state, the PCA9685 responds, and battery voltage agrees with the meter to the stated tolerance. A blocked arm command is the correct outcome while power or sensor checks fail.

9. Test drive direction with wheels lifted

Support the chassis so all tires are clear of the bench. Keep hands away from the wheels. Connect the movement supplies and explicitly arm through the interface. Apply a brief, low command to both sides. Each wheel must turn in the direction that would move the robot forward. The right-side motors are physically mirrored, so their lead polarity must be checked, not copied by wire colour.

If one motor on a side turns backward, disarm, isolate power and reverse only that motor's two output leads. Do not connect two driver outputs together. Once directions are correct, test a slow forward command, a stop, a slow reverse command and a stop. Listen for rubbing and inspect driver temperature. Do not hold a stopped wheel against power to measure stall behavior.

Done when all four directions are correct, release stops the command, reversing passes through zero, and no tire or wire rubs. Leave final turning and loaded traction tests for the floor stage.

10. Centre servos before attaching the arms

Keep the arm horns and head drive connection detached. Command the positional channels to their documented centre pulse. Check that each servo stops and does not buzz against a hard limit. After the servo mounts are secured and their retaining tools are removed, fit the supplied horns at the neutral positions shown in the mechanical chapter. Retain each horn with its own centre screw. Do not print a replacement servo spline.

Keep the combined upper shell off the skirt on a padded stand. Fit the lower bearing, spacer, shims, retainers and spindle while the bottom is accessible, following the mechanical chapter. Leave the speaker and rear display out until both arm paths are complete. Retain each servo with two M2x10 socket-head cap screws or two 2.5 mm ties; neither option replaces the mandatory horn fasteners. Feed the detached arms through the rear of the lower opening, then turn their tips outward through the front sockets. Support their exposed stems. Insert the prepared carriers from below using the mechanical chapter's complete lower-entry path before seating them on the yaw horns. Use the exact multi-step paths in the mechanical chapter. Fit opaque stretch-fabric socket liners with loose folds. Use the current nominal 30 mm fixed outer radius and 24 mm moving inner radius, keeping the fixed edge inside the smaller box. No servo case, horn, tie or lead should be visible through a finished socket.

Start circular motion at 2 degrees and 0.10 Hz. Check one arm at a time. Correct a reversed axis using the calibration setting, not a cable swap on a three-wire servo. The revised maximum is 8 degrees; the older 12-degree limit does not apply to this concealed mechanism. The tip should trace a near-circle without contact, pulling a wire or tightening the fabric. Inspect the complete motion from front, sides and below to confirm concealment.

Done when both arms can complete repeated circles quietly, every fastener remains secure, no servo stalls, and the supply stays within tolerance. Continuous buzzing, a hot servo or a distorted orbit calls for a mechanical or calibration correction before proceeding.

11. Assemble and test the head

The two 608 bearings, spacer, shims, retainers and M8 spindle belong in the integral upper bearing tower. Fit the lower hardware through the open bottom before seating the upper shell. There is no separate neck joint. Fit the fifth TT motor and wheel to the sliding carriage on the bench, then lower that assembly through the top. Install all four slide nuts and underside washers through the open bottom. Keep the tyre retracted from the drum. Fit the speaker to its rear open frame, then fit the rear display and switches after both arm insertion paths are complete, while the bottom remains accessible. Complete these steps and the arm bench tests before lowering the upper shell onto the skirt. With the head off and carriage retracted, fit four M4x20 screws with screwhead washers into the standard nuts already captured below the skirt's upper flange. Use the driver routes in the mechanical chapter.

With power isolated, fit and retain the head, then follow the jackscrew and guide-clamp sequence in `mechanical.md`. Increase contact only enough for reliable traction. The 3.6 mm slide range includes a released gap and a hard stop; it is not a calibrated force scale. Lock the setting, rotate the head through a complete turn by hand and check for binding. Use the specified access ports and short hex key. Do not tighten contact to force the head past a misaligned bearing or screw. Keep fixed motor leads clear of the rotating track.

Test low drive in each direction and zero output with the assembled robot supported. Record slip, coast angle, current and temperature. Done when the head turns through repeated full revolutions in both directions, zero removes powered drive, the tyre setting remains locked, and the head stays retained. Do not lift by the eye stalk. The controller sets motor drive level, not a measured head angle or speed.

12. Install and check audio

Confirm the speaker was fitted to its body mount after the arm assembly paths were complete and before the upper shell was seated. Keep both speaker terminals isolated from chassis ground. Wire the MAX98357A exactly as shown; its bridge speaker output has no grounded speaker lead. Confirm the gain resistor before applying power.

Play `hello.mp3` at low gain. Then test every sound button. Check for distortion, controller resets and motion pauses while audio plays. Keep gain within the speaker's 1 W limit and the logic supply budget. Warning phrases in the collection are available as sound effects; their filenames do not prove automatic event playback.

Done when all 24 files play on demand, speech is intelligible, sound does not reset the ESP32, and amplifier/speaker temperature stays acceptable during repeated use. Keep the speaker and amplifier wiring fixed in the body.

13. Run the stop tests before floor driving

Keep wheels lifted for this complete test. Arm and command slow motion. Release the drive button and confirm the command stops. Repeat and change browser tabs. Repeat and close the page. Repeat and disable the phone's Wi-Fi. Repeat and turn off the access point or station connection used for control. Measure the delay to removal of the drive signal; the firmware lease is 500 ms, and the exact task and I/O delay must be checked on hardware. Tire coasting after removal of power is not the same as a continued drive signal.

Repeat every stop case with the head rotating and arms moving. A zero head setting must remove PWM immediately. Measure at least 100ms with both head bridge inputs low on a direction reversal, including a quick move through zero. The rotating head can coast after drive ends; record its coast angle. With head motor disconnected, scope the two U11 inputs and confirm only one carries 20kHz PWM at a time, capped at 100/255 duty.

Turn the actuator switch off during a slow command. Wheel, head-motor and arm-servo power must be removed. Turn it on again. Movement must not resume until the operator explicitly arms again. Reset the ESP32 during a slow command and verify no restart. Check that a second browser cannot revive a stale command after a stop.

Test low-battery detection with a bench supply, not by repeatedly exhausting the battery. The retained limits stop motion at 11.2 V and permit explicit re-arm at 12.0 V. Firmware leaves controller/audio power on after a low-battery stop. Switch MAIN off and unplug the pack; an SLA battery has no Bioenno-style internal protection. Restore normal bench voltage and confirm recovery still requires explicit arming. Temporarily disconnect the servo-controller communication while safely supported, and confirm the documented fault response. Reconnect with power isolated.

Done only when every stop case passes. Record the measured time and test setup. Do not move to floor testing with any unexpected restart or uncontrolled motion.

14. Close the shell and run a short floor test

Check the four complete body pieces: base, skirt, combined shoulder/neck and head. There are two bolted body joints, with eight M4x20 screws, eight captive standard M4 nuts and eight screwhead washers total. No washers belong inside the nut pockets. Check registers, rear screen, switches and service access. Route fixed cables clear of wheels, arms, the head track and screw tips. Leave labelled service loops. Confirm both battery ties and all platform feet hold their loads without movement.

Weigh the complete robot. Begin on a clear, level hard floor with an operator beside the physical stop switch. Use low speed and broad forward arcs. Do not begin with a stationary pivot, carpet, ramp or threshold. The specified tires can create more steering resistance than these small motors can overcome. Stop immediately if a wheel stops

turning while commanded.

Run for one minute, isolate power and inspect joints, cables, tires, motor temperature and converter temperature. Repeat for five minutes, then fifteen minutes if the earlier test passes. Record battery voltage, mass, floor type, speed setting, current and temperatures. A conservative commissioning gate is to stop testing at 50 C measured on a motor, servo or nearby printed mount, or on any sign of softening, odour or repeated electrical shutdown. This is a build-test threshold, not a replacement for manufacturer component ratings.

Done when the fully dressed robot drives, steers in moving arcs, moves both arms, turns the head, plays audio and stops correctly under the same measured load. This is the first physical proof of the design's operating envelope.

15. Charge, maintain and store

Switch MAIN off, unplug the battery and remove it before charging. Use the Bioenno BPC-1502DC 14.6 V LiFePO4 charger only for BLF-1206A. Use the Power-Sonic PSC-121000ACX 12 V SLA charger for either specified SLA option; PSC-12500ACX is also specified for PS-1270. Label chargers by chemistry and check polarity. Never charge through the ESP32, its battery socket, an Adafruit single-cell charger or a converter output. Charge outside the body on a stable nonflammable surface with ventilation. AGM sealed lead-acid batteries must not operate or charge in an airtight enclosure. Recharge SLA after use and store it disconnected, following its manufacturer's instructions.

Before each use, check the straps, wheel fit, arm horns, head retention and exposed wires. Test the physical stop switch and release-to-stop behavior. Stop use if the battery is damaged or swollen, a connector is discoloured, a motor binds, or a printed bracket cracks. Replace a damaged structural part rather than gluing across a loaded fracture.

Do not carry the robot by an arm, the eye, dome decoration or upper body section. Lift from the single-piece base with battery power isolated. Keep the wheels free of hair and thread. Store the battery disconnected according to its manufacturer's guidance. Keep a copy of the calibrated firmware settings with the robot.

Fault guide

- Screen works but movement is blocked: check actuator switch state, pack ADC calibration, PCA9685 response, explicit arming and command timeout. Read the displayed fault before changing code.
- Controller resets when a motor starts: check rail sag, common ground, motor capacitors, converter load and connector resistance. Do not disable brownout protection.
- One wheel turns backward: isolate power and swap that motor's two bridge leads. Check all four wheels again while lifted.
- Arms buzz or fail to circle: check horn centres, direction calibration, gimbal contact, servo supply and 5 V signal buffers. Reduce amplitude while diagnosing the fit.
- Head moves under power at zero: turn ACTUATORS off. Check GPIO2 is low, the two U9 direction gates and U11 input wiring. There is no head-servo neutral adjustment. Distinguish a short coast from continued powered motion. Binding at one angle indicates track alignment, fastener intrusion or dome clearance; do not increase tyre pressure to overcome it.
- Robot moves straight but will not pivot: use moving arcs and a lower-friction floor. Measure mass and current. Do not raise motor voltage or remove current limiting.
- Audio is missing: confirm both firmware and LittleFS were flashed, inspect the sound list, verify I2S pins, and check speaker wiring and amplifier gain.

- Home network fails: join the local AP and follow the firmware recovery instructions. Reconnection must not restart motion.

Physical test record

Record date, builder, firmware commit, slicer profile, material, loaded mass and floor type. For each stage above, write PASS or FAIL with the measured result. Include supply voltages under simultaneous load, pack-voltage calibration error, stop delay, fifteen-minute temperatures and observed runtime. Until that record exists, the package remains a digitally checked design for a first prototype.

FACET-1 mechanical design and assembly

The robot has a 300 mm circular reinforced base, four ground wheels, a one-piece skirt with twelve full-height flat faces, a combined shoulder and neck, concealed arm servos and an adjustable TT-wheel head drive. The package has ten STL designs. Print the pitch carrier twice and the hook clamp four times for fourteen printed pieces. The motor-tie option omits all four hook clamps and needs ten prints. Each delivered STL is one connected part. Nominal height remains 563.8 mm, below the requested 914.4 mm limit. The 213 mm skirt and 186 mm upper shell stay below the 320 mm maximum-minus-5 mm print height target.

The reference review covers 100 distinct photographs. Retained details include the dark circular fender, four hemisphere rows, raised bezels, skirt relief, shoulder slats and rivets, smaller rounded arm boxes, spherical roots, three neck rings, dark grille liner, eyestalk discs, ribbed lamps and decorative arm stems. See [appearance-research.md](#) and [reference-image-review.csv](#).

This is a first-prototype design. No physical strength, payload, traction, runtime, bearing fit or finished robot operation has been measured. Match each generated validation and H2D report to the released FACET-1 mesh hashes. Complete the physical assembly, load and motion gates before operation.

Print inventory

Use the released STL files at 100 percent scale with a 0.4 mm nozzle, 0.20 mm layers and the recorded single-nozzle H2D profile. Its single-nozzle envelope is 325 x 320 x 325 mm. The 300 mm base plus an 8 mm brim allowance occupies 316 x 316 mm. Other parts request 5 mm brims. Normal automatic supports must be inspected in the actual sliced job. Do not substitute an orientation or auto-arrangement without checking its full toolpath bounds. [Bambu H2D specification](#).

1. `01_base.stl`: PETG, floor down. Six walls, six top/bottom layers, 40 percent gyroid. Diameter 300 mm, body 54 mm high plus 3 mm register, 6 mm floor and integral ribs. Clear supports from the wheel wells, clamp pockets, captive nuts and tie passages. The opened wheel wells admit fully seated motor/wheel units from above while the outside bumper stays continuous.
2. `02_skirt.stl`: PLA, large opening down. Four walls and 15 percent infill. Twelve flat panels taper in one straight line from 300 mm to 220 mm across opposite vertices over 210 mm. Across opposing flat faces, the body spans $300 \times \cos(15 \text{ degrees}) = 289.78 \text{ mm}$ at its lower end and $220 \times \cos(15 \text{ degrees}) = 212.50 \text{ mm}$ at its upper end. Round joint collars remain, with a maximum 300 mm outside diameter. The 3 mm top register makes a 213 mm print. Panel centers are at 15, 45 and subsequent 30-degree intervals. Each panel carries four domed details at local Z26, Z76, Z136 and Z186, with 23 mm bezels raised 1 mm from the face. Their axes follow the panel normal, tilted approximately 10.425 degrees upward from horizontal. The exterior middle band is removed. The integral internal faceted rib retains 240 mm clearance across opposing flats. The round bottom opening remains 264 mm and the main top opening 184 mm. Captive nut bosses project locally into that top opening, so do not use its nominal diameter alone to plan an insertion. Lift the whole skirt for battery/platform service. There is no bolted middle joint.
3. `04_shoulder.stl`: PETG, bottom flange down. Four walls and 25 percent infill. The shoulder and neck are one 186 mm print. Arm boxes, servo supports, open speaker mounting frame, display frame, grille, bearing tower, fixed head-drive deck and adjustment lug are integral. Clear every support through the open bottom before fitting hardware. Assemble its internal mechanisms on the bench before seating it on the skirt.
4. `06_head.stl`: PLA, drum rim down. Four walls and 15 percent infill. Friction drum, hub, spokes, dome, eye and lamps form one print. Remove supports through the drum and crown openings. Keep the inner drum track free of paint, glue and support scars.

5. `07_pitch_carrier.stl`: two PETG prints, floor down. Four walls and 30 percent infill. Each is 49.5 x 30 x 49 mm. Its Y bounds are -10 to +20 mm. Check the actual servo ears, shaft offset, ties and supplied horn before accepting fit.
6. `08_plunger_arm.stl`: PLA, as exported on the bed. Four walls and 15 percent infill. Spherical root, stem bands and cup are integral. Preserve the small servo-side stem clearance flat.
7. `09_emitter_arm.stl`: PLA, as exported on the bed. Four walls and 15 percent infill. Preserve the matching stem clearance flat. The eight rods and muzzle are decorative; there is no projectile mechanism.
8. `10_head_motor_carriage.stl`: PETG, floor down. Four walls and 30 percent infill. Retain its radial slots, motor saddle, tie routes and travel limits.
9. `11_motor_clamp.stl`: four PETG prints for the clamp option. Four walls and 30 percent infill. Print as exported with automatic supports under the bridge/toe. Omit all four for the drive-motor tie option.
10. `12_electronics_platform.stl`: PETG, deck down and legs up as exported. Four walls and 30 percent infill. The 216 mm deck, four legs and four feet are one 112 mm-high print. Clear supports and every foot fastening groove. No extra printed board spacers are required.

Identifiers 03 and 05 are retired. The four stacked body pieces are base, faceted skirt, combined upper shell and head. Use the current printed-parts manifest for all quantities and dimensions. Paint external surfaces only. Keep bearing seats, registers, sliding faces, horn seats, tie passages and the friction track clean. Use one bronze finish along each full-height skirt panel, silver stems and near-black bumper, liner, roots and cup.

The current FACET-1 skirt slice uses 1009.47 g of PLA including supports and brim, with a predicted print time of 23 hours 55 minutes 37 seconds. Its predicted installed plastic mass is 536.20 g after support removal. Have at least 1.12 kg available with a 10 percent reserve. The unchanged combined upper shell uses 1235.49 g of PETG and predicts 39 hours 10 minutes 11 seconds; allow at least 1.36 kg including reserve. Each job exceeds one 1 kg spool. Arrange compatible filament changes or sufficient spool capacity before starting. The full clamp-option package predicts 4180.82 g of filament, 2473.98 g of installed printed plastic and 124.28 hours of sequential printing. These are slicer estimates, not measured consumption or finished robot mass. Match every actual job to its released mesh hash.

Coordinate map and body joints

All coordinates are millimetres. Positive Y is rear, positive X is right and world Z0 is the floor. Base-local Z0 is world Z13.8, derived from the 31.5 mm wheel radius, 6 mm floor and nominal 11.7 mm case-bottom-to-axle datum. Measure actual motor padding and axle height.

- Base origin: world Z13.8; base body top Z67.8.
- Faceted skirt origin: Z67.8; one straight taper to body top Z277.8; register top Z280.8. Its internal rib center is Z177.8; there is no exterior slope change there.
- Combined upper origin: Z277.8. Its neck datum is an internal coordinate reference at upper-local Z120, world Z397.8; it is not a separate joint.
- Lower bearing seat: upper-local Z160..167. Upper bearing seat: Z179..186.
- Head STL origin: world Z449.8, upper-local Z172. Hub foot is world Z464.8, dome rim Z488.8 and lamp tops approximately Z563.8.
- Ground-wheel axes: X +/-98, Y +/-58, world Z31.5. The fully seated hub accommodation range is abs(X) 95.5..100.5. Never leave a wheel partly seated to obtain clearance.

- Arm yaw origins: upper-local (-47,-94,42) and (53,-94,42). Pitch-axis offsets are (-3,0,35). Neutral spherical centers are world (-50,-94,354.8) and (50,-94,354.8).
- Head carriage origin at nominal contact: upper-local (64.5,0,149.2). Head-wheel center: (64.5,0,191).
- Platform deck: base-local Z140..146; PCB undersides Z152; platform feet sit at Z36. Invert the exported platform with a 180-degree X rotation, then translate by base-local Z146.
- Battery bottom: base-local Z8 including the 2 mm pad. Maximum terminal envelope top: Z111. Keep attached contacts and leads below Z130.

There are two bolted body joints and eight M4x20 screws total. The base/skirt holes are at radius 137 mm. The skirt/upper holes are at radius 97 mm. Each set has four holes at 0, 90, 180 and 270 degrees. Use eight standard M4 hex nuts: four in the base's captive pockets and four below the skirt's upper flange. Use one washer under each screwhead, eight washers total. Do not put washers inside either set of nut pockets. Align the front relief and fully seat each register before tightening. The register locates the joint; screws retain it.

Before lowering the skirt around the chassis, turn it on a padded bench support and seat its four upper-joint nuts from below. Their reinforced supports are at skirt-local radius 97 mm, Z199..210. Seat each nut square and test it with a real M4 screw. Small insertion nibs retain the nut during handling; the pocket roof carries the screw load. Trim only a nib that prevents full seating. Do not remove the roof, widen the pocket or force a tilted nut. Use removable tape only to retain a seated nut during handling, with tape clear of the thread and bearing face. Leave the head off and keep the carriage retracted while fastening the merged upper shell through its specified driver routes.

Use a 250 mm exposed-shaft 3 mm ball-end driver rated for at least 20 degrees at the base/skirt joint. Its route tilts 16 degrees inward toward the center and clears the 184 mm upper opening. Keep bulky couplers out of the route. Captive base nuts remove the former deep underside nut-tool operation. A documented driver option is [Stahlwille 10507N KK](#), [article 73103003](#), with a 300 mm blade and listed 25-degree swivel angle. Dry-fit the actual tool before closing the shell.

Purchased part checks

The five [Adafruit 3777 TT motors](#) and five [3766 wheels](#) retain their original duties: four ground pairs and one head-drive pair. The wheel is nominally 63 mm diameter and 29 mm wide. The [motor drawing](#) gives a 22.44 mm case height, 70 mm length, 22.4 mm metal-end width and 36.6 mm across shaft ends. Measure actual cases, seated hubs, both shafts, leads and loaded tire deformation.

The original TTGO T-Display PCB is nominally 51.52 x 25.04 mm. Insert it from inside its 52.2 x 26 mm recess behind the rear lip. Keep the display, buttons, antenna and USB clear. Use the confirmed original 1.14-inch ESP32 board. [Original board files](#).

The Adafruit 1313 speaker is 77.8 x 77.8 x 25.49 mm with 60 mm mounting centers. Its rear open mounting frame is at upper-local Y52.5..56.5, with four holes at X +/-30, Z13 and Z73. The speaker envelope extends from Y56.5 to Y81.99, centered at Y69.245/Z43. Its rear edge clears the display frame at Y95. Install the speaker after the arm insertion paths are complete. Use four M3x12 screws, washers and nuts. Retain the rear board with four padded 20 x 8 x 1 mm tabs and M3x25 screws, washers and nuts. Pads touch bare PCB edges only. [Speaker drawing](#).

Base, motor fastening and battery platform

This revision keeps the four Adafruit3777 TT motors, Adafruit3766 wheels, 300mm circular base and 6mm base floor. The axle positions do not change. The battery load goes directly into that floor and its surrounding ribs. The printed electronics platform carries the boards, not the battery.

Printed parts and hardware choices

11_motor_clamp.stl is one 39.5×14×5mm design printed four times. Each clamp uses one M3×16 screw, one M3 washer and one standard M3 nut. This is four screws total for all four drive motors. The fifth, head motor has its own adjustable drive mounting in the upper body.

The alternative drive-motor fastening uses two cable ties per motor. With that option, omit all four printed clamps and their four screws/nuts/washers. Use 4.8mm ties at least 250mm long, with a manufacturer stated loop tensile rating of at least 222N. Use genuine parts with a stated rating. The rating of a cable tie is not a load rating for the printed robot.

12_electronics_platform.stl is one 216×216×112mm connected print. It includes the 6mm deck, four legs, four feet, rim, braces and mounting slots. Print it as exported, with its deck on the bed and legs up. Enable normal automatic supports for the foot overhangs. Print the clamps as exported with automatic supports under their raised bridge and toe. Use PETG, four walls and 30% infill for these two parts. Remove every support from a fastening slot. The base retains six walls, six top/bottom layers and 40% gyroid infill. These settings are a prototype specification; load strength is not yet measured. Use the current H2D slice report for actual material and time.

The platform feet use eight M3×12 screws, eight M3 washers and eight M3 nuts, two sets per foot. The alternative uses eight 4.8mm ties at least 150mm long, two per foot, with the same minimum 222N stated rating. All eight foot ties can be cut for service. Replace a cut tie; do not reuse it.

Motor assembly

1. Remove supports and smooth the edges of all tie passages. Keep the motor

disconnected. Check that the shaft turns and that no wire touches it.

2. For the screw option, insert one standard M3 nut into the side opening of

each outer boss before fitting the motor. The pocket is at base X±88.5, Y±11, Z22..24.8. Seat the nut square. A small piece of removable tape can retain it during assembly. Keep tape outside the thread and bearing face.

3. Seat each motor in its cradle at X±70.5, Y±58, with the motor case bottom

at base Z6. Put a 2mm EPDM pad under the clamp bridge. Keep the pad off the motor vents, wire terminals, shaft and moving gearbox parts.

4. Point the clamp toe inward, toward the battery. Slide the toe into the

reinforced inner rail window, then align the outer round ear with its boss. The nominal assembled clamp origin is [±70.5, ±11, 29]. Rotate the same printed clamp 180 degrees about Z on the left side of the base.

5. Fit the M3×16 screw and washer. Tighten by hand until the outer ear seats

and the pad grips the case. The nominal 1.56mm bridge-to-case gap compresses the 2mm pad by about 0.44mm. The actual case and pad control the fit. Stop if the case bends or the gearbox becomes stiff. Confirm two full exposed threads past the nut and no moving-part contact. Do not assign an untested torque value to the printed clamp.

For the tie option, omit the clamp and captive motor nut. Each motor has two longitudinal strap routes. Use |X|62 and 74 for the +Y rear motors, and |X|68 and 80 for the -Y front motors. The stagger leaves at least 1.2mm between opposed 4.8mm ties. Each route passes through the base-floor slots at local Y-0.2 and 72.2. Mirror the coordinates for the other quadrants. These 6.2×3.4mm slots put the tie ends beyond the motor case ends. They avoid the small gap between the wheel and the motor. Put a 1mm EPDM strip beneath each tie on the case. Feed the tie through one floor slot, under the floor, up through the other slot, and across the top of the motor. Place each locking head at

the inner motor end, in the central lead channel, clear of the battery and platform foot. Tighten the two loops evenly. Do not crush the case. Cut the tails flush and cover any sharp cut end. Check that all four wheels rotate freely and each motor is held in its cradle.

Battery fit and fastening

The clear cradle is 104mm across X and 157mm along Y. The three checked envelopes are Bioenno BLF1206A70×114×76mm, Power-Sonic PS1270 7Ah 67×153×102mm, and Power-Sonic PS12140 14Ah100×153×103mm. The two SLA envelopes include the selected products' dimension tolerance and terminal height. These dimensions do not describe every battery sold as 7Ah or 14Ah. Measure another product before using it. Use the battery specification and electrical instructions for its terminals, fuse, connector and charger.

Put a 2mm nonconductive pad on the base floor. Position the battery with its long dimension along Y. Cut firm, nonconductive foam blocks to remove the side/end gaps around the smaller Bioenno or 7Ah battery. Do not put loose metal spacers beside battery terminals. Keep the terminal edge and lead exit away from the two strap lines at $Y \pm 25$.

Use two 4.8mm cable ties at least 500mm long, with stated loop tensile rating at least 222N. Each loop uses the floor slots at $X \pm 51.5$ and one of $Y \pm 25$. It passes under the floor and over the battery case. Cover the terminals with correctly sized insulating boots. Route the ties over insulated case surfaces; never over a terminal, connector or wire. Use a protective pad where each tie turns over a case edge. Tighten both loops until the pack cannot slide. The foam blocks locate the smaller packs; the ties hold them down. Keep tie heads in the open end space, not beneath a platform leg. Remove and replace both battery ties whenever the battery is removed.

The maximum checked battery top is base Z111. The full 100×153mm footprint is clear to the printed platform underside at Z140. This provides 29mm before PCB hardware is fitted. Limit all PCB washer, nut and screw-tip projection to 3mm below the deck. The populated platform then has at least 26mm nominal clearance above the terminal-height envelope. Braces point outward and the underside beams lie outside that footprint. Confirm actual terminal boots and flexible leads fit in the 26mm space without a sharp bend or pressure from the platform. Disconnect the pack before handling its connectors.

Build the platform on the bench

The platform's assembled deck is Z140..146. Its legs sit on the base's raised rail seats at Z36, with centres $X \pm 56.5$, $Y \pm 48$. End keys locate each foot along its seat. Use 6mm nonconductive spacers to put the PCB underside at Z152. The drawing keeps the actual board envelope dimensions and shows their new centre positions:

- Buffer prototype: X-38,Y-29;81×51mm.
- Divider prototype: X-38,Y29;81×51mm.
- PCA9685: X39,Y-44;62.5×25.4mm.
- Motor regulator: X24,Y-10;25.4×25.4mm.
- Servo regulator: X60,Y-10;25.4×25.4mm.
- Left DRV8833: X23,Y24;26×18mm.
- Right DRV8833: X59,Y24;26×18mm.
- Head DRV8833: X23,Y53;26×18mm.
- Amplifier: X59,Y49;19.4×17.8mm.
- Logic regulator: X81,Y32;10.16×17.145mm.

Use the real boards as drilling templates. The deck has 3.2mm-wide slots on a 12mm grid. Use a slot where it matches a PCB hole. Drill a matching hole where needed. Do not enlarge any PCB hole or cut an underside beam, leg or rim. Use M2 screws for the two Pololu4091 regulator boards' three mounting holes each; use M2.5 where the other board's real mounting holes allow it. Start with 20mm screws and trim to the measured 6mm deck, 6mm spacer, board, washer and nut stack. For M2.5 use a thin nut no more than 1.6mm thick, one 0.3mm underside washer, and one 0.5mm head washer. A 1.6mm PCB gives a nominal 16.9..17.1mm screw working length. For M2 use a thin nut no more than 1.6mm, one 0.3mm underside washer and one 0.3mm head washer; that same PCB thickness gives a nominal 16.6..16.9mm length. Actual board thickness controls the cut length. Leave two full threads past each nut and keep all underside hardware within 3mm of the deck. Relocate a board or use its tie option if the real mounting holes coincide with a beam or brace. Deburr holes and remove all chips.

The alternative board fastening uses ties through adjacent deck slots and over insulated, component-free board edges, on the same 6mm nonconductive spacers. Use two independent loops for each board. A tie must not load a connector, component, solder joint or copper trace. Actual board routing is a bench-fit requirement; the drawings show envelopes, not certified connector or hole templates. Keep exposed metal off both PCB faces.

Close the chassis and retain service access

1. Fit and retain the four M4 nuts in the underside base-flange pockets before

the battery or motors prevent access. The small pocket nibs hold them during assembly; the 6.7mm roof above the pocket takes the screw load. Verify a real M4 screw threads freely. If a nib prevents full nut seating, trim only the nib and retain the nut with removable tape. Never force a tilted nut or remove the pocket roof.

2. Lower each already mated motor/wheel unit into its cradle. The wheel wells

pass through the upper flange to permit this path; the outside bumper is continuous. Retain the motors using either option. Fit, pad and tie down the selected battery. Keep the master switch off and the battery connector unplugged.

3. For platform screws, insert eight M3 nuts into the rail-seat side pockets.

A temporary tape strip can hold them during handling. Lower the populated platform over the battery and onto all four rail seats. Fit two M3×12 screws and washers per foot. Use a 2.5mm L-hex key from the side of the base; the deck prevents a straight driver from above. The checked key envelope has an 18mm exposed short leg and at least 79mm side reach. Its horizontal shaft is at base Z61 and runs along Y, away from the adjacent leg. Keep its X coordinate at the screw's $X \pm 56.5$. This approach clears the installed wheels. Check full seating and two full threads.

4. For platform ties, omit those screws and nuts. Pass each tie through its

transverse rail tunnel, around the foot, and into the foot's top groove. Use the two routes at each foot's Y centre ± 8 . Place locking heads on the outer side at Z35 or above, clear of the motor. Tighten evenly and confirm the feet cannot lift or move on the seats.

5. Check free wheel rotation. Secure leads to deck slots with service loops.

Keep wire bundles within the 216mm deck outline. Maintain the four cardinal driver corridors outside it. Before power is connected, confirm terminal boots, fuse and all ties are clear of motors, wheels and live board connections.

6. Pre-seat the four standard M4 upper-joint nuts in the skirt's underside

pockets as described in the body-joint section. Then lower the complete skirt around the assembled chassis. The 264mm lower opening passes around the 216mm platform. The platform does not pass through the 184mm upper opening. Fit the four base-to-skirt M4 screws from above, using the existing 250mm exposed 3mm ball-end driver at 16 degrees inward. The captive nuts remove the need for an underside nut tool.

7. For service, remove the four base screws and lift the whole skirt away.

Remove the platform screws or cut its eight ties, then lift the platform clear before removing the battery ties. Never pull on connected wiring.

The base retains its original 6mm floor, circular bumper and integrated ribs. The wider battery cradle, reinforced clamp rail roofs and platform seat blocks are connected to those ribs. Those are geometric load paths, not measured strength results. Use a slow first drive test on a level, smooth floor. Check wheel slip, turning, motor/driver temperature, fastener creep and emergency stopping with the selected battery. Lift the robot by its base. Do not use the arms, dome or electronics platform as lifting handles.

Verification

Run `python scripts/check_base_mounts.py` after exporting the current meshes. The result is `cad/base-mounts-checks.json`. It covers connected solids, motor cases, the four single-screw clamps and insertion paths, both tie options, three battery envelopes, 29mm printed/26mm populated terminal headroom, the populated platform, mated motor/wheel insertion, side-key access, skirt installation and the four long-driver corridors. These are finite digital samples. Physical print tolerances, board connectors, tie heads, adhesive pads, battery leads and strength still require inspection.

Battery wiring, charging and discharge handling

The specified battery-tie example is [KSS CV-500W](#): 500 mm long, 4.8 mm wide and minimum 222 N stated loop strength. The mounting samples use a ribbon no more than 1.2 mm thick. Ribbon thickness and latch size are not published for that example, so measure the actual tie and verify its complete route before use. A width or strength rating alone does not prove fit. The current slots do not accept a 7.6 mm-wide tie without a separate CAD change.

Only one battery is installed and connected at a time. The default Bioenno BLF-1206A is 12 V, 6 Ah, 72 Wh and approximately 0.7 kg. Preserve its protection circuit and factory PP30 discharge connector. Use its BPC-1502DC 14.6 V, 2 A LiFePO4 charger. [Bioenno pack and charger](#).

The specific SLA alternatives are Power-Sonic PS-1270 F2 7 Ah and PS-12140 F2 14 Ah. The maximum combined fit envelope is 100 X x 153 Y x 103 Z mm, including terminal blades and dimensional tolerance but excluding attached connectors. It does not cover every battery sold as 7 Ah or 14 Ah. The current PS-1270 sheet gives about 1.97 kg; PS-12140 gives 3.78 kg, with an older official version at 4.2 kg. Record the actual pack's dimensions and mass. These are heavier fit and electrical options whose loaded drive performance remains unproved. [PS-1270 sheet](#), [PS-12140 sheet](#).

Make one labelled SLA adapter from its matching terminals to the existing PP30 input. F2 uses a 6.35 x 0.8 mm blade; F1 uses 4.75 x 0.8 mm. Use matching insulated female contacts rated at least 10 A DC with the specified 18 AWG copper wire. Insulate each terminal separately and measure PP30 polarity before connection. Keep the single 7.5 A main fuse F1 within 100 mm of the SLA positive terminal, including all adapter and unfused lead length. For Bioenno, keep it within 100 mm of the intact factory PP30. Preserve the existing 3 A motor, 3 A servo and 1 A logic branch fuses. Do not increase motor voltage or current limits for a heavier battery. Use the five circuit sheets and wire schedule for exact connections.

Keep all battery contacts, boots and bent leads below base-local Z130. This leaves at least 10 mm below the printed deck and at least 7 mm below the permitted 3 mm PCB hardware projection. The larger battery's 29 mm bare-deck allowance is reduced to 26 mm by that hardware; it is not a verified fit for every connector. Keep ventilation open and both battery ties away from terminals. The battery always rests on the base, not on a platform leg or PCB.

Switch MAIN off, unplug and remove the battery before charging. Use the Power-Sonic PSC-121000ACX 12 V SLA charger for either selected SLA pack; PSC-12500ACX is also specified for PS-1270. Keep SLA and LiFePO4 chargers clearly labelled and separate. Follow the exact charger instructions and polarity. No onboard charging is provided. Never connect a 12 V pack to the T-Display battery socket. [PSC-121000ACX](#), [PSC-12500ACX](#).

Firmware stops motion at 11.2 V and permits explicit re-arm at 12.0 V, but the controller and audio supply remain powered. An SLA battery has no Bioenno-style internal protection. After a low-battery stop, switch MAIN off and unplug the battery. Store it unplugged and recharge SLA after use. AGM sealed lead-acid batteries are not gas-tight; do not operate or charge them in an airtight enclosure. [Power-Sonic technical manual](#). Full sources and electrical calculations are in `battery-options-research.md`.

Combined upper shell and concealed arm mounts

The shoulder and neck are one 186 mm PETG print, `04_shoulder.stl`. The former `05_neck.stl` is removed. The head, spindle, two bearings, friction drum and the separate adjustable `10_head_motor_carriage.stl` keep their existing locations. The upper shell remains removable at the skirt joint. This removes the shoulder-to-neck joint and its four M4 screws, four M4 nuts and eight washers. It does not increase overall height.

Each rounded arm box is now 90 x 52 x 103 mm, down from 94 x 58 x 106 mm. Width decreases 4.3 percent, depth 10.3 percent and height 2.8 percent. The nominal box envelope decreases 16.6 percent; that is not a measured reduction in plastic or print time. The box front moves from shoulder-local Y-115 to Y-113. Its socket centers stay X-50/+50, Y-94, Z77. The original 8-degree yaw and pitch limits stay unchanged. Do not reduce these boxes further without checking the carrier sweep, painted spherical roots and actual servos.

The fixed neck liner now has a 99.1 mm inside radius and a 100.3 mm outside radius. This leaves a nominal 1.1 mm gap around the existing 98 mm-radius carriage foot at its X64.5 contact position. The liner still ends at shoulder-local Z171. The rotating drum still starts at Z172. No new opening exposes the arm servos. The neck grille, three horizontal rings, bearing support, radial deck, jackscrew lug and travel stop are integral with the shoulder.

Servo dimensions and acceptance gates

[Tower Pro's MG92B table](#) and its [dimension diagram](#) give A35, B22.6, C31, D12, E31.5 and F22.8 mm. The diagram identifies A as base-to-shaft height, B as case length, D as width, E as the complete ear span and F as base-to-ear underside. These do not specify the mounting-hole centers, shaft offset along the case or ear thickness. [Adafruit 2307](#) also gives an overall envelope of 36 x 12 x 31 mm. These descriptions are not interchangeable detailed drawings.

The CAD preserves the existing shaft axes. It uses a nominal 5 mm offset between case center and shaft along the case length. This offset and the nominal 2 mm ear thickness must be checked against the actual four servos. Do not assume that the simplified drawing measures these dimensions. Use the real servo, horn and screws before accepting a printed fit. Align the shaft first, then check the case, ear holes, output boss and wire exit. A supplied horn and its center screw remain mandatory for every servo.

Yaw support is shoulder-local Z7. The target shaft/horn datum stays Z42. Relative to each yaw axis, the nominal case occupies X-16.3..6.3, Y-6..6 and Z7..38. The clearance pocket is X-17.3..7.3 and Y-6.5..6.5. Ear support tops are Z29.8. The two nominal ear regions are X-20.75..-16.3 and X6.3..10.75, Y-6..6, Z29.8..31.8. Pads must not move the required shaft datum. Reject an interference or a loose case that cannot be padded while retaining the shaft location.

The pitch carrier is now 49.5 x 30 x 49 mm. Its back wall starts at X32.5, leaving 0.5 mm behind the nominal servo-back plane X32. The pitch shaft remains at carrier-local (-3,0,35). The nominal case is X1..32, Y-6..6 and Z18.7..41.3. Its ear underside is X9.2. The two ear regions lie at Z14.25..18.7 and Z41.3..45.75. Again, measure

the real offset and ear geometry before accepting this fit.

Both arm stems now have a small flat on the side nearest the servo. The flat is 8 mm long and blends back to the full round stem by 14 mm. In the printed part, it leaves a nominal 9.5 mm thickness and 12 mm width before the existing horn holes are subtracted. This removes the collision between the actual case-front envelope and the former round stem without moving a shaft or socket. Use the revised arm STLs with the revised holders. The checker samples 47 cross sections between 0.25 and 11.75 mm from the root. Its minimum sampled net area is about 73.4 square millimetres, including the existing horn holes. This is a geometric measurement, not an arm-load rating.

Option A: retain the servos with screws

Use two M2x10 socket-head screws per servo: eight screws for all four servos. Use DIN 912 / ISO 4762 heads with a 1.5 mm hex socket, no more than 3.8 mm diameter and 2 mm height. Add eight M2 hex nuts no more than 1.6 mm thick and sixteen M2 washers no more than 0.3 mm thick. Use washers no more than 5 mm outside diameter. These are additional servo-ear fasteners. Keep all existing horn-to-print screws and all four supplied horn-center screws.

The yaw supports have 2.2 mm-wide slots along X, centered at X-20 and X10, Y0 relative to the yaw axis. Their straight slot length is 3 mm, plus the rounded ends. The pitch supports have 2.2 mm-wide slots along Z, centered at Z15 and Z45, Y0; their straight slot length is 2 mm. The pitch screws point along positive X, from the servo ear toward the printed carrier. Fit actual holes within these slots while keeping the shaft in its fixed position. Do not enlarge the servo ears or force a screw through an offset hole.

Fit the yaw servo screws before installing a carrier. Introduce a short L-key through the open bottom, then approach each screw from the rear inside the gunbox. A straight driver from above is blocked by the roof. [Wiha 01121](#) is the checked nominal key: 1.5 mm hex, 46 mm long leg and 15 mm short leg. Keep the long leg pointing rearward. Start with the elbow at the screw X, Y-34 and Z-20; raise it to Z51. Move forward to Y-94, then lower the elbow to Z48 so the short end enters the socket. Turn in 60-degree arcs, lift, re-index and repeat. Keep the carrier, horn and arm out during this operation.

Each ear ledge has a rear nut-tool window. Hold the nut with a fine gripping tool whose working tip, while gripping a 4 mm-across-flats nut, stays within 6 mm width and 2.4 mm thickness for the first 24 mm of reach. The tool must fit the actual printed window and clear the servo case. These are acceptance bounds, not a claim that every pair of needle-nose pliers fits. The nominal tip center is at the nut X, Y-94 and Z24.5. Introduce it through the bottom with its front at Y-34; raise the center to Z24.5, then advance it through the rear window to the nut. Keep the tool steady while the L-key turns the screw. Check actual jaws and handles on the bare shell before choosing the screw option.

Fit each pitch servo and its two screws to its carrier on the bench before inserting the carrier. The printed ledge is 4 mm thick. With a nominal 2 mm ear, two 0.3 mm washers and a 1.6 mm nut, an M2x10 screw leaves about 1.8 mm beyond the nut. Measure the actual stack. Require at least two complete exposed threads and no contact between the screw, nut, servo case, horn or moving arm. Trim and deburr a screw only when the actual stack requires it.

Tighten only until the ear is seated on the ledge. The printed pocket and pads locate the case. The screws retain the ears. Do not bend an ear to make the body touch its support. Check all fasteners through the complete 8-degree motion range before power is applied.

Option B: retain the servos with cable ties

Use two independent 2.5 mm-wide nylon cable ties per servo: eight ties total. These replace the servo-ear screws only. They do not replace the supplied horn-center screws or the fasteners that connect an arm or carrier to a horn. Use thin nonconductive pads where a tie touches the metal case. Keep tie heads and trimmed ends inward, away from the socket liner, moving horn and leads.

Each yaw pocket has two loops through its floor. The slot pairs are at X-11 and X3, with each pair at Y-10 and Y10 relative to the yaw shaft. Each loop goes under the printed floor, rises at the front and rear of the case and crosses the case top. Keep the loop below the horn and clear of the shaft. The slots accept a 2.5 mm ribbon; choose a tie whose actual thickness and latch fit the open route.

The yaw loops return through enclosed channels at shoulder-local Z2..4. Clear these channels with a blunt tie before installing a servo. The return stays above the skirt mating face. Do not route a tie below the shoulder flange, where it would stop the joint from seating.

Each pitch carrier has two loop positions at Z24 and Z36. The slots pass through its back wall at Y-7 and Y13. Wrap each loop around the fixed servo case and back wall. Keep the tie away from the shaft at (-3,0,35). Fit the ties before inserting the carrier. Tighten evenly without crushing a case or moving the shaft datum. The modeled loops establish nominal access; the actual tie head, bend radius and pad thickness need a hand fit.

Assemble the merged upper shell through its bottom

Keep the upper shell off the skirt and leave the speaker, rear board, arm carriers, head carriage and head off during this sequence. Support the shell on a padded stand that leaves its full lower opening and both front sockets accessible. Do not rest it on the display lip, neck grille or lamp parts. Remove every internal print support before fitting any hardware.

1. Clean both bearing seats. From below, insert the lower 608 bearing into shoulder-local Z160..167. Insert the upper bearing from above into Z179..186. Use the existing 12 mm metal inner-race spacer, narrow shims and six outer-race retainers. A straight lower driver approaches each of the three lower retainer positions through the open bottom. Fit the M8 spindle from below. Keep the assembly free to rotate. The lower bearing cannot be serviced with the upper shell seated on the skirt.
2. Center the four servos electrically. Introduce each yaw servo through the lower opening at its final X, with its center Y0. Raise its case base to Z45, move forward to Y-94, then lower its base to Z7 into the pocket. Retain it using Option A or B. Leave the carriers out. Fit and check the yaw horns at their existing neutral datums after the retaining tools are removed. Keep wires loose but clear of the next insertion paths.
3. Feed each detached arm through the lower opening. All following coordinates are shoulder-local. Keep its root X at -50 or +50. Start with root Y10, the stem pointing vertically down and root Z-35. Raise the root to Z45, move rearward to Y25, then raise it to Z110 with the arm still vertical. Turn the stem toward the front until it points 13.5 degrees down from horizontal, keeping the root at Y25/Z110. Raise the root to Z110.611, then move it from Y25 to Y-20 while keeping $Z = 77 + (Y + 115) \times \tan(13.5 \text{ degrees})$. At Y-20, reduce the angle from 13.5 to zero and keep $Z = 77 + 95 \times \tan(\text{angle})$. Finally, move the horizontal root from Y-20 to Y-94 at Z77. The stem and decorative tip pass outward through the front socket; the spherical root stays inside. The 40 mm plunger cup is still the limiting tip. Use a caliper on the finished cup and never force it through a painted socket.
4. Support both exposed stems on padded blocks, keeping their roots at (-50,-94,77) and (50,-94,77). Assemble each pitch servo on its carrier outside the shell. Fit the screws or ties now. Keep the arm supported while its carrier enters from below through the rear portion of the opening.
5. For both carriers, start at origin X-29/Y25 in the left part of the lower opening. The right part is blocked above by the integrated head-deck support. Raise each carrier floor from Z-55 to Z88. Move forward from Y25 to Y-44 at Z88. Lower the floor to Z46, then move X to its final value of -47 or +53. Move forward to Y-94 and lower to Z42 onto the centered yaw horn. Fit the horn-center screw and existing two horn-to-carrier M2 fasteners. This is a lower-entry path. The old straight approach through a detached neck joint does not apply.
6. Attach each supported arm to its pitch horn with its existing two M2 fasteners and supplied center screw. Remove a stem support only after the horn retains the arm. Fit the opaque stretch-fabric socket liners with loose

folds. The nominal visibility model uses a 30 mm fixed outer radius at Y-111 and a 24 mm moving inner radius. The former 33 mm outer radius would extend above the smaller gunbox. Bond the real fabric to the available inner curved surfaces around the aperture, with continuous overlap. Keep it inside the box's top edge. Keep both moving and fixed attachment edges secure. Inspect front, side and lower views through all four yaw/pitch extremes. A servo, tie, horn or lead must not be visible through either finished socket.

7. Fit the rear speaker, acoustic cloth, rear T-Display and switches after the arm routes are clear. The speaker frame is now at Y52.5..56.5, with unchanged mounting X-30/+30 and Z13/73. It has two 8 mm uprights, a bottom beam and two integral struts to the shell. The speaker faces the existing front grille. Its nominal body is X-38.9..38.9, Y56.5..81.99 and Z4.1..81.9. Keep its terminals insulated and route its leads upward. This leaves nominal 13.01 mm before the rear display recess at Y95; inspect the actual tabs, connectors and wires. The frame was moved rearward to leave an open arm-insertion route. Keep a labelled disconnect and a service loop for each harness. Run the arm motion checks on the bench before placing the upper shell on the skirt.

Fit and adjust the head drive

The fifth TT motor, existing radial carriage and 63 mm wheel stay a separate serviceable assembly. The printed fixed deck and adjustment lug are now part of the merged upper shell. Keep the original 3.6 mm slide range, the jackscrew, all four M3 slide clamps and the hard stop. Combining the fixed supports does not remove friction adjustment.

Fit the motor, pads, ties and fully seated wheel to the carriage on the bench. With the head off, lower that assembly vertically at shoulder-local carriage origin X64.5/Y0/Z149.2. The nominal head-wheel center is X64.5/Y0/Z191. The revised liner lets the complete assembly pass from above. Install the four M3x16 slide screws and washers from above, and their underside washers and nuts through the open lower shell. Fit the jackscrew nut and screw while access remains open.

Retract the carriage to X61.5 before seating the upper shell on the skirt. The skirt's four new captured M4 nuts retain the upper joint and resist tightening torque. Seat them before placing the upper shell. The captured pockets replace the need to hold loose nuts inside the closed joint. Use the specified four M4x20 screws and washers, and check that the tongue seats fully.

Keep the dome off while tightening these four screws. Use the specified 250 mm bare-shaft, ball-end 3 mm driver, rated for at least a 20-degree angle. The sampled shaft has a 1.74 mm corner radius. Its ball enters each screw socket near upper-local Z9.2. At the front and left screws, lean the shaft 16 degrees toward the center. At the rear screw, lean it only 6.5 degrees toward the front. That route passes between the speaker and the rear display frame. At the right screw, lean the shaft 16 degrees toward the rear-left diagonal: equal negative-X and positive-Y motion as the shaft rises. This route passes behind the fixed deck and clears the retracted head wheel. Keep the handle and all large bit holders above the clear top opening. Check the real tool, final wiring and actual hardware before tightening; do not force a shaft against a board, wheel or printed edge.

Then follow the original bearing, spindle and crown-retention procedure. The head STL origin remains shoulder-local Z172. Adjust contact through the existing left service port with the head in place. Nominal contact remains X64.5, with a hard outer limit of X65.1. This is a geometric position, not a force setting. Lock all four carriage screws using the original short L-key routes. Check full hand rotation before powered tests.

When servicing an arm, lower bearing or underside slide nut, remove the complete upper shell from the skirt and unplug its labelled harnesses. Isolate battery power first. The former neck joint is no longer a service opening.

Verification scope

`scripts/check_arm_assembly.py` checks the new carrier entry path against the printed upper shell and both supported arms. `scripts/check_upper_mounts.py` checks the new ear supports, optional screw stacks and tie routes, yaw-servo entry, screw and nut tools, detached-arm lower entry, complete head-drive top entry, lower-bearing tools, speaker clearance and four upper-joint driver routes. `scripts/check_mechanical.py` checks the unchanged 8-degree mechanism range against the new upper geometry. The generated reports identify the exact STL hashes used.

The common query helper compares two independent near-axis ray results and sends disagreements to the original trimesh. `contains query.py` python `scripts/mesh_queries.py` compares the method against the original on 4,400 deterministic inside, outside and surface-adjacent points. It does not change geometric tolerances or permit collisions.

These are finite digital samples. They do not prove print strength, real servo dimensions, cable flexibility, fabric folds, tire grip or physical assembly. Use the stated fit gates, then complete the bench motion, retention and stop tests on the actual robot.

Detailed bearing and friction-drive hardware

Fit the bearing stack while the merged upper shell is off the skirt. The lower 608 bearing is at upper-local Z160..167 and the upper bearing at Z179..186. Use the 12 mm-long metal spacer with 8.1 mm bore and 12 mm outside diameter between the inner races. Add the two narrow 8 x 12 x 1 mm shims, one below and one above. Six M3x8 plastic thread-forming screws and 9 mm washers retain only the outer races, three per end. Fit the lower retainers through the open bottom. Insert the M8x70 spindle from below without forcing either 8 mm bore. Bearings must rotate freely before continuing.

The following head-drive coordinates use the internal neck datum at upper-local Z120. Add 120 mm for upper-local coordinates, or 397.8 mm for world Z. The datum names geometry within the combined print. Fit the complete head carriage and all underside slide nuts before seating the upper shell. The head load passes through the spindle and bearings; the tire supplies tangential drive only.

1. Fit the fifth TT motor on its side in the head carriage. Its axle is vertical; the metal tail points toward negative Y. Use measured thin pads and two ties through the saddle. The nominal motor transform from the original drawing frame (x,y,z) is neck-local (64.5+z-11.7,y,43.5+x). The metal-end bottom is Z32.3 and the carriage floor top is 32.2. Support the narrower gearbox with measured padding rather than forcing it down. The unused lower shaft has a 15 mm-wide radial clearance slot through both carriage and fixed deck.
2. Fit its wheel fully. Nominal wheel center is neck-local (64.5,0,71). Measure actual seating; use appropriate case-height shims to keep the 29 mm tire within Z56.5-85.5 and clear of the head spokes at Z91. Both shaft ends and the hub must remain free. Keep motor wires fixed below the rotating drum.
3. Install the carriage on the fixed deck using four M3x16 screws. Fixed screw axes are (43,-50),(43,8),(83.5,-40),(83.5,8). Use 6 mm-OD, 1 mm-thick washers above the carriage and standard washers and M3 nyloc nuts below the deck. The small top washer diameter is required for the full slide travel. Leave the screws just loose enough to slide; retain all nuts. The deck is Z25.2-29.2; carriage floor is 29.2-32.2.
4. Insert one M3 nut into the open-top adjustment-lug pocket. Fit the M3x25 jackscrew along positive X through Y-35, Z38. The screw tip bears on the carriage pad. Start with carriage X61.5, nominally 3 mm clear of the inside track. Radial slots permit movement to X65.1; the fixed stop at X90.1 limits the outward plate edge. Do not enlarge slots or bypass that stop to cure a binding head.
5. Lower the head over the spindle with the tire released. The drum is 192 mm inside diameter and 198 mm outside diameter. Its bottom is neck-local Z52 and the spokes start at Z91. Fit the M8 washer above the printed

hub and the nyloc nut using a 13 mm socket no more than 23 mm outside diameter through the 24 mm crown hole. Set retention while keeping free bearing rotation. Do not tighten until the head binds.

6. Turn the dome by hand through two complete revolutions while the tire remains released. Check drum roundness, axial runout, crown retention, eye sweep and clearance from the stationary liner. The liner ends at Z51, giving nominal 1 mm clearance below the rotating drum.
7. With power off and the head seated, reach the jackscrew from the left through its service port at Y-40..-30,Z34.5..41.5. Advance it only until the tire makes light, reliable contact. Nominal geometric contact is X64.5. The final 0.6 mm travel is a geometric allowance, not a measured force setting or a required amount of compression. Excess pressure loads the motor shaft and head bearings.
8. Hold the adjustment while tightening all four slide clamps against their M3 nyloc nuts. These clamps lock the setting; there is no separate jackscrew jam nut. The clamp screws remain accessible with a 2.5 mm L-key whose short leg is 15 mm. Its horizontal shaft runs near neck-local Z49.9 through the 3.8 mm-high ports; the short leg reaches the screw socket near Z34.9. Approach the two negative-Y screws from the front and the two positive-Y screws from the rear at their specified X coordinates. Use a correctly sized tool; do not pry open a printed port.
9. Recheck two full hand turns after locking. Start powered tests at low duty in each direction. Increase contact only enough to avoid unwanted slip, then lock and repeat. Do not increase pressure to overcome an eccentric drum, tight spindle nut or snagged liner. Recheck after the motor and PETG mount warm during the short commissioning run.

The head controller commands speed and direction; it has no measured angle feedback. The rolling ratio is approximately $96/31.5=3.05$ motor-wheel turns per head turn when there is no slip. Actual slip, speed and stopping distance must be measured. The eye and dome lights are decorative and contain no moving wires, so no slip ring is required.

Physical acceptance and service

- Before motor installation, support all four motor-floor regions on equal rigid blocks. Apply twice the intended complete robot mass across the battery and stack load points for ten minutes, close to the floor. Unload and inspect for cracks or permanent distortion. This is a proposed proof test, not a verified rating.
- Verify fully seated wheels and free rotation unloaded and under the intended mass. Check clamp, tie, platform-foot, bumper and ground clearances with actual tire deformation.
- Weigh the complete robot. Four wide tires resist tight turns. Start on a level hard floor with slow starts and broad moving arcs. Measure current and temperatures. Do not assume these small TT motors can pivot in place, climb thresholds or run on carpet because the base is reinforced.
- Run twenty arm circles while inspecting every socket from low, side and frontal angles. The fabric must remain opaque, loose and clear of moving hardware at all permitted poses.
- Run a full head turn in each direction, check runout and repeat all contact/lock checks warm. A slipping drive is preferable to forcing a bound bearing with excessive pressure.
- Complete the electrical stop, Wi-Fi-loss, power-switch and simultaneous-load tests in `assembly.md`. The physical actuator switch must remain reachable.

Service and regeneration

Isolate battery power and unplug labelled harnesses before removing body screws. Remove the complete upper shell from the skirt to service an arm, lower bearing or underside slide nut. Support it on a padded stand with its lower opening free. Remove the speaker and rear display before reversing an arm insertion path. Back off head contact before removing the crown nut with a 13 mm socket no more than 23 mm outside diameter; lift the dome by its rim, never the eye or lamps.

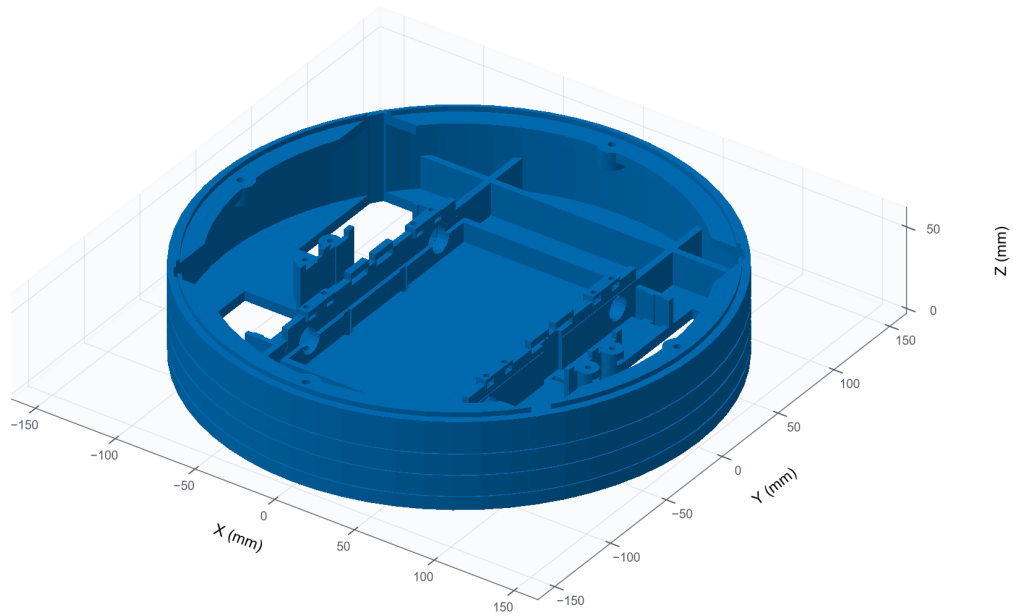
Lift the complete skirt off the base for battery or platform service. Disconnect the platform harnesses, remove its eight screws or cut its eight foot ties, and lift it clear before cutting the two battery ties. Replace every cut tie. Refit foam blocks, terminal boots and service loops, then repeat retention and motion checks. Do not lift the robot by its arms, dome or electronics platform.

Regenerate meshes with `python scripts/export_cad.py --jobs 3`, then run `python scripts/export_cad.py --check-only`. Run the current mount, arm-assembly, mechanical and H2D checks after any geometry change. Use `python scripts/render_drawings.py` to refresh the PDF illustrations from those final meshes. Never treat a report for earlier hashes as verification of a new part.

Single-piece circular motor base

Dalek FACET-1 / base

Actual STL geometry | Purchased hardware uses nominal envelopes



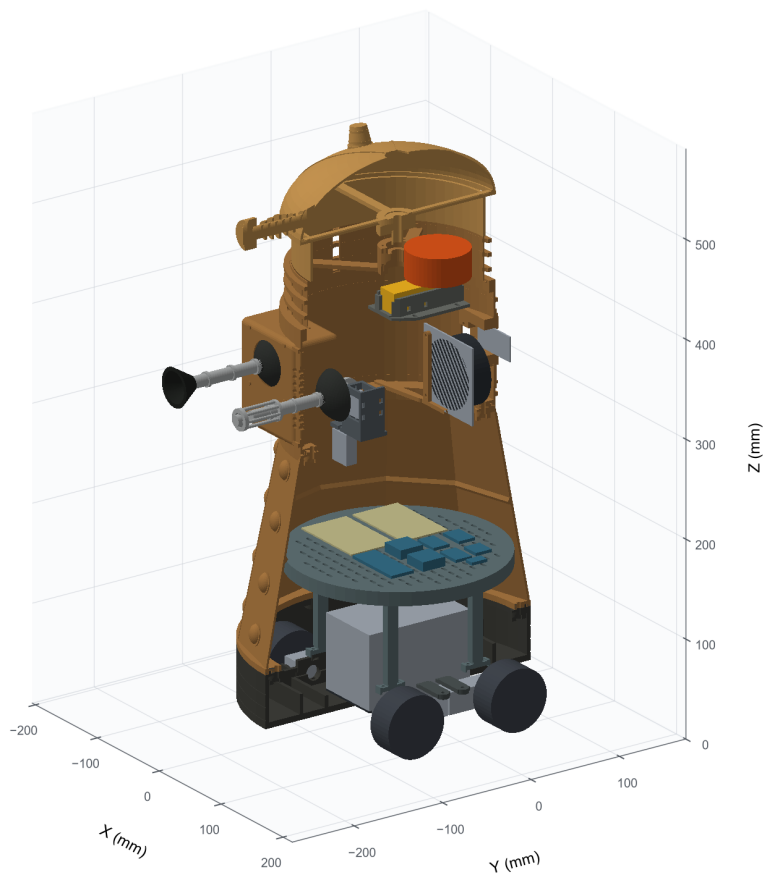
Actual FACET-1 STL geometry | Dimensions in mm | +Y is rear; -Y is front

The floor, circular perimeter, ribs, motor pockets and battery support form one connected print. The base is 300 mm in diameter before its brim.

Internal assembly view

Dalek FACET-1 / section

Right shell half omitted to expose internal components



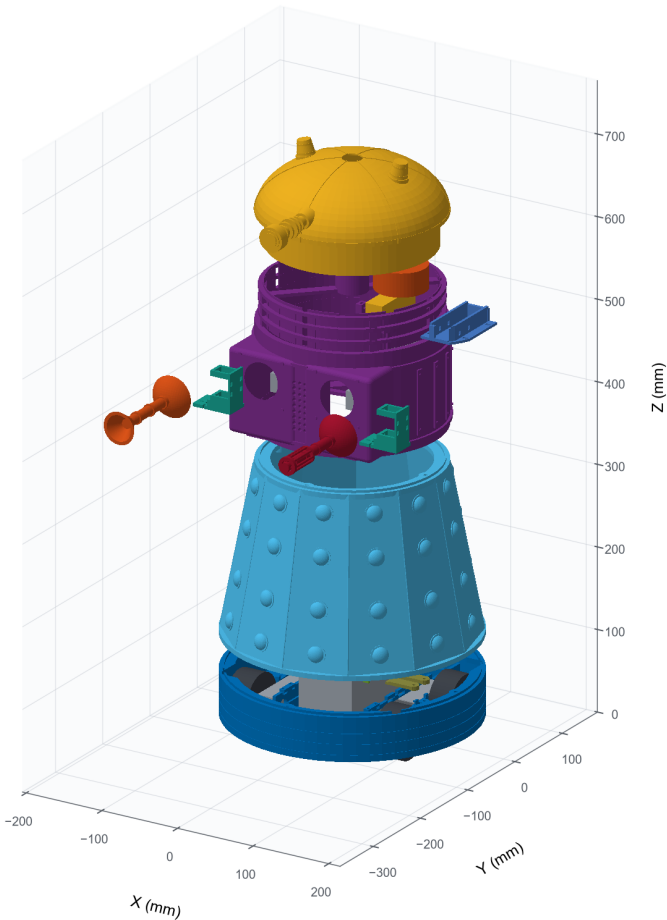
Actual FACET-1 STL geometry | Dimensions in mm | +Y is rear; -Y is front

Section through the native CAD. The continuous base supports the motor pockets, battery and vertically stacked body.

Vertical stacking order

Dalek FACET-1 / exploded

Actual STL geometry | Purchased hardware uses nominal envelopes



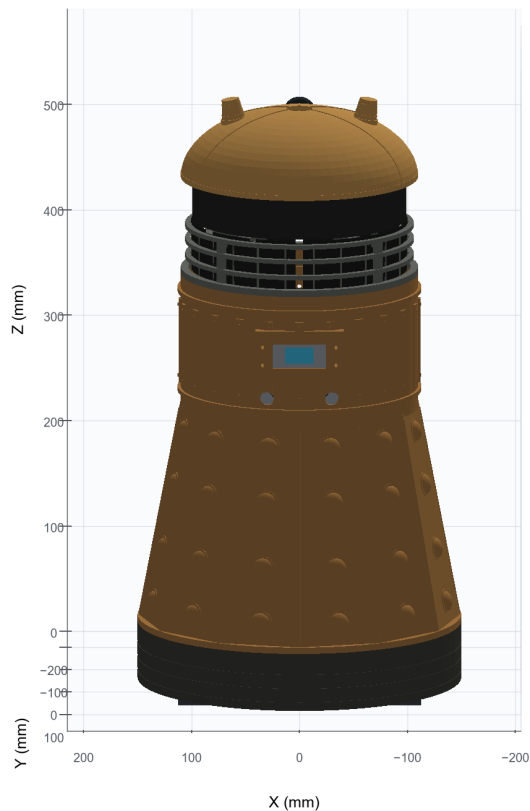
Actual FACET-1 STL geometry | Dimensions in mm | +Y is rear; -Y is front

Each body tier is one complete print. Registers align the stack; accessible fasteners retain the joints. The gaps in this view are intentional.

Rear display and access

Dalek FACET-1 / rear

Rear-mounted original TTGO T-Display



Actual FACET-1 STL geometry | Dimensions in mm | +Y is rear; -Y is front

The TTGO display faces outward. Keep the USB opening, operator switches and service fasteners accessible.

Firmware and Wi-Fi controller

This firmware targets the original 1.14-inch LilyGO TTGO T-Display with a classic ESP32, 135 × 240 ST7789 display and 4 MB flash. It does not target the T-Display S3 or the AMOLED version. The display faces outward through the robot's rear window. Landscape rotation is 1; use rotation 3 in `src/main.cpp` if the finished mount is reversed.

The software builds for the specified hardware. Its physical operation must be commissioned on the assembled robot. A successful compiler check does not establish stopping distance, servo centre, current draw, Wi-Fi range, speaker level or battery measurement accuracy.

Build and load

1. Install PlatformIO Core or the PlatformIO extension for Visual Studio Code. The pinned packages in `firmware/platformio.ini` install on the first build. Internet access is required for that build only.
2. Turn the physical actuator stop switch OFF. Disconnect the battery before connecting USB. Do not connect two 5 V sources to the board together unless the circuit diagram's power isolation is in place.
3. Use a data-capable USB cable. From the project directory run:

```
Set-Location C:\dev\robots\dalek\firmware

pio run

pio run -t buildfs

pio device list
```

4. Read the ESP32 serial port from the last command. Replace COM7 below with that port:

```
pio run -t upload --upload-port COM7

pio run -t uploadfs --upload-port COM7
```

5. Reset the board. The display must show DISARMED. If automatic flashing does not start, hold the board's BOOT button, start upload and release BOOT after the connection begins. The other button, GPIO35, is the stop/setup button.
6. Disconnect USB before restoring the normal battery supply. Keep the wheels off the floor until all commissioning checks below pass.

There is one factory application partition, 0x10000-0x1ffff (2,031,616 bytes). LittleFS occupies 0x200000-0x3ffff (2,097,152 bytes). Uploading LittleFS replaces the web assets and MP3 files in that partition. Network credentials are in a separate NVS partition. There is no wireless firmware update, cloud service, internet dependency or scheduled background job.

Pinned dependencies: Espressif32 PlatformIO platform 6.5.0, Arduino-ESP32 2.0.14, TFT_eSPI 2.5.43, Adafruit PWM Servo Driver Library 3.0.2, Adafruit BusIO 1.16.1 and ESP8266Audio 1.9.7. The audio dependency is GPL-3.0; retain its notices and supply corresponding source with redistributed firmware binaries. Later ESP8266Audio releases use different ESP32 I2S APIs, so update the platform and audio library together only after testing.

First connection

The board makes a WPA2 access point named DALEK-xxxx. It generates a device-specific 12-character key on first boot and saves it locally. The rear display shows the key. Firmware does not print keys or station passwords to the serial terminal. Join this access point using the displayed key, then open `http://192.168.4.1`. Enter the same key in the control page.

To join a router, open Wi-Fi network settings on the page. Enter the exact network name and its 8-63-byte WPA2 password. This classic ESP32 uses 2.4 GHz Wi-Fi. The password is sent in an authenticated POST request and saved in local NVS. Read the LAN address on the rear display and browse to that address from the same local network. No credentials are present in the repository.

The setup AP remains available while the robot connects to a router. This also provides fallback if the router is missing or its credentials change. Moving between AP and router connections, or losing either connection, disarms motion. Reconnecting never arms the robot. The AP supports two clients, but only one control lease can be active. Keep one control browser open.

Press the rear GPIO35 button to disarm. Hold it for five seconds to erase only the saved router credentials and return to AP operation. The device access key stays the same. The rear key also authorizes robot control for users on the router network. Do not share it widely or expose port 80 to the internet. The local interface uses HTTP; Wi-Fi encryption protects the wireless link, but this is not an end-to-end encrypted service.

Controls

Press Arm controls, then hold a movement button. Forward uses PWM 110/255, reverse 90/255 and turns use 50/255 on the inside wheel pair and 120/255 on the outside pair. Release disarms. Arm again for the next movement. Tight pivots are omitted from the supplied interface because four rubber wheels can scrub heavily against the floor. The requested TT motors have limited torque. Use a smooth, level indoor floor and gentle arcs.

The arm checkbox enables two-axis circles. Radius is 0-8 degrees and frequency is 0.10-0.80 Hz. The default is 8 degrees at 0.40 Hz. Eight degrees is the concealed gimbal's clearance limit; the browser and API reject a larger radius. One side runs half a cycle behind the other. The head slider commands the TT friction motor's drive level and direction; zero removes PWM. Displayed 100% maps to the conservative 100/255 PWM ceiling. It is not a measured speed. Hold Arms / head only to animate without ground-wheel movement. These settings also apply while a drive button is held. Sliders alone do not cause movement.

Speak plays a selected MP3 locally. Silence stops the decoder. Audio operates while the robot is disarmed. All audio stays in the fixed body, so the head can turn indefinitely without twisting an electrical cable. There is no head position sensor or automatic front-facing home position.

The browser must stay visible and focused. Releasing a pointer, losing pointer capture, cancelling a touch, changing tabs, locking the screen, closing the page or pressing Escape requests a stop. Loss of a command response also disarms the interface. Browsers can delay or discard stop packets; the board's independent timeout covers that case.

Electrical contract

Read `electronics-research.md` (`electronics-research.md`) and the power circuit (`../electronics/01-power.svg`), controller, drive and audio wiring (`../electronics/02-controller-drive-audio.svg`), servo wiring (`../electronics/03-servos.svg`) and circuit details (`../electronics/04-circuit-details.svg`) before wiring. All ESP32 signal pins are 3.3 V. Servo signals pass through the specified 74AHCT125 buffers to reach 5 V logic. Never feed 5 V into an ESP32 input.

Function | ESP32 GPIO | Connection

I2C data / clock | 21 / 22 | PCA9685 SDA / SCL, logic VCC 3.3 V

Servo disable | 27 | PCA9685 OE and U8 arm-buffer active-low output enables; U9 uses PCA4/5

Left forward / reverse | 25 / 26 | Left DRV8833 AIN1+BIN1 / AIN2+BIN2

Right forward / reverse | 32 / 33 | Right DRV8833 AIN1+BIN1 / AIN2+BIN2

Motor enable | 12 | All three DRV8833 SLP pins; 10 kΩ pull-down

Head PWM | 2 | 20 kHz to U9 inputs2/5;10 kΩ pull-down

Audio bit clock / word clock / data | 17 / 13 / 15 | MAX98357A BCLK / LRC / DIN

Actuator power sense | 36 | 5 V actuator rail through 10 kΩ top / 15 kΩ bottom divider

Battery voltage sense | 39 | Battery through 100 kΩ top / 22 kΩ bottom divider, 100 nF from ADC to ground

Stop / forget router | 35 | Existing rear board button

PCA9685 address is 0x40. Servo channels 0-3 are left yaw, left pitch, right yaw and right pitch. Channels 4/5 are static active-low head gate enables, not servo connectors. U8 buffers the four servo signals from 5V_SERVO. U9, powered from 5V_MOTOR, routes GPIO2 PWM to the selected U11 bridge input. PCA4/5 HIGH disables the corresponding U9 gate; LOW enables it. Each gate has a 10k pull-up to 3V3. U9 outputs reach U11 AIN1/AIN2 through 220ohm resistors, with 10k pull-downs at the driver. See circuit sheet 03 for physical DIP pins. The PCA stays at 50Hz; the motor receives 20kHz hardware PWM from the ESP32.

U5/U6 each run two ground motors, with input pairs joined. U11 bridge A runs the head; tie BIN1/BIN2 low and leave BOUT1/BOUT2 open. Every used bridge retains its stock 1A current limit. Never parallel bridge outputs.

Startup writes and verifies PCA MODE2=0x06: OUTDRV=1 and OUTNE=10. Global OE HIGH then makes PCA outputs high-impedance. R7/R8 pull both head-gate enables HIGH. The default OUTNE=00 would instead enable both active-low gates. A failed write or readback prevents arming. Keep this setting when changing PCA libraries.

GPIO27 needs a 10k pull-up to 3.3V. Remove the PCA breakout's onboard OE pull-down as specified in the electronics guide. GPIO12 and GPIO2 each need 10k pull-downs because they are boot strap pins and must default to disabled motion. GPIO15 connects only to the amplifier DIN input without external bias. GPIO36/39 are input-only and use the specified dividers.

The firmware reserves GPIO5/16/18/19/23/4 for the integrated display. GPIO0 is the existing BOOT button. GPIO34/14 remain reserved for the board's original battery measurement circuitry. The robot's external 12 V pack does not connect to the board's single-cell battery socket.

Stop behavior and limits

The motion task runs independently of the HTTP server and MP3 decoder, with a nominal 10 ms period. It is watched by the ESP32 two-second task watchdog. External enable resistors keep the actuator signals disabled during reset. This is a hobby control system, not a certified safety controller. The physical stop switch cuts actuator power without relying on Wi-Fi or software.

A command lease is granted only after explicit arming. The robot requires a complete valid command at least every 500 ms. The supplied browser sends a fresh command snapshot every 100 ms and aborts a command request after 300 ms. Each arm uses a new random lease; every command needs a strictly increasing sequence number. Old leases and duplicate commands do not refresh the timeout. An expired lease is checked before accepting another command, so a late request cannot revive movement. Invalid control parameters disarm immediately. Software drive values are limited to $\pm 150/255$.

Ordinary wheel speed changes ramp by at most five PWM counts per nominal 10 ms. A requested reversal first ramps to zero, waits 100 ms, then ramps the other way. A stop, fault or disarm immediately drives SLP low and both inputs low; it does not wait for the ramp. This lets the wheels coast. It does not provide a mechanical brake, and stopping distance must be measured with the final mass.

The head motor uses the same five-count ramp step with a 100/255 ceiling. Direction reversal ramps to zero, coasts for 100ms, then ramps up. A brief zero command or quick disarm/re-arm cannot bypass that interval. A zero head command immediately removes head PWM. Before changing the selected U9 gate, firmware sets GPIO2 PWM to zero, disables both gates using checked I2C writes, enables one gate and then restores bounded PWM. The shared SLP line is held steady during operation; PWM never drives SLP.

The PCA receives four checked servo writes every 20ms and checked static-gate writes when head direction changes. Any failed write latches an I2C fault, sets head PWM zero, sets all driver SLP pins low and disables servo OE. Missing PCA at boot prevents arming. No automatic motion recovery follows power, I2C, Wi-Fi or battery faults. Head PWM stops immediately on software disarm; the old head-servo neutral pulse is removed. All motors coast after electrical disable. Physical stopping distance and head coast angle still need measurement.

The specified battery is the Bioenno BLF-1206A 12 V nominal 6 Ah LiFePO4 pack. The software multiplies the ADC voltage by $(100 + 22) / 22 = 5.54545$ to recover pack voltage. It smooths readings at 10 Hz. Below 11.2 V for one second, it disarms; voltage must recover to at least 12.0 V before re-arming is allowed. A disconnected ADC or an implausible voltage also prevents arming. These thresholds are conservative operating limits, not a battery state-of-charge meter, a charger or a replacement for the pack's protection circuit. Calibrate the ADC against a multimeter before use.

Servo and sensor calibration

All calibration values are in `firmware/include/config.h`. Keep the wheels lifted and the servo horns removed for initial checks.

1. Check 5V rails with actuator power disabled. During ESP32 reset verify all three SLP pins and GPIO2 are low, and PCA OE is high.
2. Measure battery voltage with a multimeter. Read the displayed voltage. Set `PACK_ADC_CORRECTION` to measured voltage divided by displayed voltage, rebuild, and recheck at two battery voltages. Do not lower a cutoff to hide a wiring error.
3. Leave arm and head commands at zero. Enable actuator power, arm, and hold Arms / head only. The four positional servos must settle near their centres. Fit each horn with its joint centred and no mechanical load, then stop before fitting links. The initial centres are 1500 μ s; change `ARM_CENTER_US` in small steps only if needed.
4. Keep the head tyre clear of its track and command zero. GPIO2 and both U11 bridge inputs must remain low. Before attaching the motor, scope AIN1/AIN2 at a low head setting: only one input may carry 20kHz PWM. Check both directions and a reversal interval of at least 100ms with no PWM. There is no neutral pulse or neutral trimmer.
5. Start arm circles at radius 2 degrees and 0.10 Hz. Confirm that yaw and pitch are both active. Confirm that neither joint touches a stop. Increase to 8 degrees and 0.40 Hz. The pulse envelope stays within 1200-1800 μ s, but mechanical clearance is the limiting factor. Reduce the maximum radius if the actual links require it.
6. Turn power off. Set the head mount to light tyre contact, then test in each direction. Increase pressure only enough to turn the freely rotating head. Preserve slip if obstructed. `HEAD_DIRECTION` reverses the sign and `dalek: :HEAD_LIMIT` is 100/255. Do not raise this ceiling to overcome a jam. The slider is a drive-level command, not an angle or measured speed.
7. Lift all wheels. A forward command must rotate all four wheels toward forward travel. If one motor is reversed, swap only that motor's two output wires. `LEFT_DIRECTION` and `RIGHT_DIRECTION` reverse an entire side if necessary. Stop and remove actuator power before rewiring.

8. Play a phrase. The fixed digital gain is 0.28, below the design ceiling of 0.35. MAX98357A hardware gain must be 3 dB as specified in the circuit guide. Do not raise gain above 0.35 with the 8 Ω 1 W speaker and specified logic supply. Digital gain is not a calibrated wattmeter; verify clean sound, supply stability and acceptable speaker temperature.

Acceptance checks on the completed robot

Perform these with a helper at the physical stop switch, first with wheels lifted, then at low speed on a clear floor. Record the results in the assembly commissioning sheet.

- Power up, reset and reconnect USB: no wheel or servo motion before an explicit arm command.
- AP mode: router absent, page and every local voice phrase work through 192.168.4.1.
- Station mode: join the private router, read the rear LAN IP, load the page and drive through that address.
- Release the movement button: wheel outputs disable and the page becomes disarmed. Measure actual coasting distance with the final shell and battery fitted.
- While moving, disable the phone's Wi-Fi: commanded drive must cease within the 500 ms lease timeout plus control-task scheduling delay. The robot must remain disarmed after reconnection.
- While moving, lock the phone or change tabs: motion stops. Unlocking does not resume the prior command.
- Press physical stop: motor and servo supply must lose power even if the ESP32 keeps running. Releasing the switch does not restart movement.
- Press the rear GPIO35 button while holding a browser control: motion stops and does not restart without a new arm.
- With actuator power off, disconnect the PCA9685, then boot: I2C fault appears and Arm is refused. Reconnect only with power removed.
- Use an adjustable current-limited bench supply in place of the pack for the voltage check. At 11.0 V for more than one second the software must disarm. At 11.5 V it must refuse a new arm. Above 12.0 V it may accept a new explicit arm.
- Run arm circles and head motion while playing all phrases; no reset, supply sag, decoder stall or joint contact is acceptable.
- Complete gentle forward arcs on the intended floor. Measure individual motor current, case temperature and stopping distance. Do not accept a stalled inside wheel or prolonged current-limit operation.

Automated checks

From firmware/:

```
pio run
pio run -t buildfs
node --check data/app.js
node test/test_ui.js

$env:PATH = 'C:\msys64\mingw64\bin;' + $env:PATH
g++ -std=c++11 -Wall -Wextra -Werror -I include test/test_control.cpp -o $env:TEMP\dalek-control-test.exe
& $env:TEMP\dalek-control-test.exe
```

The host C++ test runs the same control-state code used by the ESP32. It covers strict parsing, boot lockout, expired/replayed leases, reconnects, unhealthy-input disarm and millisecond rollover. It accepts an 8-degree arm radius and rejects 9 degrees. Head regressions check the PWM ceiling, immediate zero/fault stop, and reversal delay through zero, re-arm and clock rollover. The Node test exercises the actual browser script with a simulated DOM and transport. Hardware routing, timing and stopping still need physical commissioning.

Sources

- [LilyGO original T-Display board, schematic links and pin allocation.](#)
- [PlatformIO LilyGo T-Display board definition.](#)
- [TFT_eSPI original T-Display setup.](#)
- [Adafruit PCA9685 driver source and checked setPWM result.](#)
- [ESP8266Audio version 1.9.7 source and license.](#)
- [ESP8266Audio current API compatibility notes.](#)

Dalek electronics design and source record

Original research: 2026-09-11. Head friction-drive revision researched 2026-09-12 against Adafruit, LILYGO, TI and NXP primary sources. This is a build design, not a claim that a robot was physically assembled or tested. Listed prices are USD before shipping and tax. Stock is a page observation, not an order reservation. The bill of materials separates checked prices from allowances.

Selected build

The controller is the original 1.14-inch TTGO T-Display ESP32. It is mounted on the rear with its screen visible. Four Adafruit 3777 motors each drive one Adafruit 3766 ground wheel. A fifth identical motor and wheel turn the head by adjustable friction contact. Each motor has a separate current-limited H-bridge. Two positional servos inside the shoulder move each arm in yaw and pitch. Their phased movement draws a circle with the arm tip. No servo is mounted on the exposed arm. The head motor stays fixed to the body. No wires enter the rotating head.

Power comes from ONE removable battery: the retained Bioenno BLF-1206A 12 V 6 Ah LiFePO4 pack, or the specified Power-Sonic PS-1270 F2 7 Ah or PS-12140 F2 14 Ah sealed lead-acid option. Never connect the options together. Three independent converters produce 5 V for drive motors, 5 V for servos, and 5 V for controller/audio. The physical ACTUATORS OFF switch removes input power from both motion converters. Logic and the display remain on. All circuit grounds join at a power distribution star. Sheet05 defines the optional SLA adapter and matched chargers.

The authoritative connection set is `electronics/wiring.csv` together with the circuit SVGs. Use the component references from those files. Pin numbers on the TTGO are GPIO numbers. Pin numbers on U8/U9 are DIP package pin numbers.

Required motors and wheels

Adafruit 3777 accepts 3-6 V. The vendor measured 160 mA unloaded and 1.5 A stalled at 6 V on a sample. At 4.5 V it measured 155 mA unloaded and 1.2 A stalled. The listing gives 0.8 kg·cm stall torque at 6 V, a 48:1 gearbox, 70×22×18 mm body and 30.6 g mass. There is no encoder. All five motors use 5 V. Five motors cost \$14.75 at the checked \$2.95 each. These values do not establish continuous load torque. A 5 V stall-current value was not published. [Adafruit 3777](#)

The required orange/clear 3766 is a press-fit TT wheel, 63 mm diameter by 29 mm wide, 38 g. The checked price is \$1.50 each. Adafruit marked it out of stock. This design retains the user's exact wheel. Obtain the wheels before accepting the final motor-to-shell fit. Do not buy a different wheel from a similar photograph. [Adafruit 3766](#)

A nominally stronger DRV8871 was rejected because its minimum motor supply is 6.5 V. That is above the required motor's 6 V maximum. Feeding the motors an overvoltage and merely lowering PWM duty is not used here. [Adafruit DRV8871](#)

Three DRV8833 boards support the 5 V motor supply. U5/U6 each run two ground motors. U11 bridge A runs the head; tie U11 BIN1/BIN2 to ground and leave its B outputs open. Each stock 0.2 ohm sense resistor sets approximately 1 A limit through 0.2 V/R. Keep the factory resistors and bypass jumpers open. On U5/U6 only, tie AIN1 to BIN1 and AIN2 to BIN2. Never join bridge outputs. All three SLP inputs share GPIO12 and its external 10k pull-down. A low SLP disables all five motors. The current limit does not make an indefinite stall acceptable. [DRV8833 product](#), [DRV8833 pinouts](#)

The four-wheel skid-steer layout has no precise speed or distance measurement. The original 3 kg planning target was not a motor manufacturer's payload rating. Use the current print report and the complete hardware mass, then weigh the finished robot. A slicer result does not prove loaded turning ability. Use hard flat indoor flooring, broad

moving arcs and the physical current, temperature and stop tests below. Wide silicone tyres can resist turning. Do not increase voltage or remove current protection to force a stalled build to move.

Servos and movement

The four arm servos are MG92B units, Adafruit 2307. The listing specifies 5 V signals as well as 5 V power. Direct 3.3 V PWM is therefore not the chosen wiring. The listed torque is 3.1 kg·cm at 4.8 V and 3.5 kg·cm at 6 V; weight 14 g; envelope 36×12×31 mm; 20-tooth horn spline. The listing also says this servo has no mechanical end stops. Use its supplied horn rather than printing a nominal spline. Begin with a small software travel window and check every linkage. [MG92B](#)

The head uses a fifth TT3777 and 63 mm wheel3766. Its tyre presses against the circular head track. The screw-adjusted mount sets contact pressure with power removed. Bearings carry the head mass. The motor mount and wire remain stationary. Begin with light contact, then increase only until the head turns reliably. The contact must still slip when obstructed; added pressure is not a substitute for correcting a tight bearing or track. There is no encoder, so the slider commands power and direction rather than an angle or measured speed. [TT3777](#), [wheel3766](#)

U10 is Adafruit 815 PCA9685. Use 3.3 V VCC and its default I2C address 0x40. The board's I2C pull-ups then stay at the ESP32 logic voltage. Leave the V+ rail disconnected because servo power is distributed separately. The board's output-enable pin is active low and normally pulled down. Remove that pull-down resistor, then fit an external 10k resistor from OE to 3.3 V. Identify it by continuity to OE and GND against the actual PCB revision; do not remove a resistor by guessed reference number. Measure the result before connecting servos. [PCA9685 pinouts](#)

U8/U9 are 74AHCT125 buffers. Power U8 from 5V_SERVO and U9 from 5V_MOTOR. Their TTL-compatible inputs recognize 3.3 V; outputs follow their own 5 V rail. U8 makes the four arm signals. U9 routes head PWM to the selected bridge input. Use AHCT, not HC, and check the actual chip marking. TI specifies input leakage with VCC=0 through 5.5 V; this avoids powering a disconnected actuator rail through controller signals. C8/C9 are 100 nF at the respective chip supplies. [Adafruit1787](#), [TI SN74AHCT125](#)

Servo channel order is 0 left yaw, 1 left pitch, 2 right yaw and 3 right pitch at 50 Hz. With horns removed, start at 1500 μ s. Fit arm horns at the mechanical centre. Begin circles at 2 degrees and 0.10 Hz. The default and maximum radius are 8 degrees after hidden-linkage clearance checks. The normal frequency is 0.40 Hz. The old 12-degree option is removed because it reduces case clearance in this concealed layout. Pitch is 90 degrees out of phase with yaw. The arm tip draws a cone; the shaft does not rotate continuously.

The original board exposes only one unused output pin, GPIO2, after the existing display, audio and controls are allocated. It feeds 20kHz PWM to U9 inputs 2/5 with a 10k pull-down. PCA channels 4/5 drive U9 active-low enables 1/4 as static full-on/full-off levels. HIGH disables a gate; LOW passes PWM. R7/R8 pull the enables to 3V3. U9 outputs 3/6 reach U11 AIN1/AIN2 through 220ohm resistors with 10k pull-downs. Spare U9 inputs 9/12 are grounded; enables 10/13 use 5V_MOTOR. Firmware writes and reads back MODE2=0x06 (OUTDRV=1, OUTNE=10). Therefore global OE HIGH makes PCA outputs high-impedance and both head gates disable through their pull-ups. The factory OUTNE=00 is unsuitable because it drives these active-low gate enables LOW. No 50Hz PWM drives the motor; GPIO2 supplies 20kHz independently. [LILYGO pin map](#), [NXP PCA9685 MODE2 and full-on/full-off controls](#)

Firmware limits head duty to 100/255. Before a direction update it removes GPIO2 PWM, writes both gates disabled and checks every I2C acknowledgement, then enables the selected gate. Reversal ramps to zero and waits 100ms before applying opposite polarity. A zero command, disarm or fault removes PWM immediately. Timeout, rear STOP, Wi-Fi loss, low battery and I2C failure disable the shared SLP line. The 100ms interval also applies after a brief zero command or quick re-arm. The driver uses fast-decay PWM, with one input pulsed and the other low.

SLP is held steady during operation; it is not used as a PWM input. [TI DRV8833 control table](#)

Power and battery

The retained BLF-1206A provides 12 V nominal, 6 Ah, 72 Wh, 12 A continuous discharge, and internal protection and cell balancing. Published mass is 0.7 kg. Its metric and inch dimensions differ: 108×64×69 mm versus 4.4×2.6×2.8 in, or 111.8×66.0×71.1 mm. Use the larger values for fit. Its padded design envelope remains 70 X ×114 Y ×76 Z mm. Use firm nonconductive pads to take up the unused space in the new universal cradle. Keep the PP30 discharge connector intact and route it away from motor ends and moving parts. [Bioenno BLF-1206A](#)

Use the matched BPC-1502DC charger for the Bioenno battery only: 14.6 V, 2 A with its factory DC charge plug. The lead-acid alternatives use their own SLA charger. Switch off, unplug the PP30 discharge connection, and remove every battery from the printed shell before charging. Charge on a nonflammable surface with the pack and charger visible. Do not charge through TTGO's lithium battery connector. This robot has no onboard charging circuit. [Bioenno matched charger](#)

The optional SLA models are PS-1270 F2 and PS-12140 F2. Their combined maximum bare envelope is 100 X 153 Y 103 Z mm, including the listed dimensional tolerance and terminal blades; attached contacts and wires need extra space. Current published masses are 1.97 kg and 3.78 kg. Some older 14 Ah units weigh 4.2 kg. See [battery-options-research.md](#) for exact model limits, revision differences, padded retention and access. The capacity name alone does not define a standard physical size. The battery rests on the base under the printed board platform and uses independent padded zip ties. [PS-1270 manufacturer sheet](#), [PS-12140 manufacturer sheet](#)

Use a short 18 AWG copper SLA adapter with fully insulated female blade contacts and a keyed PP30 connector matching the existing harness. New selected packs use 6.35×0.8 mm F2 blades. An existing F1 pack needs 4.75×0.8 mm contacts instead; never force or loosen a wrong-size contact. Contacts must be rated for at least 10 A DC and the actual wire gauge. The single existing main fuse F1 must be within 100 mm of the SLA positive terminal, counting the adapter, PP30 and both unfused leads. For the Bioenno pack, put F1 within 100 mm of its intact factory PP30. Meter-check polarity before plugging in. Sheet 05 shows this path; every downstream connection stays unchanged. [Power-Sonic terminal dimensions](#)

The selected SLA charger option is Power-Sonic PSC-121000ACX, an automatic 12 V 1 A charger with a stated minimum 3 Ah battery. It is the 14 Ah battery's recommended charger and also admits 7 Ah. The 7 Ah sheet also names PSC-12500ACX. Use the exact unit's instructions; do not substitute the Bioenno charger or a homemade voltage setting. Keep SLA ventilation paths open during use. MAIN OFF and unplugging the battery are required after a low-battery stop and between uses because the firmware stops motion while logic still draws power. SLA has no internal Bioenno-style disconnect. [SLA charger](#), [Power-Sonic technical manual](#)

Optional BOM rows have quantity 0. Their zero unquoted price fields mean a quote is required, not that the parts are free. For an SLA build, set B1 and CH1 to 0, choose exactly one alternative battery, and add CH1-SLA and ADAPTER-SLA. Do not sum three batteries or two charger chemistries into one build.

U2/U3 are Pololu 4091 D36V50F5 regulators. Their fixed 5 V outputs, input range and current capability give substantially more supply margin than the rejected low-voltage pack arrangement. The manufacturer's output-current rating depends on input voltage and temperature. The 5.5 A name is not an unconditional guarantee inside a printed enclosure. Leave the boards ventilated and solder thick power conductors directly to their duplicated power pads. Do not send more than 3 A through one ordinary header pin. The board is 25.4×25.4×9.5 mm and has M2 mounting holes. Leave EN, PG and VRP unconnected. [Pololu 4091](#)

U4 is the Adafruit 4739 MPM3610 fixed 5 V regulator. It is rated for 1.2 A and up to 21 V input. The checked price is \$6.95. This lower-power converter is sufficient only with the specified audio limit. CAD from the manufacturer's board repository gives 10.16×17.145 mm outline and a 2.5 mm mount hole at (5.08, 14.605) mm. The 5 V variant uses the same outline; verify the actual board before fastening. [MPM3610 product](#), [Adafruit board CAD](#)

Use NKK S1A switches for MAIN and ACTUATORS OFF. Their source rating is 20 A at 30 VDC; the panel opening is 12.5 mm. ON joins terminals 1 and 3. The small Adafruit covered toggle was rejected because Adafruit recommends only 1-2 A despite the 20 A marking. Each battery/branch fuse has a stated value. The main 7.5 A fuse protects the harness close to the battery. Branch fuses are 3 A at each motion converter input and 1 A at the logic converter input. These are wire-fault protection, not a precision current limiter or motor thermal model. [NKK S1A](#), [Adafruit covered toggle](#)

Current allowances at 5 V are 5 A motors (four wheel bridges plus one head bridge), 4 A for four servos, and 0.9 A logic/audio. The two discharge resistors add 0.5 W, giving 50 W total and 5.25 A estimated pack current at 11.2 V and 85% conversion efficiency. The motor regulator allowance including its bleeder is 5.05 A; the servo regulator allowance is 4.05 A. These are allowances, not measurements or unconditional regulator ratings. The 4 A servo budget uses 1 A per unit because authoritative stall-current data is missing. Measure actual units with head, arms, drive and audio active. Keep the existing 7.5 A main fuse and 3 A / 3 A / 1 A input branch fuses. Stop if a rail falls below 4.8 V, a fuse opens or a driver shuts down. Never increase fuse or sense-current limits to force movement.

The 100 ohm 1 W discharge resistors each consume 0.25 W. They reduce residual actuator voltage after S2 opens. Capacitors still contain energy for a fraction of a second. The disconnect causes coasting; it is not an instantaneous mechanical brake or a certified emergency-stop system. Prove that both actuator rails fall below 0.5 V within 1 second during the commissioning test, including the no-load case.

Controller and Wi-Fi connections

The original board uses ST7789 display pins 19, 18, 5, 16, 23 and 4. Buttons use 0 and 35. Its onboard single-cell battery measurement is GPIO34 with GPIO14 power control; that circuit must not measure the 12 V pack. The chosen separate ADC is GPIO39(SVN). GPIO36(SVP) reads actuator power state. Both are input-only. The pin map source confirms those pins are exposed. [LILYGO pin map and source](#), [manufacturer pin-map image](#)

External GPIO assignments are I2C 21/22, servo-disable 27, motor-enable 12, left drive 25/26, right drive 32/33, head PWM 2, audio 17/13/15, battery 39 and actuator sense 36. GPIO12 and GPIO2 each have a 10k pull-down to preserve their low boot strap state. GPIO15 connects only to the amplifier input without added bias. No GPIO receives 5 V. Use 2.4 GHz Wi-Fi. The firmware supplies an AP and can join an existing network. Keep the rear antenna away from the battery, motor cans, metallic paint and fasteners.

The pack divider is 100k top/22k bottom, 1% resistors, with 100 nF at GPIO39. At 14.6 V it gives 2.633 V. At the 14.7 V SLA design upper value it gives 2.651 V nominal and 2.695 V with the worst combination of 1% resistor limits. Multiplier is 122/22. A 100k/33k divider is NOT used; it would exceed the intended ADC range. A second 10k/15k divider maps 5V_MOTOR to 3.0 V at GPIO36, with 100 nF filtering. Firmware stops on a sustained 11.2 V reading and permits re-arm only above 12.0 V. These common limits remain unchanged for the selected batteries. Low voltage stops motion but leaves logic powered, so switch MAIN off and unplug the battery after a stop. Display voltage or a conservative low-battery state; neither chemistry gets a claimed precise fuel gauge.

Sound

The ESP32 decodes the supplied MP3 files and sends I2S to Adafruit 3006 MAX98357A. It needs BCLK on GPIO17, LRC on GPIO13 and DIN on GPIO15. It does not require MCLK. OUT+ and OUT- go only to the speaker. Neither speaker lead connects to ground. Use a 100k resistor from GAIN to 5V_LOGIC for 3 dB gain, and leave SD at its factory mono-mix configuration. [MAX98357 pinouts](#)

The speaker is Adafruit 1313, 8 ohm 1 W, 77.8 × 77.8 × 25.49 mm. Its four mounting tabs are 60 mm apart. It costs \$1.95 and was in stock. The supplied firmware uses fixed digital gain 0.28. This is a design setting, not an acoustic measurement. Verify that continuous test audio is below 2.83 Vrms across the 8 ohm load, using a suitable differential measurement and audio-band filtering. Never attach a grounded oscilloscope clip to an amplifier output.

Normal spoken clips have lower average power than a continuous test tone. [Adafruit 1313](#)

Board placement on the round base

Use the single printed `12_electronics_platform.stl`: 216mm diameter, 6mm deck, integral legs, deck underside at base-local Z140 and top at Z146. Boards stand on 6mm insulating spacers, giving underside Z152. Fit and wire them on the bench before lowering the platform over the battery. Use the actual PCB hole templates and the printed slot grid; drill only the insulating platform where required. Do not infer hole positions from board outlines. Fit fasteners with the plate underside accessible and limit their final underside projection as specified in the mechanical chapter.

Board centers in base-local millimetres are: buffer prototype(-38,-29), divider prototype(-38,29), PCA9685(39,-44), U2 motor regulator(24,-10), U3 servo regulator(60,-10), U5 left driver(23,24), U6 right driver(59,24), U11 head driver(23,53), U7 amplifier(59,49), and U4 logic regulator(81,32). Positive Y is rear. Keep the board long side along X except the logic regulator, whose 17.145mm side follows Y. These positions match the drawing and video layout.

The two prototype outlines are 81×51 mm; PCA is 62.5×25.4 mm; each driver is 26×18 mm; each high-current regulator is 25.4×25.4 mm; the amplifier is 19.4×17.8 mm; logic regulator is 10.16×17.145 mm. The rectangular envelopes fit without overlap. Connector bodies, solder joints, nuts and wiring still require physical clearance checks. Solder high-current leads directly to the regulator pads. Route wires above the boards with strain relief and keep heat-producing parts ventilated. No drilling may damage a trace. [Adafruit PCA9685](#), [DRV8833](#), [MAX98357A](#), [prototype board](#), [Pololu regulator](#)

Harness assembly

1. Disconnect the battery. Fit the switches, fuse holders and ground-star block. Label every rail and both switch OFF positions. Protect all exposed lugs with heat shrink.
2. Fit U2/U3 on insulated standoffs with air space around them. Fit U4 and the two prototype boards. Fit U1 in the rear holder with the display facing outward. Keep the USB connector accessible.
3. Assemble the buffer board without ICs installed. Use DIP14 sockets. Check U8 pin14 to 5V_SERVO, U9 pin14 to 5V_MOTOR, and both pin7 to GND. Check every channel against sheet03. Add one 100nF capacitor per socket. Fit signal pull-downs, gate pull-ups and 220ohm output resistors.
4. Modify PCA OE as described above. With the board unpowered and isolated, check that OE is not still tied to ground through its original resistor. Add the external 10k to 3.3 V. Record the actual board revision and removed resistor location for later service.
5. Wire the power branches in 18 AWG. Wire one 22 AWG positive/ground pair from the servo distribution block to each servo plug. Do not route aggregate servo current through the PCA header rail or prototype-board strips.
6. Wire each motor to its own bridge. Add 100 nF directly at each motor's solder tabs. Retain only short lengths of the supplied 28 AWG leads. Twist the two motor conductors. Add strain relief near the plastic gearbox, not on the motor tabs.
7. Wire signals from the CSV. Keep I2C and I2S away from motor wiring. Add service loops between vertically stacked sections and the rear board mount. Keep wires clear of wheels, the head friction track and hidden arm links. The rotating head has no wiring. U9 and the head motor both use 5V_MOTOR; U8 and the four servos use 5V_SERVO.
8. Fit and label the removable J_USB_ISO run-power connection. To program: switch MAIN off, unplug the battery, open the run-power link, then attach USB. Unplug USB before reconnecting the link and battery. No

external battery is connected to the TTGO JST socket.

Bench acceptance before the shell is closed

Use a current-limited bench supply for first power where available. Otherwise leave motors and servos unplugged until polarity and rail checks pass. Raise all four ground wheels. Keep hands clear of hidden links and the head friction contact.

1. With no power, check for shorts between each positive rail and GND. Check for no direct connection between the three 5 V outputs. Check capacitor polarity and fuse values. Check each power connection by meter, not wire color alone.
2. MAIN on and ACTUATORS off: measure logic 5 V; verify rear screen boots. Verify both actuator rails discharge below 0.5 V within 1 second. Check no servo power appears through a signal wire.
3. ACTUATORS on but disarmed: measure all three rails 4.8-5.2 V. Confirm the battery reading agrees with a meter within 0.2 V after calibration. Verify GPIO36 is about 3.0 V.
4. Hold TTGO reset. Verify all three SLP pins and GPIO2 below 0.3V, and PCA OE above 3.0V. Motors must not run. Release reset; the robot stays disarmed. Repeat power cycles with ACTUATORS on.
5. Fit one motor at a time. Verify direction and stop at low duty, with ground wheels raised and head tyre clear of its track. Before attaching the head motor, scope U11 AIN1/AIN2 at low command. Only one input may have 20kHz PWM. Maximum duty must be 100/255 and a direction reversal must include at least 100ms with both inputs low. Do not deliberately stall a motor.
6. Fit four arm servos with horns removed. Check centres and mapping. Attach centred horns, then make slow small circles while the shoulder is open. Fit the head track after checking bearings rotate freely. With power off, adjust tyre pressure only enough for reliable drive. Zero head command must remove PWM; there is no servo-neutral adjustment.
7. At low speed, close the browser, turn off Wi-Fi and disconnect the control device in separate tests. The 500ms timeout must remove wheel and head drive and disable arm signals. Reconnect requires explicit arming. Test physical ACTUATORS OFF while driving the head and ground wheels; both motion supplies must discharge while the display remains on. Measure actual head coast angle.
8. Simulate low battery with an adjustable supply rather than over-discharging the pack. Confirm sustained 11.2 V causes stop, 12.0 V is required before re-arm, and an invalid ADC prevents arming.
9. Weigh the finished robot. First drive on a hard flat floor at low duty. Test straight motion and broad forward turns. The supplied control interface intentionally uses arcs rather than in-place turns. Log loaded pack voltage, each 5 V rail minimum and motor-driver temperature. No rail may drop below 4.8 V; there must be no reset or repeated thermal shutdown. Stop if the robot binds while turning. A smaller/lighter shell or lower-friction floor is required if the specified motors cannot turn it reliably.
10. Run 15 minutes of representative driving and animation with pauses. Confirm regulator and driver package cases remain below 60°C with a thermometer. Motors, servos and printed mounts must stay below the assembly guide's more conservative 50°C limit. Wire joints must remain cool and wheels must not rub. These are project acceptance limits, not claimed vendor maximums. Test audio with drive and arms active. Check there is no reboot or audible power noise.

Calculations and limits

electronics/calculations.json is generated from the documented assumptions by python electronics/generate.py. At the 1 A current limit, a simple linear interpolation from the vendor's 6 V current/torque data estimates 0.0492 N·m per motor, or 6.25 N combined wheel force at 31.5 mm radius. Gearbox losses, current ripple and unit variation reduce confidence in this estimate. It is not a continuous-torque rating. The unloaded 6 V, 250 RPM sample would imply 0.825 m/s wheel speed; the actual 5 V loaded robot will differ. PWM limits are not a speed governor without encoders.

With 75% of nominal 72 Wh treated as useful energy, Bioenno runtime estimates are 3.6 hours at 15 W average and 2.16 hours at 25 W average. These are planning values for that pack only. SLA capacity falls as discharge current rises, and the robot's cutoff stops before the datasheet capacity endpoint. Do not scale the Bioenno estimate directly by SLA amp-hours. The builder must measure loaded current and an actual discharge cycle to state runtime. [Power-Sonic capacity and discharge guidance](#)

The generated battery_options results give a separate partial mass for each choice, using the installed-plastic value and revision named in mass_baseline. They add five motors, five wheels and four servos, then one battery. Boards, cables, fasteners, bearings and finish remain additional mass. The 14 Ah option adds 3.08 kg over the 0.7 kg Bioenno pack, or 3.5 kg for an older 4.2 kg SLA unit. Weigh the finished assembly and repeat the loaded tests for each installed battery. Electrical and physical fit do not establish a TT motor payload rating.

Obtain five required 3766 wheels before accepting wheel and friction-track fit. The previous head servo is removed from the BOM. Recheck current stock and regulator availability before ordering. The firmware and circuit files are checked digitally; head slip setting, loaded turning, noise, servo current and runtime require physical commissioning.

Battery options for the printed platform revision

Research date: 2026-09-12. Scope: retain the Bioenno battery and define two specific 12 V sealed lead-acid alternatives. This note supplies dimensions and electrical requirements for the new CAD. It does not certify a printed mount or loaded robot. The batteries, chargers and connectors have not been purchased or physically tested.

The outcome is one removable battery below one printed board platform. Changes to motor type, motor voltage, current limits and onboard charging are outside this work. The battery adapter, optional BOM entries and calculation cases are also integrated into the existing electrical generator and notes. Proof consists of primary manufacturer specifications, the existing circuit and firmware, the unchanged downstream connection endpoints, and explicit calculations below. Final mesh and print checks belong to the CAD revision.

Supported battery models

Keep the existing Bioenno BLF-1206A: 12 V, 6 Ah, 72 Wh, 0.7 kg and 12 A continuous discharge. Its page gives both $108 \times 64 \times 69$ mm and $4.4 \times 2.6 \times 2.8$ inches; these are not exact equivalents. The larger converted dimensions are $111.76 \times 66.04 \times 71.12$ mm. The existing padded envelope, with the long dimension along robot Y, is $70 X \times 114 Y \times 76 Z$ mm. Its protection circuit and PP30 discharge connector remain intact. Use its BPC-1502DC 14.6 V, 2 A LiFePO₄ charger. [Bioenno battery and charger specification](#)

The defined 7 Ah alternative is Power-Sonic PS-1270 F2. Its current sheet gives $151 \times 65 \times 94$ mm case dimensions and 100 mm total height over the terminals, with ± 2 mm dimensional tolerance. The standard battery is approximately 1.97 kg; the sheet lists 2.3 kg for the VDS variant. Capacity is 7 Ah at 0.35 A to 10.5 V, but only 4.27 Ah at the stated one-hour rate. The sheet recommends PSC-12500ACX. Its published short-circuit current is 196 A. Use the F2 version when buying a new pack. An existing F1 version needs matching F1 contacts. [PS-1270 datasheet, version 1.2, 2026-08-18](#)

The defined 14 Ah alternative is Power-Sonic PS-12140 F2. The current sheet gives $151 \times 98 \times 95$ mm case dimensions and 101 mm total height, with ± 2 mm dimensional tolerance. Its approximate mass is 3.78 kg. Capacity is 14 Ah at 0.7 A to 10.5 V, and 8.5 Ah at the stated one-hour rate. The recommended charger is PSC-121000ACX. Its published short-circuit current is 392 A. [PS-12140 datasheet, version 1.2, 2026-08-18](#)

An older official summary lists the PS-12140 at 4.2 kg and 100 mm total height. Use the newer, taller dimensions for the CAD and allow the older, heavier battery in the load assessment. Record the label, measured dimensions and mass of the actual purchased unit. A 14 Ah label alone does not establish fit. [Power-Sonic earlier summary](#)

These are AGM valve-regulated sealed lead-acid batteries. The design does not cover flooded motorcycle batteries, two batteries connected together, or every battery sold as a standard 7 Ah or 14 Ah unit. Only one battery is connected at a time.

CAD envelope and access requirements

Use robot X for width and Y for front-to-rear length. Rotate both SLA packs so their 151 mm dimension follows Y. The combined maximum bare-battery envelope is $100 X \times 153 Y \times 103 Z$ mm. The 103 mm height already includes terminal blades and dimensional tolerance. It excludes attached contacts, boots and bent wire.

The proposed clear cradle is $104 X \times 157 Y$ mm. This admits the maximum case with 2 mm foam on each side and each end. Foam pieces stay on the cradle. Fill unused space around the smaller 7 Ah or Bioenno pack with firm nonconductive pads. The pads must prevent lateral movement without pressing on terminals, seams or valves. The platform legs carry the boards directly to the base. The battery does not support the platform.

With the base floor at local Z6 and a 2 mm bottom pad, the largest terminal envelope ends at Z111. The platform underside at Z140 leaves 29 mm above the bare terminal envelope before PCB hardware is fitted. PCB screws, nuts and washers must project no more than 3 mm below the deck, leaving at least 26 mm above the bare terminal envelope. Keep attached battery contacts, boots, tie heads and wire bends below Z130, giving at least 7 mm clearance to the permitted PCB hardware. These are geometric allowances, not a verified connector fit. Change the connector routing if the actual hardware fails that check; do not raise or modify the printed platform without new fit checks.

The existing skirt's top opening is 184 mm. A 100 × 153 mm rectangle has a 182.7813 mm diagonal. That leaves only 0.6094 mm radial clearance when centered. Paint, print error, tilted insertion and terminal boots can remove that margin. The supported service procedure must therefore allow the whole skirt to lift off the base. Remove the printed platform before removing the battery. Passing the battery through the top is an optional measured fit, with foam left behind and contacts fully within the outline. Do not force the pack through the opening.

The new platform's PCB supports must keep board undersides and solder joints clear of the deck. Provide ventilation below the regulators and access for both battery terminals. Zip ties can retain boards only across suitable bare board edges or insulated saddles. They must not cross hot inductors, connectors, soldered leads or components. The regulator mounting holes remain M2; do not enlarge them to accept M2.5 hardware. [Pololu regulator outline and mounting holes](#)

Battery retention

Provide at least two independent transverse tie paths at separate Y positions. Use padded contact with the battery case and rounded printed slot edges. Locate the paths away from terminal blades, vents and cables. The loops pull the battery toward the base; separate pads and stops resist sliding. Do not make the board platform the battery restraint.

Allow at least 500 mm tie length. The bare maximum transverse path is already approximately $2 \times (100 + 103) = 406$ mm before the floor path, pads and threading tail. A 350 mm strap from the earlier BOM is too short. A design minimum is a 4.8 mm-wide nylon tie with a published loop strength of at least 222 N; slot width must match the chosen product. A larger product is acceptable if its strap and head clear the CAD.

A concrete size/rating reference is KSS CV-500W: 500 mm long, 4.8 mm wide and 222 N minimum loop strength. Its published bundle limit is 147 mm. The manufacturer page does not dimension ribbon thickness or latch size, so measure the actual tie before accepting it. The digital route uses a 1.2 mm-thick ribbon; thicker products need a successful full-path hand fit and must not force or scrape a printed slot. The rating belongs to the tie, not the printed slots or complete restraint. Do not join short ties or substitute a 7.6 mm strap into the 6.2 mm slots. Replace ties removed during service. [KSS CV-500W manufacturer specification](#)

For the soft Bioenno pouch, use broad nonconductive padding beneath each tie. Tighten only enough to prevent movement. Do not dent or crease its wrap. The base proof load and restraint check must use the actual installed battery mass. No tie or printed-slot load rating has been verified here.

Adapter, fuses and charging

Keep the existing keyed PP30 robot input. Make one labeled SLA adapter from the selected terminal type to the same PP30 polarity. F2 is a 6.35 mm-wide, 0.8 mm-thick blade; F1 is 4.75 mm wide and 0.8 mm thick. Use matching insulated female quick-disconnects rated for at least 10 A DC and the selected 18 AWG copper wire. Cover each terminal independently. Measure positive polarity at the assembled PP30 before connecting it. Do not leave alligator clips inside the robot. [Power-Sonic terminal definitions](#)

Locate the single existing 7.5 A main fuse F1 within 100 mm of the SLA positive terminal. This maximum includes the short adapter, PP30 connection and unfused lead to F1. F1 stays after the PP30 interface in the existing

connection schedule. Fully insulate and restrain the short unfused path. For the Bioenno pack, place F1 within 100 mm of its intact factory PP30. Do not add a second main fuse. Keep the 3 A motor-converter, 3 A servo-converter and 1 A logic-converter branch fuses. The larger battery does not justify larger fuses or motor-current limits. These are project harness requirements based on the existing 50 W allowance in `electronics/calculations.json`.

The Power-Sonic PSC-121000ACX is an automatic 12 V, 1 A SLA charger with a stated minimum battery capacity of 3 Ah. It is the recommended 14 Ah charger and its capacity range also admits 7 Ah. The 7 Ah battery's own suggested PSC-12500ACX is another supported choice. Use the exact charger instructions and terminal polarity; charge with the battery removed and disconnected from the robot. The current charger page has inconsistent generic voltage fields, so this design does not turn those fields into a homemade charging profile. [PSC-121000ACX charger specification](#), [PSC-12500ACX charger](#)

Keep the LiFePO₄ and SLA chargers separately labeled. Never use the TTGO battery socket as a charge connection for either 12 V pack. No battery is charged inside the body. Keep ventilation paths open during operation; sealed lead-acid does not mean gas-tight. The manufacturer prohibits use in an airtight enclosure and calls for load disconnection at the end of discharge. Recharge after use rather than storing a discharged SLA pack. [Power-Sonic technical manual](#), pages 12 and 21

Existing power and firmware compatibility

The two Pololu D36V50F5 regulators accept 5.5-50 V and retain their 5 V outputs. The MPM3610 board accepts 4.5-21 V and also stays at 5 V. Thus the selected SLA batteries do not need a different regulator voltage. Actual current capacity still depends on heat and airflow. Preserve the existing regulator temperature and rail-voltage tests after moving the boards. [Pololu 4091](#), [Adafruit MPM3610](#)

The existing 100k/22k divider produces 2.6508 V at a 14.7 V input. With the top resistor 1% low and the bottom 1% high, the result is 2.6946 V. The existing firmware allows plausible readings up to 15.0 V. No charging source is connected while operating. The divider and this operating range therefore remain suitable for the selected batteries. These values are calculated, not ADC measurements.

`firmware/include/config.h` currently stops motion at 11.2 V and permits re-arm at 12.0 V. Keep these common limits; they stop before the selected batteries' stated 10.5 V low-rate capacity endpoint. This does not provide a charge percentage or guarantee a particular depth of discharge. The current firmware disables motion but keeps the controller and audio supply powered. An SLA pack has no Bioenno-style internal protection. After any low-battery stop, switch MAIN off and unplug the battery. Leave it unplugged between uses. Do not claim that the current firmware gives an unattended full-battery disconnect.

Finished mass and drive limits

The final FACET-1 slice report predicts 2.47398 kg of installed printed plastic for the motor-clamp option. It excludes support waste and purchased hardware. Five 30.6 g motors, five 38 g wheels and four 14 g servos add 0.399 kg, for a 2.87298 kg partial subtotal. These terms are also recorded in `electronics/calculations.json`. [Required motor mass and torque](#), [required wheel mass](#), [servo mass](#)

Adding the battery gives incomplete robot totals of 3.57298 kg with Bioenno, 4.84298 kg with PS-1270, and 6.65298 kg with the current PS-12140. The older 4.2 kg battery makes that partial total 7.07298 kg. An explicit 1 kg allowance for all remaining electronics, bearings, fasteners, wiring, pads and finish gives roughly 4.57, 5.84 and 7.65 kg respectively, or 8.07 kg with the older heavy pack. The allowance is not a measured BOM sum or a payload rating. Weigh the finished robot.

The existing calculation estimates 6.245 N of combined ground-wheel force at the driver current limits, using a linear current/torque approximation. It is not a continuous rating. At the illustrative 7.65 kg mass, the force is only about 0.0832 times robot weight; steering scrub uses part of that force. Neither motor nor wheel listing supplies a

robot payload or allowable gearbox shaft load. SLA support establishes a defined fit and electrical option. Loaded driving remains a physical acceptance gate.

Test the complete mass on hard, level indoor flooring with slow starts and broad moving turns. Retain the existing 5 V supply and 1 A bridge limits. Reject a build that binds, repeatedly stops, drops a rail below 4.8 V, overheats, or deforms its printed supports. Do not increase motor voltage or defeat current limiting to move the heavier battery. Use the retained Bioenno pack if the SLA build fails the loaded tests. The requested drive parts stay unchanged.

Research checks

Read directly: `bom/electronics.csv`, `bom/hardware.csv`, `cad/dalek.scad`, `cad/h2d-slice-check.json`, `docs/electronics-research.md`, `docs/mechanical.md`, `electronics/generate.py`, `electronics/calculations.json`, `firmware/include/config.h` and the battery handling logic in `firmware/src/main.cpp`.

Primary PDFs above establish specific battery revisions and dimensions. Manufacturer pages establish the retained regulators, connectors, charger choices and drive components. Python arithmetic returned a 182.7812900709479 mm battery diagonal, 2.6508196721311474 V nominal divider output and 2.694555353901996 V at worst 1% resistor limits. No physical fit, load, runtime or charging test was performed.

`python electronics/generate.py` passed with 185 wiring rows, five parseable SVG sheets and three alternative battery cases. Comparison with the prior Git wiring CSV retained all 185 endpoints, nets, gauges and colors; only the BAT_POS adapter note changed. The default electronics BOM remained \$443.23 because all four optional rows have quantity 0. All five SVGs rendered with their text inside the page bounds. The changed power, circuit-detail and battery-option pages were visually checked. `git diff --check` passed for the assigned files. Rerun the generator after final CAD slicing so `mass_baseline` names the new revision and its installed-plastic mass.

Design research and engineering decisions

This package describes a small, original Dalek-shaped indoor robot. The design uses the requested Adafruit TT motors and orange wheels, an original TTGO T-Display ESP32, a printed shell, two moving arms, and a head that can turn through repeated full revolutions. It is a fabrication design for a first prototype. The CAD and software can be checked digitally. Their operation under actual load still needs the commissioning tests in `assembly.md`.

Source method and date

Product facts were checked against manufacturer product pages, dimensional drawings, component datasheets, and official board source files on 11 September 2026. The required motor and wheel pages were checked first. Component research and exact links are also recorded in `electronics-research.md`, `mechanical.md`, and the BOM CSV files. Prices exclude shipping, tax and regional differences. A product being listed does not guarantee stock when an order is placed.

Quoted electrical ratings describe a specific part and operating condition. Design targets, calculated limits, and assumptions below are identified separately. Neither a successful firmware compile nor a closed STL mesh establishes that a physical robot is safe to drive.

Size and shape

FACET-1 changes the skirt to twelve full-height flat panels arranged around the circular 300mm base. Each main face follows one linear taper from 150mm vertex radius at Z0 to 110mm at Z210. The old middle bend and exterior round band are removed. The four hemisphere rows sit normal to these faces, with raised circular bezels. The true panel wall thickness is 1.8mm; circular joint collars and a faceted internal reinforcing rib retain the existing assembly datums. The skirt remains one 213mm print and the robot remains 563.8mm tall.

Four body sections stack vertically: base, faceted skirt, combined 186mm shoulder/neck, and head. The other nine STL designs, all mounts and the 90 x 52 x 103mm arm boxes retain their geometry. There are still ten STL files: fourteen prints with the four motor clamps, or ten when motor ties replace those clamps. The existing 100-image inventory remains the appearance reference; this revision follows the user's specific request for actual flat panels while keeping the circular base.

Each body section is designed to print upright on the H2D. Bambu specifies 325 x 320 x 325 mm for single-nozzle printing and 300 x 320 x 325 mm for dual-nozzle printing. The 350 mm combined width is not available to either nozzle individually. The circular base plus an 8 mm brim occupies 316 x 316 mm, so it requires the single-nozzle area and an explicit placement check. The actual exported dimensions and slice placement are recorded for each tested part. Use the supplied print manifest at 100% scale. Scaling changes motor pockets, holes and the stacking registers. [1]

The base uses PETG, a 6mm floor, continuous perimeter walls and ribs, and integrated motor pockets. The selected structural slice uses six perimeter lines, six top and bottom layers and 40% gyroid infill. Wheel wells extend through the inner top flange so the motor/wheel pairs can enter vertically; the outer 3mm bumper band remains continuous. Four printed hook clamps use one screw per motor, or two independent ties per motor use dedicated staggered floor routes. The battery rests on the base. The single printed board platform stands above it and supports only the electronics. These are design specifications, not measured load ratings. The assembly guide specifies a physical proof-load test with the small gearmotors removed from the load path.

The reinforcement has a weight cost. The current `cad/h2d-slice-check.json` separates predicted installed model mass from removable support and brim. Earlier rectangular-body mass values do not describe this revision. Add the battery, all five motors and wheels, four servos, electronics, bearings, fasteners, wires and finish to the

current installed-plastic prediction. The original small motors remain as requested; no payload rating is inferred from a successful slice. Their ability to drive and turn the completed assembly must be established with programmed commands and current and temperature measurements on the intended floor.

PLA is suitable for indoor cosmetic body sections, while the base and loaded actuator structures use PETG. The skirt and shoulder are complete rings, with register features that align each tier and fasteners that retain it. Small functional parts are printed first for fit checks. No separate coupon STL is required. Paint remains clear of the registers, bearing seats and motor pockets.

The requested drive parts set the operating limit

Adafruit 3777 is a low-cost 1:48 plastic gearbox motor. Its allowed supply is 3-6 V. Adafruit reports 1.5 A stalled at 6 V and 0.8 kg.cm stall torque at that voltage. Adafruit 3766 is a 63 mm diameter, 29 mm wide press-fit wheel for the TT shaft. Four separate motors drive four wheels. The wheel page was out of stock when checked; the design retains that exact requested wheel. [2,3]

Stall torque is a brief limiting value, not a continuous operating rating. At the 6 V specification, 0.8 kg.cm is approximately 0.0785 N.m. With a 31.5 mm wheel radius, four motors give an ideal combined stall force of $4 \times 0.0785 / 0.0315 = 9.97$ N. At the chosen 5 V supply and with current limiting, the available force is lower. Gear loss, shaft friction, unequal load and warm windings lower it further. This calculation is an upper bound; it is not a payload rating.

For a design example, a 3 kg robot on a smooth floor with an assumed rolling resistance coefficient of 0.03 requires about 0.88 N just to roll. An acceleration of 0.15 m/s² adds 0.45 N. That 1.33 N example is plausible for gentle straight motion but does not include steering scrub. The coefficients and acceleration are engineering assumptions, not measurements of the supplied wheels. Carpet, door thresholds and slopes are outside the initial operating envelope.

With four fixed wheels, steering requires the tires to slide sideways. Moving the wheel mounts inward to fit the circular base reduces the available steering lever arm. Stationary pivot turns cannot be promised on a high-grip floor. Use broad moving arcs first; release the control immediately if a wheel stops. The exact track and wheelbase appear in the mechanical guide. Final track also depends on wheel insertion depth. No earlier steering calculation for the wider rectangular chassis establishes performance of this revision.

Motor speed is also open loop. The motors have no encoders. A PWM percentage controls electrical drive, not a measured speed in metres per second. Motors may need polarity and trim checks. The firmware ramps drive commands and limits their range. Wheel diameter and unloaded motor speed can estimate an upper speed, but loaded speed and stopping distance must be measured on the completed robot. [2]

Driver selection

Three Adafruit DRV8833 boards provide five used H-bridges: four wheel channels and one head-motor channel. The remaining bridge is unused. Each pair of wheel motors on a side receives the same logical direction command. Motor outputs are not wired together. The Adafruit breakout has a nominal 1 A current limit per bridge. Retain its current-sense resistors. A current limit reduces damage during a fault but does not make a stalled motor acceptable for continuous operation. [4]

The DRV8833 operating supply range includes 5 V. The TI datasheet distinguishes continuous, peak and package-dependent current values. The actual breakout's default limit and heat dissipation govern this build. It is not valid to choose a driver merely because a catalogue headline says 2 A peak. Reset-state wiring also matters: the shared sleep line must be low at ESP32 reset, not left to firmware that has not yet started. [4,5]

Battery and power distribution

The selected pack is the Bioenno BLF-1206A, a protected 12 V, 6 Ah LiFePO₄ pack with a stated 72 Wh energy rating. The manufacturer lists a 12 A continuous discharge limit, approximately 0.7 kg mass, and the BPC-1502DC 14.6 V, 2 A charger. Its internal protection and large energy reserve are useful for a mobile prop. It supplies converters; it must never connect directly to a 5 V rail, a TT motor, or the T-Display battery socket. [6]

The design also accepts one Power-Sonic PS-1270 F2 12V7Ah or PS-12140 F2 12V14Ah lead-acid battery through the documented fused PP30 adapter. The combined supported bare envelope is 100X x 153Y x 103Zmm, including terminals and manufacturer tolerance. Capacity alone does not specify case size. A 157Y x 104Xmm cradle provides room for fixed padding, and the platform leaves clearance for insulated terminals and wiring. Two long padded zip-tie loops retain the selected pack; foam blocks locate the smaller cases. Charge lead-acid batteries with their matched lead-acid charger, outside the robot. The existing LiFePO₄ charger remains dedicated to its original pack. Exact data, sources and mass limits are in `battery-options-research.md` and the fifth circuit sheet.

The 14Ah pack adds several kilograms compared with the Bioenno. Fit and electrical compatibility do not prove loaded traction, gearbox capacity or turning performance. Retain the existing 5V motor supply and current limits. Weigh and test the finished robot on a level hard floor. After any low-battery stop, switch MAIN off and unplug the battery: firmware stops motion but does not disconnect the logic supply, and an SLA battery has no pack BMS to prevent continued discharge.

This is the main deliberate exception to an Adafruit-only shopping list. Adafruit's 3.7 V, 6600 mAh pack has a much smaller energy reserve and a stated 3.3 A maximum discharge. Its own page recommends a substantially lower sustained draw. Boosting its output to 5 V while driving five motors and four servos would demand more battery current than the output current. Capacity in mAh alone is not a sufficient selection rule. A protected ready-built pack avoids assembling and balancing loose cells. [7]

Separate regulated 5 V supplies feed the wheel motors, servos, and logic/audio. This prevents a servo pulse or motor start from drawing through the ESP32 board's small power path. All returns meet at the common ground distribution point. The physical actuator switch removes energy from both movement supplies while leaving the controller display available. The main fuse is close to the battery. Wire size, connector ratings, branch fuses and the actual switch DC rating appear in the electrical package.

Runtime is an estimate until measured. With 72 Wh nominal and an assumed 75% usable energy allowance, 54 Wh remain. At an estimated average load of 15 W, that is 3.6 hours; at 25 W, 2.16 hours. These values cover energy only. Temperature, battery condition, low-voltage shutdown, repeated starts and actual servo load change the result. Do not call any of these a guaranteed runtime. Record the first full discharge test with voltage, duration, activity and temperature. [6]

Arms and head

Each arm has two perpendicular positional servo axes. The controller varies the two angles using sine and cosine signals. For small angles, the arm tip follows a near-circular orbit. At large angles, the geometry becomes a spatial curve, so the firmware uses a modest amplitude. Equal pulse amplitudes do not guarantee equal physical angles until the servo centres and directions have been calibrated. The two assemblies can share a phase offset without needing a cam mechanism.

Four MG92B servos provide the arm axes from inside the shoulder. Spherical arm roots cover the sockets; cases, horns and cables remain behind the armor. Their power and signal requirements must both be met: the Adafruit product page specifically calls for 5 V control signals. A PCA9685 running at 3.3 V does not provide a 5 V HIGH just because its separate servo power terminal has 5 V. U8 translates the four servo signals. U9 routes the head motor's 20 kHz PWM under static direction enables. The electrical chapter specifies enable pins, pull-downs,

bypassing and unused inputs. [8]

The head uses a fifth Adafruit 3777 TT motor and 3766 wheel. The horizontal tyre contacts the inside of an integral drum beneath the dome. A sliding motor carriage and adjustment screw set contact; four guide clamps with nyloc nuts hold the setting. The tyre supplies compliance. Start with clearance, introduce light contact, lock the setting and check a complete hand revolution for binding and runout. The adjustment is not a calibrated contact-force control. The bearings carry the dome load; the drive wheel supplies tangential force. [2,3]

Head PWM controls drive level and direction, not a measured angle or speed. The design has no head encoder or home sensor. It therefore cannot promise a repeatable face-forward command. Rubber traction, print finish, runout and temperature change the friction behavior. The manual requires loaded rotation and stopping checks at the final adjustment. The controller applies a zero-output interval before reversing the head motor.

The eye and dome decorations contain no moving electrical wires. Audio comes from a speaker fixed in the body. This permits repeated complete rotations without a slip ring. Bearing, shaft, carriage, adjustment and retention hardware is enumerated in the mechanical bill of materials.

Controller and network

The design is for the original ESP32 TTGO T-Display with a 1.14-inch ST7789 display and 4 MB flash. LilyGO publishes the TFT pins, button pins, battery-monitor pins and board drawings. The S3 and other T-Display models have different dimensions and pin assignments and cannot use this firmware and rear mount unchanged. The screen faces outward at the back. Leave its antenna area clear of wires, metal tape and the battery. [10]

ESP32 Wi-Fi supports station mode, access-point mode and their combined use. The robot serves its own browser interface. A nearby phone can connect directly without an Internet connection, or the robot can join an existing 2.4 GHz network. Station credentials are entered locally. No cloud broker or paid service is needed. The exact setup, fallback behavior and displayed addresses are described in `firmware.md`. [11]

Motion needs a current command stream and an explicit arm action. A lost connection, stale command sequence, browser release, physical actuator power loss or safety fault must remove drive commands. Software timeout is an additional control, not a substitute for the physical power switch. The network is local and HTTP is unencrypted; do not expose it through router port forwarding. The Wi-Fi and application password must remain private to the operator.

Voice collection

The package includes 24 newly synthesized MP3s, about 99 seconds total. Their measured file sizes and durations are recorded in `audio/catalog.json`. They use original short transcripts, the installed desktop speech engine, and a reproducible metallic voice effect. They do not contain ripped television clips. The source list is editable and the generation scripts are supplied. All clips are stored in ESP32 flash, so a memory card or network download is unnecessary.

A MAX98357A converts digital audio to speaker power. The selected 8 ohm, 1 W speaker and limited playback gain keep demand within the logic supply budget. The two speaker terminals form a bridge output: neither is a ground terminal. Fix the speaker to the stationary body and route its wires away from the analog voltage-sense leads. Refer to the electrical source notes for amplifier gain and speaker limit calculations.

What is established, and what must be measured

Digital checks establish that the delivered files exist, mesh volumes are positive, exported parts fit the selected print envelope, audio decodes, and firmware compiles. Inspection checks that connector names and pins agree across code and diagrams. These checks are valuable but cannot detect actual servo backlash, moulding variation, printer shrinkage, real traction, bearing friction, battery noise or a loose screw.

The first build must measure fit, loaded mass, wheel clearance, free head travel, arm current, supply sag, temperature and stopping behavior. The assembly guide makes those checks before closing the shell and before the first floor run. If the requested drive parts cannot move the actual mass on the actual surface, the correct response is to reduce mass or change the operating surface. Do not remove current protection or raise motor voltage above its rating to force a result.

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Dalek appearance reference review

The strongest appearance target is the bronze Time War Dalek: a low dark fender, a tapering body with ordered rows of gold hemispheres, a detailed shoulder band, a dark ringed neck, and a domed head with a raised eye cowl. The revised robot should retain that visual order while using a circular base and fully concealed arm actuators. The circular footprint and adjustable TT-motor head drive are project requirements. They are not claims about the construction of a television prop.

This report defines visual requirements for the mechanical revision. It does not certify the final CAD, printed parts, motor performance, bearing fit, or completed robot. The separate mechanical verification records must establish those points.

Reference inventory and evidence quality

The accompanying reference-image-review.csv (reference-image-review.csv) records 100 selected, visually inspected reference images. Each row contains its source page, original image URL, pixel dimensions, downloaded-byte SHA256, decoded-pixel SHA256, a 256-bit difference hash, an individual observation, and a proposed design implication. Reference IDs D001-D100 are stable within that inventory. All images were accessed and reviewed on 2026-09-12.

The collection contains 50 photographs from an original-prop restoration, 23 bronze-era construction, exhibition and screen images in the Dalek 63•88 archive, 13 official BBC Studios character/story images, 10 BBC America design-history images, one additional original-prop owner photograph, and three official BBC Studios/Propstore bronze auction photographs. These are 100 distinct images, not 100 distinct props. Several images document separate components and successive assembly stages of the same original prop. That coverage is useful for the hidden joints and stacking requirements as well as the exterior finish.

Seventy-seven selected images come from official BBC publications or the owner/restorer of an original prop. Twenty-three come from a specialist historical archive; that archive's historical account is secondary, and original photographer attribution varies. The archive images support visible shape and finish observations, not independent proof of a manufacturing dimension. No dimension in this robot should be justified by counting pixels in an uncalibrated photograph.

The final set has 100 unique image URLs, 100 unique downloaded-byte hashes and 100 unique decoded-pixel hashes. Visual screening also removed repeated crops, non-Dalek character portraits, packaging images, concept collages and weak views. For example, the narrow bronze publicity crop in the archive was omitted in favor of its wider BBC version, and the wide crop of the official execution Dalek was omitted in favor of the larger portrait. Hash uniqueness alone would not establish that those were different photographs. The closest pair among the selected 256-bit difference hashes differed in 89 positions. That calculation is supporting evidence, not a substitute for the visual review.

The reference images and contact sheets remain in temporary analysis storage. They are not included in the repository, the PDF, the video, or the S3 deliverables. The CSV links to the exact originals so the evidence can be inspected without redistributing a third-party image collection.

Overall shape and the circular base

The low fender and broad lower skirt carry much of the recognizable Dalek outline. Official bridge and classic-prop images show the wheels hidden behind the lower body, while the upper body becomes narrower toward the neck. The restored original also shows a small outward ledge below its lowest hemisphere row. These observations support a substantial perimeter bumper, concealed wheels and a clear taper. [D089, BBC bridge photograph](#); [D050, original restoration front view](#).

The photographed bronze props generally have faceted fenders and skirt panels. A perfectly circular base is therefore a deliberate adaptation. It should read as a round version of the same broad fender: a continuous dark lower band, a shallow ledge or bevel above it, and the decorated skirt rising inside that perimeter. A round cylinder with no transition to the body would lose this structure. [D057, bronze full-body view](#).

For this robot, the circular base must be a real geometric boundary in the STL and plan view. Rounded corners on a rectangular base do not meet the requirement. Wheel openings should face downward and inward so they do not create four large exterior notches. The base must still provide independent motor saddles, a continuous structural floor and a clear load path to the stack fasteners. Decorative roundness does not establish strength; the base still needs the mechanical checks and physical commissioning specified elsewhere.

FACET-1 follows the user's request for actual flat panels. The skirt has twelve continuous planar faces and one linear taper, with four aligned hemisphere rows and narrow relief at the panel edges. It remains one complete print. The base and mating collars stay circular. This replaces the earlier curved skirt skin; the reference images themselves were not re-reviewed for this shape-only revision.

Arms and concealed servos

The visible appendage joint is a ball or rounded pivot in a socket surround. An original restoration close-up shows a tube emerging from a ball inside a raised box; another view shows that the surrounding shoulder conceals the rest of the joint. The newer bronze design adds an ornate square plate around a dark circular pivot. [D013, original arm socket](#); [D037, shoulder socket box](#); [D080, official bronze front view](#).

All four arm servo cases, horns, fixing straps and wire loops should sit behind the shoulder skin. The finished exterior should expose only the round socket, the arm stem and the plunger or emitter. A servo-shaped cover on the end of an arm would hide its label but would still spoil the silhouette. The mechanism belongs inside the body. A socket surround integral to the shoulder and a moving spherical cover integral to an existing moving part can preserve the ten-STL limit.

Concealment must hold while the arms move. At every permitted yaw and pitch combination, the socket cover must overlap the shoulder opening. The visible joint must not reveal a servo through an open side or underneath the arm. It must also clear the shoulder, the adjacent arm and the central body panel. The arm stems should remain slim; placing an entire servo in an enlarged external arm shaft would not match the reference.

The proposed circular wiggle needs two coordinated internal axes for each arm. The appearance evidence does not establish the linkage, angle limit or torque. Those are engineering decisions. The operating animation should use the same pivot locations and limits as the CAD and firmware. Show internal mechanisms only during an explicitly marked assembly cutaway, then restore the complete shoulder surface before the exterior operating sequence.

Detail priorities by component

The skirt should retain four ordered levels of hemispheres. Several construction images show that these are repeated shell details, while the bronze examples add dark rims around their roots. Shallow integral annular bezels offer much of that distinction without requiring separate printed rings. Use a consistent diameter within a row and align the columns through the stack joints. The golden dome of each hemisphere should stand out against its darker root and bronze body panel. [D004, skirt construction](#); [D056, nearly finished bronze body](#).

The shoulder benefits most from physical relief. Bronze references show thick rectangular slats with inset faces and round fastener heads. The front central panel is wider than the flanking slats. Side slats are longer where the shoulder offers more height. Their spacing should continue around the back, with only the necessary interruption for the confirmed TTGO T-Display opening. [D058, close slat and appendage photograph](#); [D086, official side photograph](#).

The neck should have three clearly projecting rings over a darker recessed drum. Thin vertical posts and horizontal grille bars give depth to the gaps. Open black space alone can make the neck appear hollow; an unbroken solid bronze drum loses the contrast. A printed dark grille or a dark finish behind integral bars can supply the visual effect. Keep enough service and ventilation access for the actual mechanism. [D015, restored neck structure](#); [D039, neck ring and grille detail](#).

The dome needs a smooth crown, a short lower lip and a raised eye cowl. Shallow panel grooves can run from the cowl region toward the sides. The eye should have a narrow stem, a group of pale concentric discs and a darker larger lens head. A small central recess and several concentric rings give the lens a convincing face. Keep the two dome lamps visibly separate and angled outward. Fine lamp ribbing should be robust enough to survive support removal and handling. [D063, bronze cowl close-up](#); [D100, official lens and lamp close-up](#).

The plunger should have a shallow cup, a defined rim and a smooth stem. The emitter should read as a central tube surrounded by an open cage of rods. Thin movie-prop rods cannot simply be scaled down without regard to print strength. Use fewer or thicker robust rods if needed, but preserve the open cage and muzzle silhouette. Do not add projectile or effect hardware to this decorative component. [D033, assembled cage](#); [D034, muzzle close-up](#).

These details have a clear priority. First preserve the body taper, fender, ringed neck and dome outline. Then add socket covers, hemispheres, slat relief and the eye cowl. Finally add shallow grooves, bolt-head cues and lamp ribs. Small detail should not create more STL files, obstruct a fastener, expose a servo, block airflow or reduce an essential wall thickness.

Color and finish

Use a darker bronze body with warmer gold slats and hemisphere faces. Use near-black in the neck grille, socket balls, hemisphere bezels, plunger cup and lower bumper. Silver appendage stems and cage rods provide a separate material cue. Pale eyestalk discs and a dark lens ring keep the eye legible. These distinctions are directly visible in the official bronze auction side view. [D099, BBC Studios/Propstore side photograph](#).

Use restrained shading in grooves and around bolt heads. Avoid large artificial damage holes or exposed cables. Different studio lighting can make the same bronze prop appear yellow, brown or purple, so the photographs do not justify a single measured paint color. Select actual paints from dry sample pieces under ordinary room light. A decorative blue eye color can suggest the reference without implying that the current head contains a powered lamp. Any functioning light requires a separately documented electrical implementation.

Hidden adjustable friction head drive

The new TT motor and tyre must remain inside the neck and head envelope. Their external visibility is not required to communicate head motion. The prop reference confirms the broader appearance principle: its dome can be lifted separately, and its internal manual control disappears when assembled. [D068, interior dome photograph](#).

The adjustable friction contact is a custom robot mechanism. A service drawing should show the motor, wheel, smooth driven surface, adjustment direction and locking fasteners. The assembled exterior should show none of those details. The adjustment needs a repeatable way to move the tyre into contact and then lock the setting without distorting the neck or forcing the bearing axis off-center. A separate bearing path must carry the dome's weight; tyre pressure should not become the only head support.

The reference review does not determine the correct tyre compression or slip torque. Those depend on the actual Adafruit tyre, printed surface finish, assembled head mass and bearing friction. Commission the mechanism at low power, increase contact only enough for reliable motion, and verify it in both directions. The assembly PDF and video should show how the builder reaches and locks the adjustment after the surrounding parts are fitted. A decorative slot that cannot be reached with a tool does not satisfy adjustability.

Print and assembly implications

Keep each body level as one complete vertically stackable print. Integrate shallow detail into those existing pieces. Preserve a single strong base that fits wholly inside the H2D's usable length and width, including the selected brim and print orientation. A dimensioned plan view must show the new circular outside diameter and the actual wheel envelopes. Slicer checks should be run on the released STL hashes, not on a separate attractive preview model.

Use ordinary purchased fasteners and measured mounting clearances for the hidden hardware. The ten-file limit allows repeated instances of one universal moving part, but the inventory must state both the number of distinct STL designs and the number of physical prints. Avoid using invisible extra printed brackets to make the mechanism work. The video, CAD and assembly instructions must agree about which part carries each servo, head motor and bearing.

Finish recessed neck and socket areas before final stacking. Confirm motor access before installing electronics above the base. Align bump columns and shoulder details before tightening stack bolts. Route the arm leads entirely inside the shoulder, with slack for the permitted motion. Keep the original rear T-Display readable and its controls accessible. The display opening is a functional adaptation and should be a neat rear panel rather than a feature on the front chest.

Visual acceptance for the revised deliverables

The exterior views should prove the intended result from the front, rear, both sides, above and a low angle. A plan view must show a circular base. The wheels should remain concealed in ordinary standing views. The rear view must show the original TTGO display. The shoulders must show only the intended socket surfaces and arm stems at neutral and at the extremes of the complete allowed motion.

An exploded or cutaway view should show both internal arm mechanisms and the head friction drive. A close view should show the adjustment fasteners and tool path. The completed operating view must restore all covers and shoulder walls. Its arm, head and chassis movements must come from the actual revised geometry. No external servo should disappear merely because a camera angle hides it.

Digital visual acceptance is separate from physical proof. The design remains subject to printed fit, supported load, head traction, motor current, heat, clearance and stop-response tests. The report provides a specific and traceable appearance standard; it does not report any of those tests as completed.

Source register

1. The Prop Gallery, [Doctor Who - restoring an original Dalek](#). Primary owner/restoration account; 50 individual restoration photographs selected. Publication date not stated on the page. The CSV links to each full image.
2. Dalek 63•88, [The New Series: Series One](#). Specialist archive; 23 individual bronze-era construction, exhibition and screen images selected. Photographer and copyright holder vary. Publication date not treated as a manufacturing date.
3. BBC Studios, [Daleks - Explore the Whoniverse](#). Official character and story image index; 13 selected images. Each exact original asset URL is recorded.
4. BBC America Editors, [Doctor Who Gallery: The Evolution of Dalek Design](#), 2020-12-31. Ten selected photographs and screen stills; credited BBC.
5. The Prop Gallery, [Doctor Who - Original Dalek](#). One selected close photograph of the restored original prop. Publication date not stated on the page.
6. BBC Studios / Propstore, [Everything available in the Doctor Who x Propstore auction](#), 2026 auction coverage. Three selected photographs of the bronze prop, including front, side and lens detail.

All source pages and individual image assets were accessed on 2026-09-12. Direct visible observations are distinguished from proposed design implications in the CSV. The reference count describes the reviewed inventory and does not imply permission to redistribute its copyrighted contents.

Original robot voice collection

The 24 MP3 files are in `../firmware/data/audio/`. They are included there so the same files can be uploaded to the robot's LittleFS flash filesystem. `catalog.json` records each exact transcript, duration, size and SHA-256. `phrases.csv` is the editable source list.

These are newly synthesized robot lines, not audio extracted from Doctor Who, a BBC recording, or a particular actor's performance. The text was written for this build. The source is the installed Microsoft David Desktop voice through Windows System.Speech. A low oscillator adds a metallic sound. The collection includes brief familiar words such as "Exterminate" alongside original control and character phrases. No downloaded sound samples are used.

The supplied format is mono MP3, 22,050 Hz, 32 kbit/s. Playback uses the ESP32 and the MAX98357A amplifier in the body. The speaker and all its wires stay stationary during head rotation. The controller lists files automatically. The files are manual sound buttons; the presence of a warning phrase does not mean firmware automatically plays that warning.

To recreate the collection on this Windows machine:

```
powershell -NoProfile -ExecutionPolicy Bypass -File C:/dev/robots/dalek/scripts/generate_voices.ps1
```

This requires the installed Microsoft desktop voice, Python with NumPy, and FFmpeg/FFprobe on PATH. It runs once. It registers no scheduled task. It creates a unique temporary WAV directory and removes only that directory when finished. The persistent outputs are the MP3s and catalog.

To replace a line, edit its row in `phrases.csv`, regenerate, rebuild the filesystem with PlatformIO, and upload that filesystem while actuator power is off. The waveform effect uses a 0.86 playback-rate multiplier, nonlinear amplitude compression, 30 Hz ring modulation, a quiet 45 ms echo, and a 170-3400 Hz frequency band. Peak amplitude is limited before MP3 encoding. Start with amplifier volume low and check the speaker for distortion.

To add your own recordings, use lowercase filenames with letters, numbers, underscores or hyphens and the `.mp3` suffix. Keep the entire flash filesystem inside its partition. Use only recordings you have permission to use. Character names and third-party voice/tool software retain their respective ownership; this package does not grant rights in third-party material.

Digital verification and physical limits

FACET-1 has twelve full-height flat skirt panels in one print, within the same ten STL designs: fourteen prints with the four motor clamps, or ten when motor ties replace them. The platform and its legs form one print. Shoulder and neck form one 186mm print. The skirt is 213mm and the robot remains 563.8mm tall. Both motor and servo-case retention options are documented.

Geometry and access

The following commands passed from C:/dev/robots/dalek. Their reports identify exact current mesh hashes.

- python scripts/export_cad.py --check-only: ten consistently wound, watertight meshes, each one connected positive solid. All fit the stock H2D single-nozzle envelope and recorded brim allowances. Solid-material volume bound: 3,202.8g, not sliced or measured mass.
- python scripts/check_base_mounts.py: 346 checks and 1,117,327 evaluations. Adds actual panel-plane, normal, 1.8mm wall and 240mm rib measurements to the existing mounting tests. Covers motor/wheel installation, four hook clamps/screws, alternative ties, both platform-foot options, batteries, populated terminal clearance, captured nuts, tools and skirt installation.
- python scripts/check_upper_mounts.py: retained 68 checks and 1,650,576 samples. Covers servo ears and retention, yaw-servo entry, bounded screw/nut tools, detached-arm paths, head/bearing entry, speaker clearance and all four body-joint driver approaches.
- python scripts/check_arm_assembly.py: retained 8 checks and 417,648 samples. Both carriers enter through the left lower opening before moving to their final positions. Both supported arms are included.
- python scripts/check_mechanical.py: 110 checks and 374,697 samples. Includes wheels, batteries, skirt interfaces, arm extremes and head adjustment. The 30mm outer fabric radius gives zero visible servo targets in 4,800 current-body rays.
- python scripts/mesh_queries.py: retained 4,400 samples and zero differences from the original query on the final upper mesh. Two near-axis rays must agree; disagreements use the original query. Tolerances are unchanged.

Only the skirt STL changed. The nine other STL hashes match the MOUNT-1 release; upper, carrier-entry and query-comparison evidence is retained for those exact unchanged meshes. All current general/base/facet checks were rerun. The old curved skirt failed all 36 panel-plane, normal and wall tests.

Finite sampling does not prove every possible continuous pose. Skirt lowering separately uses a conservative continuous projection bound. Actual servo shaft offsets and ear holes are not fully dimensioned by the maker. The guide specifies measured fit gates and slotted supports.

Printing and mass

The H2D script uses isolated installed Bambu Studio settings and never contacts a printer. The complete faceted skirt was newly sliced. Nine unchanged slice records were retained only after verifying their exact STL hashes. All ten source hashes and deposited heights match. All generated deposition fits the stock left 325 x 320mm area. Requested brim settings do not prove a separate brim was generated; the report records actual feature output.

The clamp option predicts 4,180.82g filament, 2,473.98g installed plastic and 124.28 hours of sequential printing. The skirt uses 1,009.47g PLA and predicts 23 hours 55 minutes 37 seconds: allow 1.12kg with reserve. The upper shell uses 1,235.49g PETG and predicts 39 hours 10 minutes 11 seconds: allow 1.36kg. Both need sufficient capacity or compatible filament changes.

The electrical generator reads the current installed-plastic prediction. Including only plastic, five motors/wheels, four servos and the battery gives incomplete totals of 3.573kg Bioenno, 4.843kg 7Ah SLA and 6.653kg 14Ah SLA. Other electronics, bearings, fasteners, wiring, padding and finish must be added. Actual weighing is required.

Circuits, control and media

python electronics/generate.py passed 185 wiring rows, five parseable circuit sheets and three battery cases. All pre-existing nets, endpoints, wire gauges and colors remain unchanged. The battery-input note and fifth sheet explain the single fused SLA adapter and matched chargers. Regulator outputs and the 11.2V stop/12.0V re-arm thresholds remain unchanged. Low voltage stops motion but leaves logic powered: switch MAIN off and unplug after a low-battery stop and between uses.

Firmware, pins and 24 original MP3 files are unchanged. Prior compile/control-test evidence applies to those same bytes. The earlier build reported 45,416 bytes RAM, 995,025 bytes application flash and a 2,097,152-byte filesystem image. No new physical control result is claimed.

python scripts/render_drawings.py passed 16 engineering PNGs, five CAD previews and the ZIP. It checks quantities, transforms, board separation and unchanged source hashes. All final views were inspected. Paint boundaries use render-plane cuts without changing STL surfaces.

python scripts/render_video.py passed the new 120-second assembly and simulated-operation video: 2,880 frames, 1920 x 1080, 24fps, H.264 video, AAC audio, full FFmpeg decode and 47 unchanged source hashes. Narration timing rejects overrunning clips. The video includes both motor options, battery choices, the platform, lower-entry upper assembly, adjustable head drive and simulated motion. It is not physical test footage.

The PDF is built from current guides, BOM and five circuit sheets, then rendered and visually checked before publication. The existing GitHub Actions/OIDC workflow publishes exact committed PDF/video bytes to private S3 and checks full downloaded hashes. The requested HTML email contains the PDF and a tested video link with explicit expiry. Workflow and SES acceptance identifiers are recorded in the root execution plan.

Physical acceptance

The design is not physically validated. Complete actual fit, supported base proof-load, retention, connector-clearance, opaque-liner, hand-rotation, stop and loaded-driving tests. Four small TT gearmotors have no published robot payload rating. A stronger base and a fitting 14Ah case do not establish traction, gearbox life or turning ability. Keep the 5V supplies and current limits. Use the lighter battery if the SLA build fails its loaded tests.

Electrical bill of materials

bom\electronics.csv

- 1. reference:** M1-M5 | **quantity:** 5 | **item:** TT geared motor for four wheels and head friction drive | **manufacturer part:** Adafruit 3777 | **unit usd:** 2.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock on2026-09-12 | **url:** <https://www.adafruit.com/product/3777> | **notes:** Required motor;3-6V operated at5V;each uses its own1A limited bridge
- 2. reference:** W1-W5 | **quantity:** 5 | **item:** Orange clear TT wheel | **manufacturer part:** Adafruit 3766 | **unit usd:** 1.50 | **price basis:** listed | **availability checked 2026 09 11:** Out of stock on2026-09-11;recheck before purchase | **url:** <https://www.adafruit.com/product/3766> | **notes:** Four ground wheels plus one head friction wheel;63x29mm;obtain exact part before fit signoff
- 3. reference:** U1 | **quantity:** 1 | **item:** Original TTGO T-Display ESP32 1.14 inch | **manufacturer part:** LILYGO TTGO T-Display | **unit usd:** 25.00 | **price basis:** allowance | **availability checked 2026 09 11:** Exact revision and stock must be checked | **url:** <https://github.com/Xinyuan-LilyGO/TTGO-T-Display> | **notes:** Existing user board can remove this cost; not T-Display-S3 or AMOLED
- 4. reference:** U5-U6 U11 | **quantity:** 3 | **item:** DRV8833 dual motor driver | **manufacturer part:** Adafruit 3297 | **unit usd:** 5.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock on2026-09-12 | **url:** <https://www.adafruit.com/product/3297> | **notes:** Keep stock0.2ohm sense resistors;U5/U6 four wheel bridges;U11A head;U11B unused
- 5. reference:** U10 | **quantity:** 1 | **item:** PCA9685 servo PWM board | **manufacturer part:** Adafruit 815 | **unit usd:** 14.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/815> | **notes:** Remove OE pull-down; VCC3.3V; leave V+ unconnected
- 6. reference:** SV0-SV3 | **quantity:** 4 | **item:** MG92B positional metal gear micro servo | **manufacturer part:** Adafruit 2307 | **unit usd:** 11.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/2307> | **notes:** Includes horns and horn screws; needs 5V control signals
- 7. reference:** U8-U9 | **quantity:** 2 | **item:** 74AHCT125 DIP14 level buffer | **manufacturer part:** Adafruit 1787 | **unit usd:** 1.50 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/1787> | **notes:** U8 four5V servo signals;U9 headPWM routing;useAHCT powered-off-input tolerance;verify actual IC marking
- 8. reference:** U2-U3 | **quantity:** 2 | **item:** 5V5.5A buck regulator | **manufacturer part:** Pololu D36V50F5 4091 | **unit usd:** 39.95 | **price basis:** listed | **availability checked 2026 09 11:** Rationed active and preferred; backorders allowed | **url:** <https://www.pololu.com/product/4091> | **notes:** Separate motor and servo regulators;25.4x25.4x9.5mm; no outputs paralleled
- 9. reference:** U4 | **quantity:** 1 | **item:** 5V1.2A logic buck regulator | **manufacturer part:** Adafruit MPM3610 4739 | **unit usd:** 6.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/4739> | **notes:** Input max21V; solder power pins; EN left open
- 10. reference:** B1 | **quantity:** 1 | **item:** Protected12V6Ah72WhLiFePO4 battery | **manufacturer part:** Bioenno BLF-1206A | **unit usd:** 79.99 | **price basis:** listed | **availability checked 2026 09 11:** Add to cart offered | **url:** <https://www.bioennopower.com/products/12v-6ah-lifepo4-battery-pvc> | **notes:** Default ONE battery;12A continuous;0.7kg;keep PP30;pad smaller pack in universal cradle;replace quantity with0 if selecting one SLA option
- 11. reference:** CH1 | **quantity:** 1 | **item:** Matched14.6V2A LiFePO4 charger | **manufacturer part:** Bioenno BPC-1502DC | **unit usd:** 21.99 | **price basis:** listed | **availability checked 2026 09 11:** Add to cart offered | **url:** <https://www.bioennopower.com/products/lithium-12v-2a-amp-lifepo4-battery-charger> | **notes:** Bioenno ONLY;remove pack to charge;set quantity0 when selecting SLA charger
- 12. reference:** B1-ALT7 | **quantity:** 0 | **item:** Optional replacement12V7Ah sealed AGM battery | **manufacturer part:** Power-Sonic PS-1270 F2 | **unit usd:** 0 | **price basis:** Optional quote required not free | **availability checked 2026 09 11:** Datasheet checked2026-09-12;quote and stock needed | **url:** <https://www.power-sonic.com/wp-content/uploads/datasheets/ps-1270.pdf> | **notes:** Choose instead of B1 or ALT14;set ONE battery quantity1 after quote;F2 terminals;max bare67X153Y102Zmm;1.97kg standard;actual fit and loaded drive test required
- 13. reference:** B1-ALT14 | **quantity:** 0 | **item:** Optional replacement12V14Ah sealed AGM battery | **manufacturer part:** Power-Sonic PS-12140 F2 | **unit usd:** 0 | **price basis:** Optional quote required not free | **availability checked 2026 09 11:** Datasheet checked2026-09-12;quote and stock needed | **url:** <https://www.power-sonic.com/wp-content/uploads/datasheets/ps-12140.pdf> | **notes:** Choose instead of B1 or ALT7;set ONE battery quantity1 after quote;F2 terminals;max bare100X153Y103Zmm;3.78kg current or4.2kg older;actual fit and loaded drive test required
- 14. reference:** CH1-SLA | **quantity:** 0 | **item:** Optional SLA charger for selected7Ah or14Ah battery | **manufacturer part:** Power-Sonic PSC-121000ACX | **unit usd:** 0 | **price basis:** Optional quote required not free | **availability checked 2026 09 11:** Manufacturer specification checked2026-09-12;quote and stock needed | **url:** <https://www.power-sonic.com/product/psc-121000acx/?preview=true> |

notes: 12V1A automatic SLA charger;minimum3Ah battery;set quantity1 only for SLA;remove battery to charge;never use on Bioenno

15. reference: ADAPTER-SLA | **quantity:** 0 | **item:** Optional insulated18AWG SLA blade-to-PP30 adapter | **manufacturer part:** Matched F2 female terminals plus PP30 pair and boots | **unit usd:** 12.00 | **price basis:** allowance optional | **availability checked 2026 09 11:** Terminal size verified;local components need quote | **url:**

<https://www.power-sonic.com/wp-content/uploads/datasheets/ps-12120-12121-2.pdf> | **notes:** Set quantity1 with SLA;two6.35x0.8mm F2 female contacts rated>=10A DC matched to18AWG;use4.75x0.8mm F1 contacts only for actualF1 pack;existingF1 fuse within100mm of battery positive including connector path;no extra fuse

16. reference: U7 | **quantity:** 1 | **item:** MAX98357A mono I2S amplifier | **manufacturer part:** Adafruit 3006 | **unit usd:** 5.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/3006> | **notes:** GAIN100k to5V gives3dB; fixed digital gain0.28

17. reference: SPK1 | **quantity:** 1 | **item:** 8ohm1W3inch speaker | **manufacturer part:** Adafruit 1313 | **unit usd:** 1.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/1313> | **notes:** 77.8x77.8x25.49mm; never ground either speaker lead

18. reference: S1-S2 | **quantity:** 2 | **item:** Panel SPST ON-OFF disconnect | **manufacturer part:** NKK S1A | **unit usd:** 12.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** https://www.nkkswitches.com/wp-content/themes/impress-blank/search/inc/part.php?part_no=S1A | **notes:** 20A30VDC;12.5mm hole; includes M12 panel hardware; OFF label required

19. reference: FH1-FH4 | **quantity:** 4 | **item:** Inline ATO fuse holder16AWG | **manufacturer part:** Littelfuse 0FHA0001ZXJ | **unit usd:** 4.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** <https://info.littelfuse.com/ato-fha-series-landing-page> | **notes:** 20A holder; insulate leads and restrain wiring

20. reference: F1 | **quantity:** 1 | **item:** ATO7.5A32V fuse | **manufacturer part:** Littelfuse 025707.5 | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** <https://www.littelfuse.com/products/fuses-overcurrent-protection/fuses/automotive-passenger-car/blade-fuses> | **notes:** Single main fuse;SLA positive terminal toF1<=100mm including adapter and PP30;Bioenno F1<=100mm from intact factory PP30

21. reference: F2-F3 | **quantity:** 2 | **item:** ATO3A32V fuse | **manufacturer part:** Littelfuse 0257003 | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** <https://www.littelfuse.com/products/fuses-overcurrent-protection/fuses/automotive-passenger-car/blade-fuses> | **notes:** Input branch fuses for motor and servo converters

22. reference: F4 | **quantity:** 1 | **item:** ATO1A32V fuse | **manufacturer part:** Littelfuse 0257001 | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** <https://www.littelfuse.com/products/fuses-overcurrent-protection/fuses/automotive-passenger-car/blade-fuses> | **notes:** Input fuse for logic converter

23. reference: PP30 | **quantity:** 1 | **item:** Anderson Powerpole mating red-black pair with contacts | **manufacturer part:** PP30 genuine housings and30A contacts | **unit usd:** 6.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** <https://www.andersonpower.com/us/en/resources/PowerpoleResourcesPage.html> | **notes:** Use contacts matched to18AWG; do not cut factory battery lead

24. reference: PROTO | **quantity:** 2 | **item:** Half-size solderable prototype board | **manufacturer part:** Adafruit 1609 | **unit usd:** 4.50 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/1609> | **notes:** One buffer board; one low-current divider board; power distribution uses separate heavy wire

25. reference: SOCKET | **quantity:** 2 | **item:** DIP14 IC socket | **manufacturer part:** 14pin0.3inch socket | **unit usd:** 0.75 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Protects buffers during soldering

26. reference: R1-R8 R12 R20-R24 R40-R45 | **quantity:** 20 | **item:** 10kohm1percent0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Input pull-downs;PCA OE pull-up;headPWM pull-down;two headgate pull-ups

27. reference: R10 R14 | **quantity:** 2 | **item:** 100kohm1percent0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Battery divider top and amplifier gain resistor

28. reference: R11 | **quantity:** 1 | **item:** 22kohm1percent0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Battery divider bottom; never replace with33k

29. reference: R13 | **quantity:** 1 | **item:** 15kohm1percent0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Actuator sense bottom resistor

30. reference: R30-R35 | **quantity:** 6 | **item:** 220ohm0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Four servo signal and two head bridge input series resistors

31. reference: R15-R16 | **quantity:** 2 | **item:** 100ohm1W resistor | **manufacturer part:** Metal film flameproof axial | **unit usd:** 0.40 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Actuator rail discharge; dissipates0.25W each during use

32. reference: C8-C11 MOTOR_CAPS | **quantity:** 9 | **item:** 100nF25V or higher ceramic capacitor | **manufacturer part:** X7R radial | **unit usd:** 0.15 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Five motor suppression; two buffer bypass; two ADC filters

33. reference: C1-C2 | **quantity:** 2 | **item:** 220uF25V lowESR capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 0.75 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Motor and servo converter input; polarity critical

34. reference: C3 | **quantity:** 1 | **item:** 100uF25V capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 0.50 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Logic converter input

35. reference: C4-C5 C7 C12 | **quantity:** 4 | **item:** 470uF10V lowESR capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 0.75 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** One at each of three drivers and one at logic output

36. reference: C6 | **quantity:** 1 | **item:** 1000uF10V lowESR capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Servo distribution block; polarity critical

37. reference: POWER_WIRE | **quantity:** 4 | **item:** 18AWG flexible stranded copper wire metre | **manufacturer part:** Silicone insulation red-black-orange | **unit usd:** 1.50 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Combined total 4m; trim to actual path plus service loop

38. reference: BRANCH_WIRE | **quantity:** 5 | **item:** 22AWG flexible stranded wire metre | **manufacturer part:** Red and black copper | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Individual servo and logic branches

39. reference: SIGNAL_WIRE | **quantity:** 5 | **item:** 26AWG stranded signal wire metre | **manufacturer part:** Assorted color copper | **unit usd:** 0.60 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Keep I2C and I2S runs short under 200mm where possible

40. reference: CONNECTORS | **quantity:** 1 | **item:** Connector and termination allowance | **manufacturer part:** Insulated terminal blocks and ferrules and servo header extensions | **unit usd:** 18.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Power blocks rated $\geq 10A30VDC$; four servo 3pin sockets; removable run-power link; five 2pin motor plugs

41. reference: INSULATION | **quantity:** 1 | **item:** Electrical assembly consumables | **manufacturer part:** Heatshrink cable ties labels solder and foam | **unit usd:** 12.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Insulation and strain relief are required

Mechanical hardware bill of materials

bom\hardware.csv

- 1. category:** fastener | **item:** M4x20 socket screw | **quantity:** 8 | **unit:** piece | **location:** Two body interfaces | **specification:** Four base-to-skirt and four skirt-to-upper; no separate neck joint
- 2. category:** fastener | **item:** M4 plain washer | **quantity:** 8 | **unit:** piece | **location:** Two body interfaces | **specification:** 8-9mm OD; one under each of eight body screwheads; no washers in any captive nut pocket
- 3. category:** fastener | **item:** M3 hex nut | **quantity:** 9 | **unit:** piece | **location:** Speaker display and jackscrew | **specification:** Four speaker plus four display plus one captured jackscrew nut; separate option rows cover motors and platform
- 4. category:** fastener | **item:** M3 nyloc nut | **quantity:** 4 | **unit:** piece | **location:** Head slide clamps | **specification:** Locks four guide screws after setting contact; no separate jackscrew jam nut
- 5. category:** fastener | **item:** M3 plain washer | **quantity:** 20 | **unit:** piece | **location:** Speaker display and slide underside | **specification:** Eight speaker plus eight display plus four head-slide underside; separate option rows cover motors and platform
- 6. category:** fastener | **item:** M3x12 socket screw | **quantity:** 4 | **unit:** piece | **location:** Speaker | **specification:** Open printed mounting frame plus speaker tab washers and nut; verify actual stack and flange
- 7. category:** fastener | **item:** M3x25 socket screw | **quantity:** 4 | **unit:** piece | **location:** Display retaining tabs | **specification:** Through integrated 16mm frame
- 8. category:** retainer | **item:** Nonconductive retaining tab 20x8x1mm | **quantity:** 4 | **unit:** piece | **location:** Display | **specification:** Polycarbonate or nylon strip; 3.4mm hole 3mm from outer end; 17mm inward reach; pad bare PCB edge
- 9. category:** fastener | **item:** M3x8 plastic thread-forming screw | **quantity:** 6 | **unit:** piece | **location:** Bearing retainers | **specification:** Three top and three bottom; 2.6mm pilots
- 10. category:** retainer | **item:** M3 steel washer 9mm OD 1mm thick | **quantity:** 6 | **unit:** piece | **location:** Bearing retainers | **specification:** At 14mm screw radius; overlap outer race only
- 11. category:** fastener | **item:** M8x70 hex-head bolt | **quantity:** 1 | **unit:** piece | **location:** Head spindle | **specification:** Must pass 8mm bearing bore freely
- 12. category:** fastener | **item:** M8 nyloc nut | **quantity:** 1 | **unit:** piece | **location:** Head spindle | **specification:** 13mm socket via 24mm crown hole
- 13. category:** fastener | **item:** M8 washer 16mm OD | **quantity:** 1 | **unit:** piece | **location:** Head hub top | **specification:** Only above printed hub
- 14. category:** bearing | **item:** 608-2RS bearing | **quantity:** 2 | **unit:** piece | **location:** Neck tower | **specification:** 22mm OD 8mm ID 7mm width
- 15. category:** spacer | **item:** Metal spacer 8.1x12x12mm | **quantity:** 1 | **unit:** piece | **location:** Between bearing inner races | **specification:** 8.1mm ID 12mm OD 12mm long; square parallel ends
- 16. category:** spacer | **item:** Precision shim 8x12x1mm | **quantity:** 2 | **unit:** piece | **location:** Bearing inner races | **specification:** One below and one above; narrow OD essential
- 17. category:** fastener | **item:** M3x16 socket screw | **quantity:** 4 | **unit:** piece | **location:** Head radial slide | **specification:** Through 3mm carriage and 4mm neck deck; clamp after adjustment
- 18. category:** fastener | **item:** M3 washer 6mm OD 1mm thick | **quantity:** 4 | **unit:** piece | **location:** Head slide screw heads | **specification:** Small OD clears motor-saddle walls; do not substitute wide washers
- 19. category:** fastener | **item:** M3x25 socket screw | **quantity:** 1 | **unit:** piece | **location:** Head contact jackscrew | **specification:** Through captured M3 nut; four slide clamps lock the chosen setting
- 20. category:** fastener | **item:** M2x10 machine screw | **quantity:** 8 | **unit:** piece | **location:** Four arm servo horns | **specification:** Two each plus supplied spline screws
- 21. category:** fastener | **item:** M2x20 machine screw | **quantity:** 6 | **unit:** piece | **location:** Two Pololu 4091 regulators | **specification:** Three per board; trim to measured deck/spacer/PCB stack; at least two full threads and at most 3mm projection below deck; never force M2.5 through M2 holes

- 22. category:** fastener | **item:** M2 hex nut maximum1.6mm thick | **quantity:** 14 | **unit:** piece | **location:** Arm horns and Pololu regulators | **specification:** Eight mandatory horn nuts plus six regulator nuts fitted on the bench
- 23. category:** fastener | **item:** M2 plain washer0.3mm thick | **quantity:** 28 | **unit:** piece | **location:** Arm horns and Pololu regulators | **specification:** 16 for mandatory horns plus12 for regulators; one per regulator screwhead and one below printed deck
- 24. category:** fastener | **item:** M2.5x20 machine screw | **quantity:** 32 | **unit:** piece | **location:** Electronics boards | **specification:** Trim to actual6mm deck/6mm spacer/PCB stack; at least two full threads and at most3mm projection below deck; unused pieces are spares
- 25. category:** spacer | **item:** Nylon spacer6mm long M2.5 bore | **quantity:** 38 | **unit:** piece | **location:** Electronics boards | **specification:** PCB undersides at base Z152;32 general board mounts plus six Pololu M2 screws using freely passing bores; unused pieces are spares; no printed spacers
- 26. category:** fastener | **item:** M2.5 thin hex nut maximum1.6mm thick | **quantity:** 32 | **unit:** piece | **location:** PCB nuts below printed deck | **specification:** DIN439 or ISO4035;5mm wrench size; fit on bench; complete hardware projection below deck at most3mm; unused pieces are spares
- 27. category:** fastener | **item:** M2.5 plain washer0.5mm thick | **quantity:** 32 | **unit:** piece | **location:** PCB screwheads | **specification:** One under each of32 general PCB screwheads; unused pieces are spares
- 28. category:** retainer | **item:** 2.5mm cable tie200mm long | **quantity:** 40 | **unit:** piece | **location:** Optional arm/board retention head motor and wiring | **specification:** Eight arm-servo ties replace ear fasteners only; two head-motor ties mandatory; up to20 board ties replace PCB screws; balance for routing/spares; fit actual latch and pads
- 29. category:** padding | **item:** EPDM foam sheet300x200mm | **quantity:** 1 | **unit:** set | **location:** Motors battery servos display | **specification:** Nonconductive1mm2mm3mm assortment; keep off hot components
- 30. category:** consumable | **item:** Removable thread-locking compound | **quantity:** 1 | **unit:** bottle | **location:** Metal-to-metal only | **specification:** Keep off plastic bearings and nyloc nuts
- 31. category:** consumable | **item:** PETG filament | **quantity:** 3 | **unit:** kg | **location:** Structural prints | **specification:** Current model and support estimate plus10percent allowance rounded up to1kg spools
- 32. category:** consumable | **item:** PLA filament | **quantity:** 3 | **unit:** kg | **location:** Cosmetic prints | **specification:** Supply allowance retained for model and supports plus a large-skirt retry contingency; not predicted consumption
- 33. category:** consumable | **item:** Paint and finish supplies | **quantity:** 1 | **unit:** set | **location:** Integrated details | **specification:** Paint bumps eye and lamps; no separate decoration meshes
- 34. category:** accessory | **item:** Opaque removable tape30x30mm | **quantity:** 1 | **unit:** piece | **location:** Optional crown cover | **specification:** Remove for spindle service; no printed cap
- 35. category:** accessory | **item:** Acoustic cloth100x100mm | **quantity:** 1 | **unit:** piece | **location:** Speaker grille guard | **specification:** Attach inside central front grille without touching cone
- 36. category:** accessory | **item:** Opaque black stretch fabric80x80mm | **quantity:** 2 | **unit:** piece | **location:** Hidden arm socket liners | **specification:** 44mm centre opening and overlapped radial slit; bond inside spherical-root rear rim and fixed socket; retain loose folds
- 37. category:** accessory | **item:** Flexible fabric adhesive | **quantity:** 1 | **unit:** tube | **location:** Socket liners and acoustic cloth | **specification:** Compatible with printed plastic and fabric; keep away from moving horns
- 38. category:** padding | **item:** Nonconductive1mm tab shim | **quantity:** 4 | **unit:** piece | **location:** Speaker | **specification:** Use only if surround clearance requires; cut from purchased sheet
- 39. category:** retainer | **item:** Hook-and-loop cable strap | **quantity:** 6 | **unit:** piece | **location:** Service harness | **specification:** Disconnectable loops at body interfaces
- 40. category:** tool | **item:** 250mm3mm ball-end hex driver rated20degrees | **quantity:** 1 | **unit:** piece | **location:** Base to one-piece skirt bolts | **specification:** 16degree inward approach through184mm top opening;250mm exposed shaft required
- 41. category:** tool | **item:** 2.5mm L-hex key short leg15mm | **quantity:** 1 | **unit:** piece | **location:** Head slide clamp screws | **specification:** Short leg may be cut and deburred; horizontal shaft passes neck-localZ49.9 through service ports
- 42. category:** tool | **item:** Metric hex keys and nut drivers | **quantity:** 1 | **unit:** set | **location:** Remaining fasteners | **specification:** 13mm socket outside diameter<=23mm for head access
- 43. category:** tool | **item:** Caliper and2.1mm2.8mm3.4mm drills | **quantity:** 1 | **unit:** set | **location:** Fit and hardware preparation | **specification:** Measure actual parts; deburr carefully

- 44. category:** tool | **item:** Digital scale and DC current meter | **quantity:** 1 | **unit:** set | **location:** Commissioning | **specification:** Measure finished mass and drive load
- 45. category:** fastener | **item:** M4 standard hex nut | **quantity:** 8 | **unit:** piece | **location:** Captive base and skirt joint nuts | **specification:** Four underside base pockets and four underside skirt-top pockets; seat before skirt installation; no washers inside pockets
- 46. category:** fastener | **item:** M3x16 socket screw | **quantity:** 4 | **unit:** piece | **location:** Optional printed drive-motor clamps | **specification:** One per hook clamp; omit all four clamps and these screw sets for motor ties
- 47. category:** fastener | **item:** M3 hex nut | **quantity:** 4 | **unit:** piece | **location:** Optional drive-motor clamp bosses | **specification:** One per motor; fit side pocket before motor; omit for motor ties
- 48. category:** fastener | **item:** M3 plain washer | **quantity:** 4 | **unit:** piece | **location:** Optional drive-motor clamp screws | **specification:** One under each of four screwheads; omit for motor ties
- 49. category:** retainer | **item:** 4.8mm cable tie250mm minimum | **quantity:** 8 | **unit:** piece | **location:** Alternative drive-motor retention | **specification:** Two per motor instead of clamp prints and screws; stated loop tensile rating at least222N; pad case and keep heads clear; modeled ribbon maximum1.2mm; actual latch fit required
- 50. category:** fastener | **item:** M3x12 socket screw | **quantity:** 8 | **unit:** piece | **location:** Optional board-platform feet | **specification:** Two per foot; side2.5mm L-key along Y; omit for foot ties
- 51. category:** fastener | **item:** M3 hex nut | **quantity:** 8 | **unit:** piece | **location:** Optional board-platform rail seats | **specification:** Two per foot in side-loaded pockets; omit for foot ties
- 52. category:** fastener | **item:** M3 plain washer | **quantity:** 8 | **unit:** piece | **location:** Optional board-platform foot screws | **specification:** One under each screwhead; omit for foot ties
- 53. category:** retainer | **item:** 4.8mm cable tie150mm minimum | **quantity:** 8 | **unit:** piece | **location:** Alternative platform-foot retention | **specification:** Two per foot through rail tunnels and top grooves; stated loop tensile rating at least222N; replace after cutting; modeled ribbon maximum1.2mm; actual latch fit required
- 54. category:** fastener | **item:** M2x10 socket-head screw | **quantity:** 8 | **unit:** piece | **location:** Optional four servo-ear mounts | **specification:** DIN912 or ISO4762;1.5mm hex socket;head at most3.8mm diameter and2mm high;two per servo;ear option only;mandatory horn fasteners remain
- 55. category:** fastener | **item:** M2 hex nut maximum1.6mm thick | **quantity:** 8 | **unit:** piece | **location:** Optional servo-ear mounts | **specification:** One per ear screw; actual ear holes and shaft position must fit slots; omit for servo ties
- 56. category:** fastener | **item:** M2 plain washer0.3mm maximum | **quantity:** 16 | **unit:** piece | **location:** Optional servo-ear mounts | **specification:** Maximum5mm OD; one below each head and nut; omit for servo ties
- 57. category:** retainer | **item:** KSS CV-500W cable tie500x4.8mm | **quantity:** 2 | **unit:** piece | **location:** Battery restraint | **specification:** Two padded battery loops;222N stated minimum; modeled ribbon maximum1.2mm; measure actual ribbon/latch fit; https://www.kss.com.tw/index_en.php?action=products4&cid=17&fid=3&id=8461&sid=14
- 58. category:** padding | **item:** Firm nonconductive foam blocks | **quantity:** 1 | **unit:** set | **location:** Battery cradle | **specification:** Cut to fill side/end gaps around Bioenno or7Ah pack within104x157mm cradle; keep valves terminals and leads clear
- 59. category:** tool | **item:** 2.5mm L-hex key18mm exposed short leg | **quantity:** 1 | **unit:** piece | **location:** Platform-foot screw access | **specification:** Approach along Y from outside base with wheels fitted; at least79mm side reach; verify actual tool envelope
- 60. category:** fastener | **item:** M2.5 plain washer0.3mm thick | **quantity:** 32 | **unit:** piece | **location:** PCB nuts below printed deck | **specification:** One underside washer per general board screw; complete washer/nut/tip projection at most3mm; unused pieces are spares
- 61. category:** tool | **item:** Wiha01121 1.5mm L-hex key | **quantity:** 1 | **unit:** piece | **location:** Optional yaw-servo ear screws | **specification:** 46mm long leg15mm short leg;rear approach and60degree reindex arcs; https://www2.wiha.com/files/asia-pacific/EN-Asia-Pacific_catalogue.pdf
- 62. category:** tool | **item:** Fine gripping tool for M2 nuts | **quantity:** 1 | **unit:** piece | **location:** Optional yaw-servo ear nut windows | **specification:** Gripped4mm-AF nut and tip must fit6mm width and2.4mm thickness for first24mm reach; measure real jaws and handles

Printed part quantities and orientations

bom\printed-parts.csv

1. part: 01_base | **quantity:** 1 | **material:** PETG | **x mm:** 300.0 | **y mm:** 300.0 | **z mm:** 57.0 | **volume mm3:** 721610.71 | **solid mass g:** 916.45 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 8 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Floor down; single nozzle | **notes:** Diameter300 circular unibody;6mm floor;4mm wall/ribs;6walls6top/bottom40%gyroid;8mmbrim;normal auto supports for captured pockets and ledges;clear through wheel wells

2. part: 02_skirt | **quantity:** 1 | **material:** PLA | **x mm:** 300.0 | **y mm:** 300.0 | **z mm:** 213.0 | **volume mm3:** 523169.79 | **solid mass g:** 648.73 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Large opening down | **notes:** Twelve full-height planar faces;linear300-to220mm vertex-diameter taper over210mm plus3mm register;four hemisphere rows;faceted240mm-clear rib;captured M4 nuts;4walls15%infill;normal auto supports

3. part: 04_shoulder | **quantity:** 1 | **material:** PETG | **x mm:** 226.0 | **y mm:** 228.2 | **z mm:** 186.0 | **volume mm3:** 629663.67 | **solid mass g:** 799.67 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Bottom flange down | **notes:** Shoulder and neck one print; concealed compact servo boxes; integrated bearing tower and fixed friction deck; assemble through open bottom before seating;4walls25%infill;normal auto supports

4. part: 06_head | **quantity:** 1 | **material:** PLA | **x mm:** 220.0 | **y mm:** 285.667 | **z mm:** 114.025 | **volume mm3:** 254128.48 | **solid mass g:** 315.12 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Drum rim down | **notes:** Dome eye discs tilted lamps hub friction drum spokes integrated; normal auto supports inside dome under eye and hub;4walls15%infill

5. part: 07_pitch_carrier | **quantity:** 2 | **material:** PETG | **x mm:** 49.5 | **y mm:** 30.0 | **z mm:** 49.0 | **volume mm3:** 13386.43 | **solid mass g:** 17.0 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Floor down | **notes:** Print this same file twice;4walls30%infill; actual horn and servo fit gate

6. part: 08_plunger_arm | **quantity:** 1 | **material:** PLA | **x mm:** 138.0 | **y mm:** 50.447 | **z mm:** 50.593 | **volume mm3:** 25738.27 | **solid mass g:** 31.92 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** As exported; ensure on bed | **notes:** Spherical root and telescoping rings integrated; support cup and root; supplied servo horn; decorative arm only

7. part: 09_emitter_arm | **quantity:** 1 | **material:** PLA | **x mm:** 139.0 | **y mm:** 50.447 | **z mm:** 50.593 | **volume mm3:** 24124.18 | **solid mass g:** 29.91 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** As exported; ensure on bed | **notes:** Spherical root eight rods muzzle rings integrated; normal auto supports; supplied servo horn; decorative arm only

8. part: 10_head_motor_carriage | **quantity:** 1 | **material:** PETG | **x mm:** 51.0 | **y mm:** 80.0 | **z mm:** 21.0 | **volume mm3:** 21985.9 | **solid mass g:** 27.92 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Floor down | **notes:** Radial slots padded TT saddle and cable-tie slots;4walls30%infill

9. part: 11_motor_clamp | **quantity:** 4 | **material:** PETG | **x mm:** 39.5 | **y mm:** 14.0 | **z mm:** 5.0 | **volume mm3:** 2131.4 | **solid mass g:** 2.71 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** As exported | **notes:** Four identical hook clamps;one M3 screw each;omit all four clamps for alternative two-ties-per-motor retention;4walls30%infill

10. part: 12_electronics_platform | **quantity:** 1 | **material:** PETG | **x mm:** 216.0 | **y mm:** 216.0 | **z mm:** 112.0 | **volume mm3:** 305710.02 | **solid mass g:** 388.25 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Deck down; legs up | **notes:** One216mm platform with integral legs above battery;invert for assembly;6mm deck;4walls30%infill;four feet retained by screws or ties

Point-to-point wiring schedule

electronics\wiring.csv

1. **net:** BAT_POS | **origin:** B1.PP30+ | **destination:** F1.IN | **wire awg:** 18 | **color:** red | **note:** Select ONE battery on sheet05; single F1 7.5A within 100mm of SLA positive terminal including adapter or Bioenno factory connector; never connect packs together
2. **net:** BAT_FUSED | **origin:** F1.OUT | **destination:** S1.1 | **wire awg:** 18 | **color:** red
3. **net:** PACK_SW | **origin:** S1.3 | **destination:** S2.1 | **wire awg:** 18 | **color:** red | **note:** NKK S1A ON connects terminals1 and3
4. **net:** PACK_SW | **origin:** S1.3 | **destination:** F4.IN | **wire awg:** 18 | **color:** red | **note:** NKK S1A ON connects terminals1 and3
5. **net:** PACK_SW | **origin:** S1.3 | **destination:** R10.1 | **wire awg:** 18 | **color:** red | **note:** NKK S1A ON connects terminals1 and3
6. **net:** ACT_IN | **origin:** S2.3 | **destination:** F2.IN | **wire awg:** 18 | **color:** orange | **note:** Physical actuator power disconnect; logic stays powered
7. **net:** ACT_IN | **origin:** S2.3 | **destination:** F3.IN | **wire awg:** 18 | **color:** orange | **note:** Physical actuator power disconnect; logic stays powered
8. **net:** MOTOR_IN | **origin:** F2.OUT | **destination:** U2.VIN | **wire awg:** 18 | **color:** orange | **note:** F2 3A
9. **net:** SERVO_IN | **origin:** F3.OUT | **destination:** U3.VIN | **wire awg:** 18 | **color:** orange | **note:** F3 3A
10. **net:** LOGIC_IN | **origin:** F4.OUT | **destination:** U4.IN | **wire awg:** 22 | **color:** red | **note:** F4 1A
11. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** U5.VMotor | **wire awg:** 18 | **color:** orange
12. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** U6.VMotor | **wire awg:** 18 | **color:** orange
13. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** U11.VMotor | **wire awg:** 18 | **color:** orange
14. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** U9.14 | **wire awg:** 18 | **color:** orange
15. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** R12.1 | **wire awg:** 18 | **color:** orange
16. **net:** 5V_SERVO | **origin:** U3.VOUT | **destination:** J_SERVO.2 | **wire awg:** 18 | **color:** red | **note:** Four separate arm-servo branches; never feed through PCA9685 V+ traces
17. **net:** 5V_SERVO | **origin:** U3.VOUT | **destination:** U8.14 | **wire awg:** 18 | **color:** red | **note:** Four separate arm-servo branches; never feed through PCA9685 V+ traces
18. **net:** 5V_LOGIC | **origin:** U4.OUT | **destination:** J_USB_ISO.1 | **wire awg:** 22 | **color:** red
19. **net:** 5V_LOGIC | **origin:** U4.OUT | **destination:** U7.VIN | **wire awg:** 22 | **color:** red
20. **net:** 5V_LOGIC | **origin:** U4.OUT | **destination:** R14.1 | **wire awg:** 22 | **color:** red
21. **net:** ESP_5V | **origin:** J_USB_ISO.2 | **destination:** U1.5V | **wire awg:** 22 | **color:** red | **note:** Removable run-power link; OPEN and unplug battery before USB
22. **net:** 3V3 | **origin:** U1.3V3 | **destination:** U10.VCC | **wire awg:** 26 | **color:** red
23. **net:** 3V3 | **origin:** U1.3V3 | **destination:** R6.1 | **wire awg:** 26 | **color:** red
24. **net:** GND | **origin:** B1.PP30- | **destination:** U2.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board
25. **net:** GND | **origin:** B1.PP30- | **destination:** U3.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board
26. **net:** GND | **origin:** B1.PP30- | **destination:** U4.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board
27. **net:** GND | **origin:** B1.PP30- | **destination:** U5.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board

28. **net:** GND | **origin:** B1.PP30- | **destination:** U6.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board

29. **net:** GND | **origin:** B1.PP30- | **destination:** U11.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board

30. **net:** GND | **origin:** B1.PP30- | **destination:** J_SERVO.1 | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board

31. **net:** GND | **origin:** GROUND_STAR | **destination:** U1.GND | **wire awg:** 22 | **color:** black

32. **net:** GND | **origin:** GROUND_STAR | **destination:** U7.GND | **wire awg:** 22 | **color:** black

33. **net:** GND | **origin:** GROUND_STAR | **destination:** U8.7 | **wire awg:** 22 | **color:** black

34. **net:** GND | **origin:** GROUND_STAR | **destination:** U9.7 | **wire awg:** 22 | **color:** black

35. **net:** GND | **origin:** GROUND_STAR | **destination:** U10.GND | **wire awg:** 22 | **color:** black

36. **net:** GND | **origin:** B1.PP30- | **destination:** GROUND_STAR | **wire awg:** 18 | **color:** black

37. **net:** LEFT_IN1 | **origin:** U1.GPIO25 | **destination:** U5.AIN1 | **wire awg:** 26 | **color:** signal

38. **net:** LEFT_IN1 | **origin:** U1.GPIO25 | **destination:** U5.BIN1 | **wire awg:** 26 | **color:** signal

39. **net:** LEFT_IN2 | **origin:** U1.GPIO26 | **destination:** U5.AIN2 | **wire awg:** 26 | **color:** signal

40. **net:** LEFT_IN2 | **origin:** U1.GPIO26 | **destination:** U5.BIN2 | **wire awg:** 26 | **color:** signal

41. **net:** RIGHT_IN1 | **origin:** U1.GPIO32 | **destination:** U6.AIN1 | **wire awg:** 26 | **color:** signal

42. **net:** RIGHT_IN1 | **origin:** U1.GPIO32 | **destination:** U6.BIN1 | **wire awg:** 26 | **color:** signal

43. **net:** RIGHT_IN2 | **origin:** U1.GPIO33 | **destination:** U6.AIN2 | **wire awg:** 26 | **color:** signal

44. **net:** RIGHT_IN2 | **origin:** U1.GPIO33 | **destination:** U6.BIN2 | **wire awg:** 26 | **color:** signal

45. **net:** LEFT_IN1 | **origin:** LEFT_IN1 | **destination:** R1.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

46. **net:** GND | **origin:** GND | **destination:** R1.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

47. **net:** LEFT_IN2 | **origin:** LEFT_IN2 | **destination:** R2.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

48. **net:** GND | **origin:** GND | **destination:** R2.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

49. **net:** RIGHT_IN1 | **origin:** RIGHT_IN1 | **destination:** R3.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

50. **net:** GND | **origin:** GND | **destination:** R3.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

51. **net:** RIGHT_IN2 | **origin:** RIGHT_IN2 | **destination:** R4.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

52. **net:** GND | **origin:** GND | **destination:** R4.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

53. **net:** MOTOR_ENABLE | **origin:** MOTOR_ENABLE | **destination:** R5.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

54. **net:** GND | **origin:** GND | **destination:** R5.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

55. **net:** MOTOR_ENABLE | **origin:** U1.GPIO12 | **destination:** U5.SLP | **wire awg:** 26 | **color:** signal

56. **net:** MOTOR_ENABLE | **origin:** U1.GPIO12 | **destination:** U6.SLP | **wire awg:** 26 | **color:** signal

57. **net:** MOTOR_ENABLE | **origin:** U1.GPIO12 | **destination:** U11.SLP | **wire awg:** 26 | **color:** signal

58. **net:** SDA | **origin:** U1.GPIO21 | **destination:** U10.SDA | **wire awg:** 26 | **color:** signal

59. **net:** SCL | **origin:** U1.GPIO22 | **destination:** U10.SCL | **wire awg:** 26 | **color:** signal

60. **net:** SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U10.OE | **wire awg:** 26 | **color:** signal | **note:** R6 10k to 3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

61. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** R6.2 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

62. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.1 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

63. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.4 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

64. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.10 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

65. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.13 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

66. net: I2S_BCLK | **origin:** U1.GPIO17 | **destination:** U7.BCLK | **wire awg:** 26 | **color:** signal

67. net: I2S_LRC | **origin:** U1.GPIO13 | **destination:** U7.LRC | **wire awg:** 26 | **color:** signal

68. net: I2S_DATA | **origin:** U1.GPIO15 | **destination:** U7.DIN | **wire awg:** 26 | **color:** signal | **note:** GPIO15 strap: amplifier INPUT only; no pull-up/down added

69. net: BAT_ADC | **origin:** R10.2 | **destination:** R11.1 | **wire awg:** 26 | **color:** signal

70. net: BAT_ADC | **origin:** R10.2 | **destination:** U1.GPIO39 | **wire awg:** 26 | **color:** signal

71. net: BAT_ADC | **origin:** R10.2 | **destination:** C10.1 | **wire awg:** 26 | **color:** signal

72. net: GND | **origin:** GROUND_STAR | **destination:** R11.2 | **wire awg:** 26 | **color:** signal

73. net: GND | **origin:** GROUND_STAR | **destination:** C10.2 | **wire awg:** 26 | **color:** signal

74. net: ACT_OK | **origin:** R12.2 | **destination:** R13.1 | **wire awg:** 26 | **color:** signal

75. net: ACT_OK | **origin:** R12.2 | **destination:** U1.GPIO36 | **wire awg:** 26 | **color:** signal

76. net: ACT_OK | **origin:** R12.2 | **destination:** C11.1 | **wire awg:** 26 | **color:** signal

77. net: GND | **origin:** GROUND_STAR | **destination:** R13.2 | **wire awg:** 26 | **color:** signal

78. net: GND | **origin:** GROUND_STAR | **destination:** C11.2 | **wire awg:** 26 | **color:** signal

79. net: AMP_GAIN | **origin:** R14.2 | **destination:** U7.GAIN | **wire awg:** 26 | **color:** signal | **note:** R14 100k to5V_LOGIC sets3dB

80. net: SPK_P | **origin:** U7.OUT+ | **destination:** SPK1.+ | **wire awg:** 24 | **color:** yellow | **note:** Floating bridge output; NEVER ground speaker leads

81. net: SPK_N | **origin:** U7.OUT- | **destination:** SPK1.- | **wire awg:** 24 | **color:** blue

82. net: LEFT_FRONT_+ | **origin:** U5.AOUT1 | **destination:** M_LEFT_FRONT.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

83. net: LEFT_FRONT_- | **origin:** U5.AOUT2 | **destination:** M_LEFT_FRONT.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

84. net: LEFT_FRONT_+ | **origin:** LEFT_FRONT_+ | **destination:** C_LEFT_FRONT.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

85. net: LEFT_FRONT_- | **origin:** LEFT_FRONT_- | **destination:** C_LEFT_FRONT.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

86. net: LEFT_REAR_+ | **origin:** U5.BOUT1 | **destination:** M_LEFT_REAR.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

87. net: LEFT_REAR_- | **origin:** U5.BOUT2 | **destination:** M_LEFT_REAR.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

88. net: LEFT_REAR_+ | **origin:** LEFT_REAR_+ | **destination:** C_LEFT_REAR.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

89. net: LEFT_REAR_- | **origin:** LEFT_REAR_- | **destination:** C_LEFT_REAR.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

90. net: RIGHT_FRONT_+ | **origin:** U6.AOUT1 | **destination:** M_RIGHT_FRONT.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

91. net: RIGHT_FRONT_- | **origin:** U6.AOUT2 | **destination:** M_RIGHT_FRONT.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

92. net: RIGHT_FRONT_+ | **origin:** RIGHT_FRONT_+ | **destination:** C_RIGHT_FRONT.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

93. net: RIGHT_FRONT_- | **origin:** RIGHT_FRONT_- | **destination:** C_RIGHT_FRONT.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

94. net: RIGHT_REAR_+ | **origin:** U6.BOUT1 | **destination:** M_RIGHT_REAR.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

95. net: RIGHT_REAR_- | **origin:** U6.BOUT2 | **destination:** M_RIGHT_REAR.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

96. net: RIGHT_REAR_+ | **origin:** RIGHT_REAR_+ | **destination:** C_RIGHT_REAR.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

97. net: RIGHT_REAR_- | **origin:** RIGHT_REAR_- | **destination:** C_RIGHT_REAR.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

98. net: PWM0 | **origin:** U10.PWM0 | **destination:** U8.2 | **wire awg:** 26 | **color:** signal

99. net: PWM0 | **origin:** PWM0 | **destination:** R20.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

100. net: GND | **origin:** GND | **destination:** R20.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

101. net: PWM5V_0 | **origin:** U8.3 | **destination:** R30.1 | **wire awg:** 26 | **color:** signal

102. net: SERVO_LEFT_YAW | **origin:** R30.2 | **destination:** SV0.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

103. net: SERVO_LEFT_YAW | **origin:** SERVO_LEFT_YAW | **destination:** R40.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

104. net: GND | **origin:** GND | **destination:** R40.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

105. net: 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV0.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

106. net: GND | **origin:** J_SERVO.1 | **destination:** SV0.GND | **wire awg:** 22 | **color:** black/brown

107. net: PWM1 | **origin:** U10.PWM1 | **destination:** U8.5 | **wire awg:** 26 | **color:** signal

108. net: PWM1 | **origin:** PWM1 | **destination:** R21.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

109. net: GND | **origin:** GND | **destination:** R21.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

110. net: PWM5V_1 | **origin:** U8.6 | **destination:** R31.1 | **wire awg:** 26 | **color:** signal

111. net: SERVO_LEFT_PITCH | **origin:** R31.2 | **destination:** SV1.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

112. net: SERVO_LEFT_PITCH | **origin:** SERVO_LEFT_PITCH | **destination:** R41.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

113. net: GND | **origin:** GND | **destination:** R41.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

114. net: 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV1.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

115. net: GND | **origin:** J_SERVO.1 | **destination:** SV1.GND | **wire awg:** 22 | **color:** black/brown

116. net: PWM2 | **origin:** U10.PWM2 | **destination:** U8.9 | **wire awg:** 26 | **color:** signal

117. **net:** PWM2 | **origin:** PWM2 | **destination:** R22.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

118. **net:** GND | **origin:** GND | **destination:** R22.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

119. **net:** PWM5V_2 | **origin:** U8.8 | **destination:** R32.1 | **wire awg:** 26 | **color:** signal

120. **net:** SERVO_RIGHT_YAW | **origin:** R32.2 | **destination:** SV2.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

121. **net:** SERVO_RIGHT_YAW | **origin:** SERVO_RIGHT_YAW | **destination:** R42.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

122. **net:** GND | **origin:** GND | **destination:** R42.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

123. **net:** 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV2.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

124. **net:** GND | **origin:** J_SERVO.1 | **destination:** SV2.GND | **wire awg:** 22 | **color:** black/brown

125. **net:** PWM3 | **origin:** U10.PWM3 | **destination:** U8.12 | **wire awg:** 26 | **color:** signal

126. **net:** PWM3 | **origin:** PWM3 | **destination:** R23.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

127. **net:** GND | **origin:** GND | **destination:** R23.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

128. **net:** PWM5V_3 | **origin:** U8.11 | **destination:** R33.1 | **wire awg:** 26 | **color:** signal

129. **net:** SERVO_RIGHT_PITCH | **origin:** R33.2 | **destination:** SV3.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

130. **net:** SERVO_RIGHT_PITCH | **origin:** SERVO_RIGHT_PITCH | **destination:** R43.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

131. **net:** GND | **origin:** GND | **destination:** R43.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

132. **net:** 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV3.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

133. **net:** GND | **origin:** J_SERVO.1 | **destination:** SV3.GND | **wire awg:** 22 | **color:** black/brown

134. **net:** HEAD_PWM | **origin:** U1.GPIO2 | **destination:** U9.2 | **wire awg:** 26 | **color:** signal | **note:** 20kHz 8bit PWM; low before direction updates; cap100/255

135. **net:** HEAD_PWM | **origin:** U1.GPIO2 | **destination:** U9.5 | **wire awg:** 26 | **color:** signal | **note:** 20kHz 8bit PWM; low before direction updates; cap100/255

136. **net:** HEAD_PWM | **origin:** HEAD_PWM | **destination:** R24.1 | **wire awg:** component | **note:** 10k pull-down: GPIO2 boot-safe and head PWM off on reset

137. **net:** GND | **origin:** GND | **destination:** R24.2 | **wire awg:** component | **note:** 10k pull-down: GPIO2 boot-safe and head PWM off on reset

138. **net:** HEAD_DISABLE1 | **origin:** U10.PWM4 | **destination:** U9.1 | **wire awg:** 26 | **color:** signal | **note:** STATIC HIGH disables this gate; LOW enables; never servo pulses

139. **net:** HEAD_DISABLE1 | **origin:** HEAD_DISABLE1 | **destination:** R7.1 | **wire awg:** component | **note:** 10k pull-up to3V3; AHCT TTL threshold

140. **net:** 3V3 | **origin:** 3V3 | **destination:** R7.2 | **wire awg:** component | **note:** 10k pull-up to3V3; AHCT TTL threshold

141. **net:** HEAD_BUFFER1 | **origin:** U9.3 | **destination:** R34.1 | **wire awg:** 26 | **color:** signal

142. **net:** HEAD_IN1 | **origin:** R34.2 | **destination:** U11.AIN1 | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor; only one input gets PWM

143. **net:** HEAD_IN1 | **origin:** HEAD_IN1 | **destination:** R44.1 | **wire awg:** component | **note:** 10k pull-down at driver input

144. **net:** GND | **origin:** GND | **destination:** R44.2 | **wire awg:** component | **note:** 10k pull-down at driver input

145. net: HEAD_DISABLE2 | **origin:** U10.PWM5 | **destination:** U9.4 | **wire awg:** 26 | **color:** signal | **note:** STATIC HIGH disables this gate; LOW enables; never servo pulses

146. net: HEAD_DISABLE2 | **origin:** HEAD_DISABLE2 | **destination:** R8.1 | **wire awg:** component | **note:** 10k pull-up to3V3; AHCT TTL threshold

147. net: 3V3 | **origin:** 3V3 | **destination:** R8.2 | **wire awg:** component | **note:** 10k pull-up to3V3; AHCT TTL threshold

148. net: HEAD_BUFFER2 | **origin:** U9.6 | **destination:** R35.1 | **wire awg:** 26 | **color:** signal

149. net: HEAD_IN2 | **origin:** R35.2 | **destination:** U11.AIN2 | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor; only one input gets PWM

150. net: HEAD_IN2 | **origin:** HEAD_IN2 | **destination:** R45.1 | **wire awg:** component | **note:** 10k pull-down at driver input

151. net: GND | **origin:** GND | **destination:** R45.2 | **wire awg:** component | **note:** 10k pull-down at driver input

152. net: GND | **origin:** GROUND_STAR | **destination:** U11.BIN1 | **wire awg:** 26 | **color:** signal | **note:** Unused bridge B inputs low; BOUT1/BOUT2 unconnected

153. net: GND | **origin:** GROUND_STAR | **destination:** U11.BIN2 | **wire awg:** 26 | **color:** signal | **note:** Unused bridge B inputs low; BOUT1/BOUT2 unconnected

154. net: HEAD_MOTOR+ | **origin:** U11.AOUT1 | **destination:** M_HEAD.+ | **wire awg:** 22 | **color:** yellow

155. net: HEAD_MOTOR- | **origin:** U11.AOUT2 | **destination:** M_HEAD.- | **wire awg:** 22 | **color:** blue

156. net: HEAD_MOTOR+ | **origin:** HEAD_MOTOR+ | **destination:** C_HEAD.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

157. net: HEAD_MOTOR- | **origin:** HEAD_MOTOR- | **destination:** C_HEAD.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

158. net: GND | **origin:** GROUND_STAR | **destination:** U9.9 | **wire awg:** 26 | **color:** black | **note:** Unused buffer INPUT tied low

159. net: GND | **origin:** GROUND_STAR | **destination:** U9.12 | **wire awg:** 26 | **color:** black | **note:** Unused buffer INPUT tied low

160. net: 5V_MOTOR | **origin:** U2.VOUT | **destination:** U9.10 | **wire awg:** 26 | **color:** orange | **note:** Unused buffer OE high

161. net: 5V_MOTOR | **origin:** U2.VOUT | **destination:** U9.13 | **wire awg:** 26 | **color:** orange | **note:** Unused buffer OE high

162. net: MOTOR_IN | **origin:** MOTOR_IN | **destination:** C1.1 | **wire awg:** component | **note:** 220uF 25V low-ESR

163. net: GND | **origin:** GND | **destination:** C1.2 | **wire awg:** component | **note:** 220uF 25V low-ESR

164. net: SERVO_IN | **origin:** SERVO_IN | **destination:** C2.1 | **wire awg:** component | **note:** 220uF 25V low-ESR

165. net: GND | **origin:** GND | **destination:** C2.2 | **wire awg:** component | **note:** 220uF 25V low-ESR

166. net: LOGIC_IN | **origin:** LOGIC_IN | **destination:** C3.1 | **wire awg:** component | **note:** 100uF 25V

167. net: GND | **origin:** GND | **destination:** C3.2 | **wire awg:** component | **note:** 100uF 25V

168. net: 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** C4.1 | **wire awg:** component | **note:** 470uF 10V low-ESR

169. net: GND | **origin:** GND | **destination:** C4.2 | **wire awg:** component | **note:** 470uF 10V low-ESR

170. net: 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** C5.1 | **wire awg:** component | **note:** 470uF 10V low-ESR

171. net: GND | **origin:** GND | **destination:** C5.2 | **wire awg:** component | **note:** 470uF 10V low-ESR

172. net: 5V_SERVO | **origin:** 5V_SERVO | **destination:** C6.1 | **wire awg:** component | **note:** 1000uF 10V low-ESR

173. net: GND | **origin:** GND | **destination:** C6.2 | **wire awg:** component | **note:** 1000uF 10V low-ESR

174. net: 5V_LOGIC | **origin:** 5V_LOGIC | **destination:** C7.1 | **wire awg:** component | **note:** 470uF 10V low-ESR

175. net: GND | **origin:** GND | **destination:** C7.2 | **wire awg:** component | **note:** 470uF 10V low-ESR

176. net: 5V_SERVO | **origin:** 5V_SERVO | **destination:** C8.1 | **wire awg:** component | **note:** 100nF ceramic

177. **net:** GND | **origin:** GND | **destination:** C8.2 | **wire awg:** component | **note:** 100nF ceramic

178. **net:** 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** C9.1 | **wire awg:** component | **note:** 100nF ceramic

179. **net:** GND | **origin:** GND | **destination:** C9.2 | **wire awg:** component | **note:** 100nF ceramic

180. **net:** 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** C12.1 | **wire awg:** component | **note:** 470uF 10V low-ESR at head driver

181. **net:** GND | **origin:** GND | **destination:** C12.2 | **wire awg:** component | **note:** 470uF 10V low-ESR at head driver

182. **net:** 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** R15.1 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

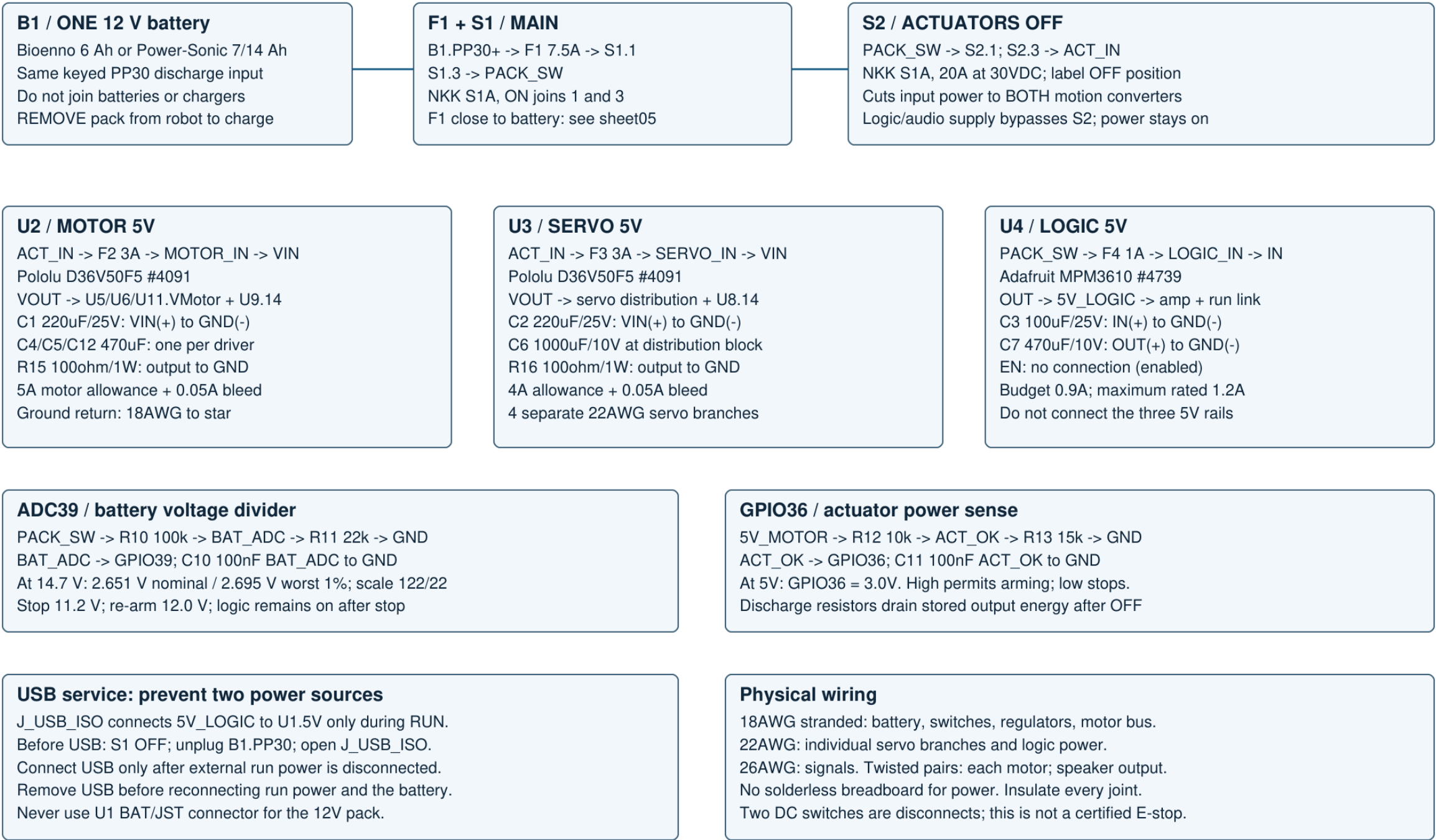
183. **net:** GND | **origin:** GND | **destination:** R15.2 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

184. **net:** 5V_SERVO | **origin:** 5V_SERVO | **destination:** R16.1 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

185. **net:** GND | **origin:** GND | **destination:** R16.2 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

01 / Power distribution and isolation

B1 = ONE battery: Bioenno 6 Ah OR SLA 7 Ah OR SLA 14 Ah. Adapter and charger choices: sheet 05.



02 / Controller, drive motors and sound

U1 = original LILYGO TTGO T-Display ESP32, 1.14 inch ST7789. GPIO numbers, not header positions.

U1 / exact external pin assignment

GPIO21	_____	SDA -> U10.SDA
GPIO22	_____	SCL -> U10.SCL
GPIO27	_____	SERVO_DISABLE / OE
GPIO12	_____	MOTOR_ENABLE / SLP
GPIO25	_____	LEFT_IN1
GPIO26	_____	LEFT_IN2
GPIO32	_____	RIGHT_IN1
GPIO33	_____	RIGHT_IN2
GPIO17	_____	I2S_BCLK
GPIO13	_____	I2S_LRC
GPIO15	_____	I2S_DATA
GPIO39 (SVN)	_____	BAT_ADC
GPIO36 (SVP)	_____	ACT_OK
GPIO2	_____	HEAD_PWM -> U9.2 + .5

Reserved board connections

TFT: MOSI19, SCLK18, CS5, DC16, RST23, BL4
Buttons: GPIO0 and GPIO35; reserve both.
Onboard battery ADC34 and power14 are NOT pack sense.
GPIO12 and GPIO2 each have a 10k boot-safe pull-down.
GPIO15 connects only to amplifier DIN; no external bias.
GPIO37, 38 and UART pins: no external connection.

Hardware stopped on reset

R1..R4: 10k from each wheel input net to GND.
R5: 10k MOTOR_ENABLE to GND; ALL THREE SLP pins join.
R6: 10k SERVO_DISABLE to3V3; R24: HEAD_PWM to GND.
Firmware: SLP low, PWM low, OE high before devices init.
U1.3V3 -> U10.VCC; all GND pins join common ground.
Physical power OFF removes wheel, head and arm power.

U5 / Adafruit DRV8833 #3297

LEFT_IN1 -> AIN1 AND BIN1 (INPUTS ONLY)
LEFT_IN2 -> AIN2 AND BIN2 (INPUTS ONLY)
SLP <- MOTOR_ENABLE; VMotor <- 5V_MOTOR; GND <- star
AOUT1/AOUT2 -> M_LEFT_FRONT +/-
BOUT1/BOUT2 -> M_LEFT_REAR +/-
Keep BOTH default0.2ohm sense resistors:1A per bridge.
DO NOT parallel the output bridges or two motors on one.
FLT, VM, ASEN, BSEN: no external connection.

U6 / Adafruit DRV8833 #3297

RIGHT_IN1 -> AIN1 AND BIN1 (INPUTS ONLY)
RIGHT_IN2 -> AIN2 AND BIN2 (INPUTS ONLY)
SLP <- MOTOR_ENABLE; VMotor <- 5V_MOTOR; GND <- star
AOUT1/AOUT2 -> M_RIGHT_FRONT +/-
BOUT1/BOUT2 -> M_RIGHT_REAR +/-
Keep BOTH default0.2ohm sense resistors:1A per bridge.
DO NOT parallel the output bridges or two motors on one.
FLT, VM, ASEN, BSEN: no external connection.

U7 / MAX98357A + SPK1

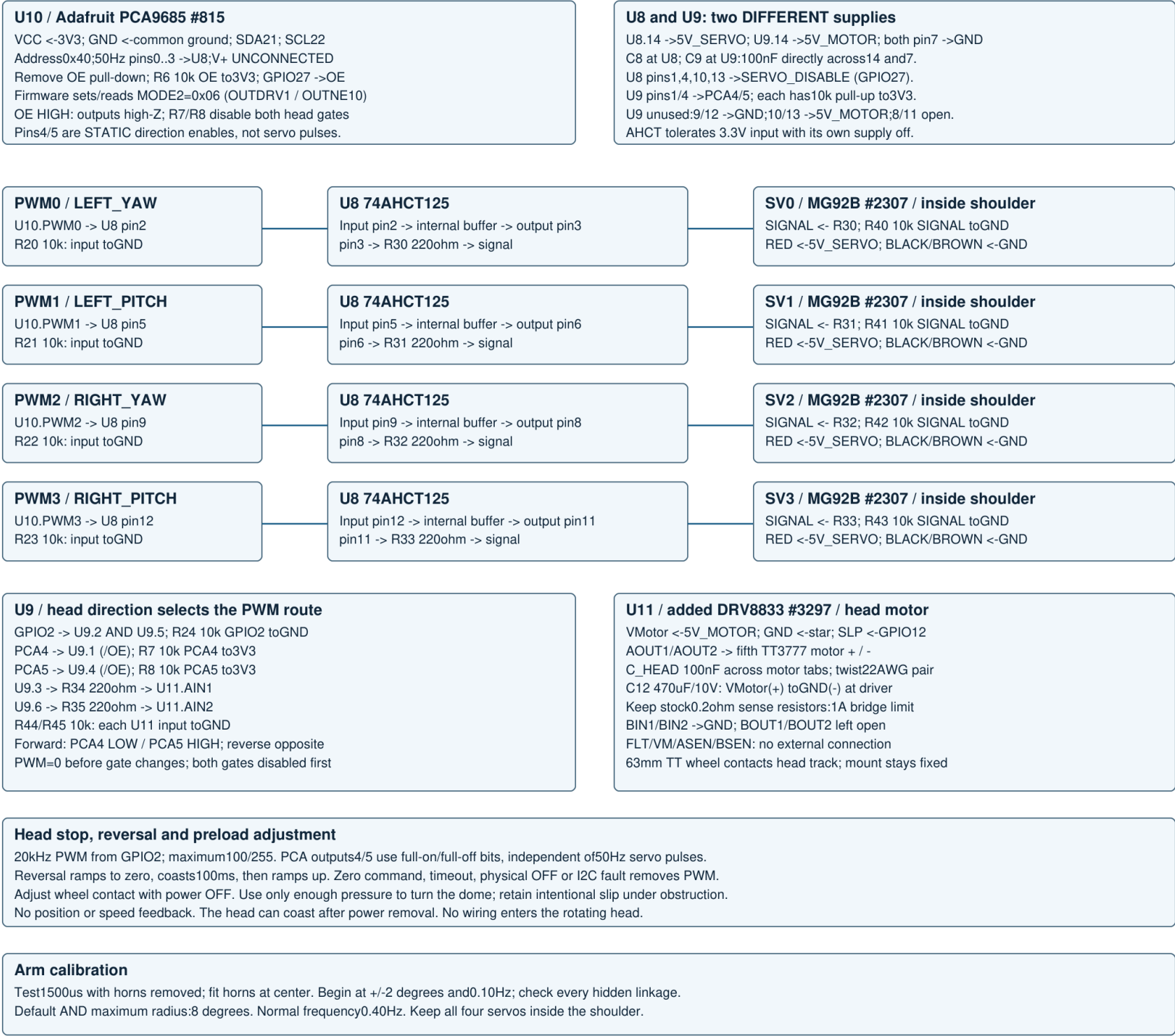
VIN <- 5V_LOGIC; GND <- common ground
BCLK <- GPIO17; LRC <- GPIO13; DIN <- GPIO15
GAIN -> R14 100k -> 5V_LOGIC (3dB)
SD: leave factory 1Mohm resistor configuration (mono mix)
OUT+ -> SPK1+; OUT- -> SPK1-; NEITHER lead to GND
SPK1: Adafruit 1313, 8ohm 1W; fixed digital gain 0.28
No external DAC, SD card or MP3 trigger module required.

Drive commissioning

100nF ceramic across each motor's metal tabs.
Use fast-decay PWM: one input PWM, other input low.
Stop both inputs low and SLP low; this is coast, not a brake.
Check each motor forward direction with wheels in air.
Swap that motor's two leads if its direction is wrong.
1A limit protects the driver, not indefinite motor stalls.
Indoor flat surface; no carpet, stairs or sustained wheel jams.

03 / Four arm servos and head PWM routing

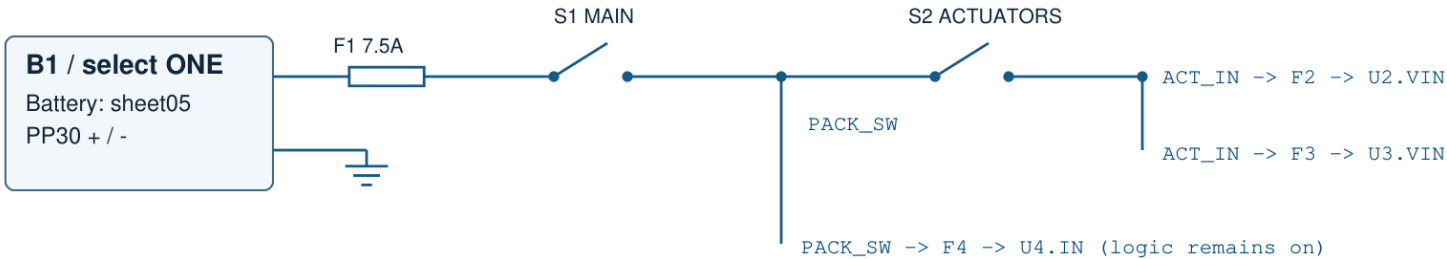
All chip numbers below are DIP-14 PHYSICAL pin numbers. U8/U9 are 74AHCT125, not 74HC125.



04 / Circuit details: power, sensing and bridge outputs

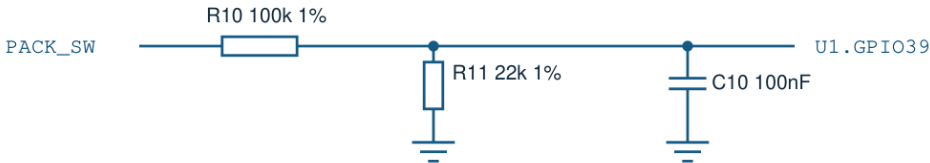
Conventional connection symbols. Dots are junctions. Matching net labels join all five sheets.

A. Battery fuse and independent motion disconnect



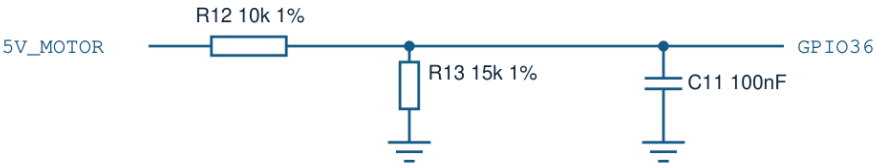
S1/S2 are shown OFF. NKK S1A ON joins terminals1 and3. Ground is never switched.

B. Battery measurement / GPIO39



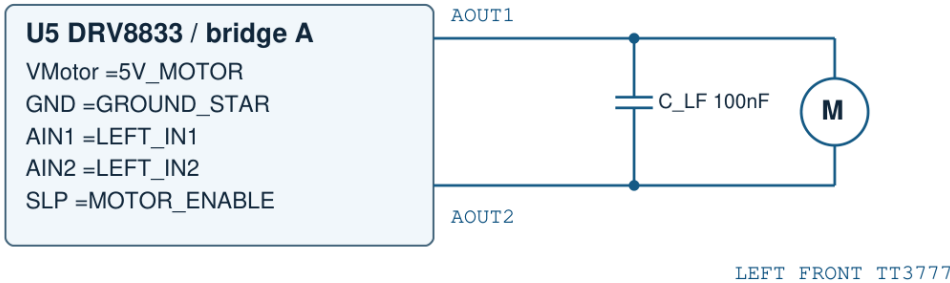
14.7 V: 2.651 V nominal / 2.695 V worst 1%; multiplier 5.54545

C. Motion power sense / GPIO36



5V x15/(10+15) =3.0V; low means physical power OFF

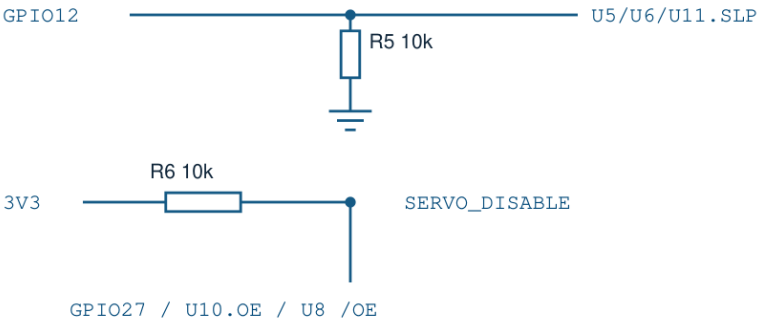
D. Motor output circuit (repeat for all5 motors)



U5.B ->LEFT REAR; U6.A/B ->RIGHT FRONT/REAR; U11.A ->HEAD (routing on sheet03).

Bridge outputs never join one another. The only shared connections are INPUTS and supplies.

E. Reset default states



Remove originalPCAOE pull-down.

GPIO2 and head AIN1/AIN2:10k pull-downs.

POWER NOTES: F2/F3=3A; F4=1A. Correct capacitor polarity. All connection values also appear in sheets01-03.

DALEK ROUND / FRICTION | 2026-09-12 | Circuit + wiring.csv form one drawing set | Physical validation required

electronics\04-circuit-details.svg | Drawing page: 17 x 11 inches. Use named nets and the wiring schedule together.

05 / Battery choices and removable SLA adapter

Choose ONE B1 option. These are alternative batteries, not three batteries wired together.

Option A / retained Bioenno

B1: BLF-1206A, 12 V 6 Ah LiFePO4
0.7 kg; 72 Wh; 12 A continuous rating
Keep factory PP30 discharge connector
Keep separate factory charge connector
CH1: BPC-1502DC, 14.6 V 2 A LiFePO4
This charger is for Option A ONLY.
Foam pads take up space in large cradle.

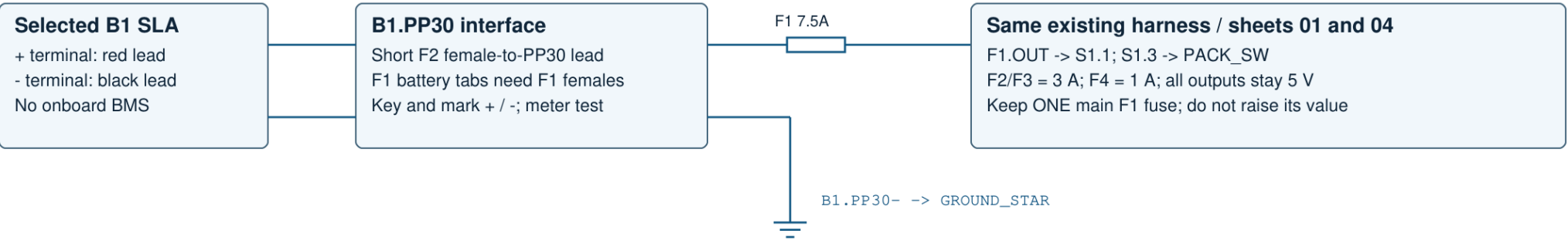
Option B / Power-Sonic 7 Ah

B1: PS-1270 F2, 12 V 7 Ah sealed AGM
151 x 65 x 94 mm case; 100 mm over tabs
Dimensions +/-2 mm; standard 1.97 kg
F2 blades: 6.35 x 0.8 mm
Use insulated blade-to-PP30 adapter
CH1-SLA: PSC-121000ACX, 12 V 1 A
PS-1270 also lists PSC-12500ACX.

Option C / Power-Sonic 14 Ah

B1: PS-12140 F2, 12 V 14 Ah sealed AGM
151 x 98 x 95 mm case; 101 mm over tabs
Dimensions +/-2 mm; 3.78 kg current
Older units can weigh 4.2 kg; weigh yours
Use insulated blade-to-PP30 adapter
CH1-SLA: PSC-121000ACX, 12 V 1 A
Loaded TT drive test is REQUIRED.

Selected SLA battery -> insulated adapter -> existing run harness (18 AWG copper)



Adapter assembly and fuse location

Cover each battery blade and contact separately.
Use female contacts rated >=10 A DC and matched to 18 AWG.
SLA positive terminal to F1: <=100 mm TOTAL lead/connector path.
For Bioenno, put F1 <=100 mm from its intact factory PP30.
Strain-relieve leads; no live clips or exposed metal in the body.

Charger and low-battery rules

MAIN OFF; unplug PP30; remove battery before any charging.
SLA uses the SLA charger; Bioenno uses its LiFePO4 charger.
Never join packs, charge through TTGO, or run while charging.
Motion stop at 11.2 V leaves logic on: MAIN OFF and UNPLUG.
Keep the SLA unplugged between uses; do not seal its vents.

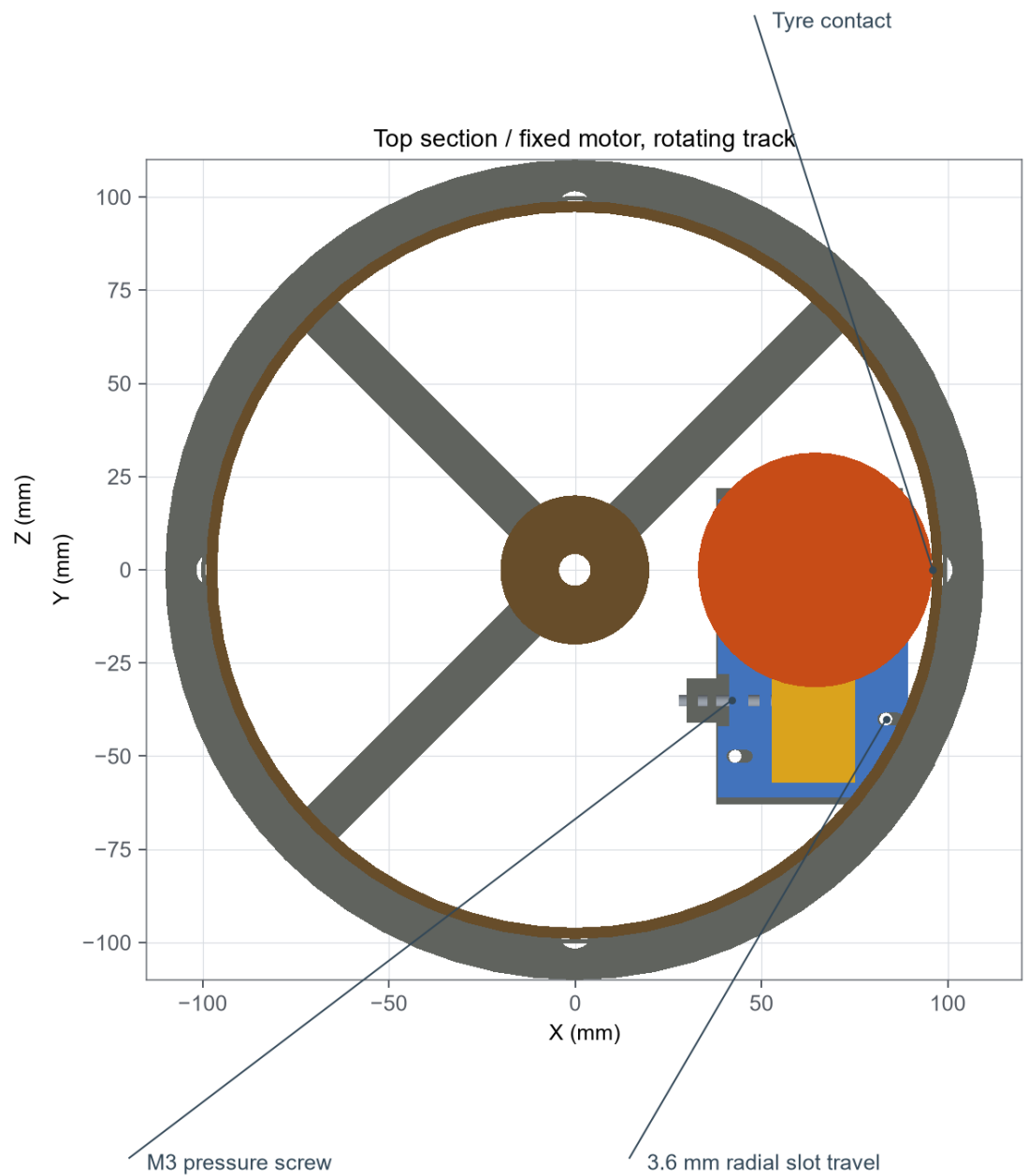
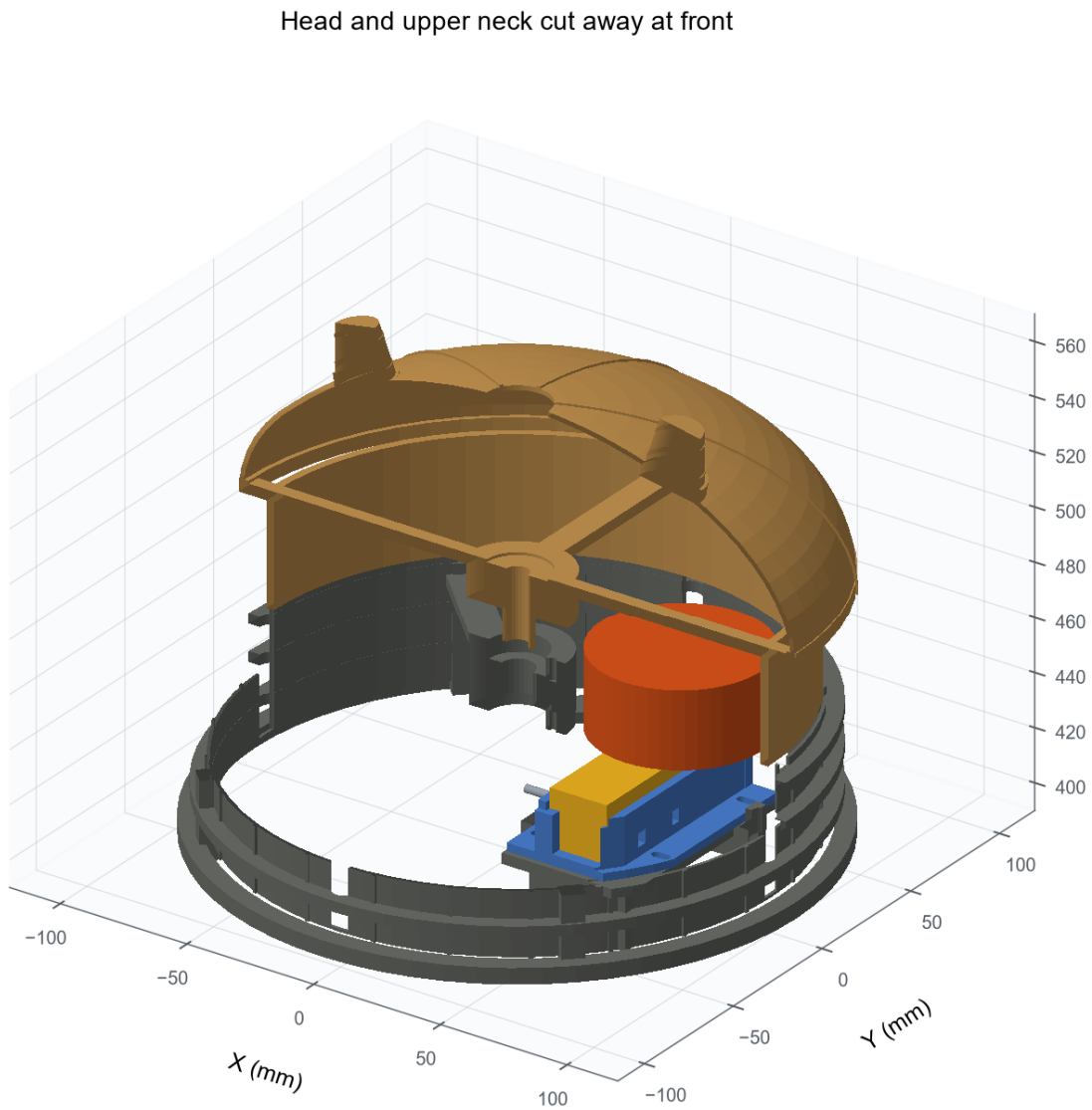
Mechanical and electrical acceptance

Maximum bare SLA envelope: 100 X 153 Y 103 Z mm including dimensional tolerance and terminal blades.
Contacts, insulating boots, tie heads and wire bends need additional free space below the printed platform.
Battery rests on the base; platform legs carry the boards. Independent padded ties restrain the battery.
Measure actual battery mass and dimensions; capacity labels do not establish universal fit or motor payload.
Check all 5 V rails at 4.8-5.2 V under load; repeat hard-floor turning and temperature tests for each battery option.
The stock 1 A motor bridge limits and 5 V motor supply stay unchanged. No greater drive rating is claimed.
Read docs/battery-options-research.md for sources, fit limits and the removable-skirt service procedure.

Adjustable head friction drive

Dalek FACET-1 / adjustable head friction drive

Fixed drive support is integral with shoulder | Separate sliding carriage preserves 3.6 mm adjustment



Blue: sliding carriage | Orange: 63 mm tyre | Yellow: motor envelope | Set light contact; preserve slip if obstructed.

Actual printed geometry shown in cutaway. Follow the mechanical adjustment and locking sequence with power isolated.